



NIST AI RMF 600-1

VMware Cloud Foundation 9.1

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Disclaimer

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Version

910-20260623-01

Introduction

This document describes how VMware Cloud Foundation (VCF) 9.1 and the separately licensed advanced services assessed alongside it, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, and VMware Avi Load Balancer, can support organizations aligning with the NIST Generative Artificial Intelligence Profile (NIST AI 600-1). The Generative AI Profile is a cross-sectoral profile of the NIST AI Risk Management Framework. It identifies risks that are unique to or exacerbated by generative AI and suggests actions for managing them, organized under the AI RMF functions, categories, and subcategories. Each assessed product is reported separately for every control, with VCF covering the full platform, including its consumption services and Private AI Services capabilities. The responsibility for identifying the organization's AI risk profile and applying the profile's suggested actions remains with the organization.

<https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf>

This guidance evolves. Please check the current revision at: <https://brcm.tech/vcf-compliance>

Additional Resources

This is one of a library of companion guides that map VMware Cloud Foundation and its advanced services to security and compliance frameworks. For each framework, the guide is published as a spreadsheet, a CSV data file, a Markdown document, and a PDF, together with a Security Configuration Guide crosswalk that identifies the VCF technical hardening settings relevant to that framework. The companion guide library, the VMware Cloud Foundation Security Configuration Guide, and related security documentation, sample scripts, and Q&A are available at <https://brcm.tech/vcf-security>.

Third Party Identifiers and Mappings

This document includes regulatory compliance and security control identifiers from external sources as a convenience to end users. This does not constitute endorsement, in either direction.

There is not a one-to-one mapping of product capabilities to third-party controls. A product capability, or set of capabilities, may be applicable to multiple controls. Conversely, a control may be satisfied with the use of multiple capabilities.

Control identifier numbers have been included from the Secure Controls Framework, version 2026.1, under the terms of the Creative Commons Attribution-NoDerivatives 4.0 International Public License. No modifications have been made to the control identifier numbers under the terms of the license.

<https://securecontrolsframework.com/>

Coverage Levels

Each control entry below carries the framework control text, the SCF control or controls it maps to, and an Aggregate Coverage line showing the strongest position any one of the assessed products holds for that control. Products that meet or contribute to the control follow as subsections, each with its own coverage level and response. Products the control does not apply to are listed together at the end of the entry, with a brief explanation. The coverage levels are:

- **Meets** -- the product directly provides the capability described in the control. No additional organizational process, external product, or human activity is needed beyond configuring the product itself.
- **Contributes** -- the product provides technical capabilities that help satisfy the control, but the control also requires something outside the product to be fully met, such as an organizational process, human activity, or an additional tool.
- **Not Applicable** -- the control does not apply to the product. Physical facility controls, staffing decisions, organizational governance processes, and domains the product does not operate in fall here.

- **Not Supported** -- the control is relevant to the product's domain (infrastructure, networking, security), but the product lacks the specific capability the control requires.
- **Not Assessed** -- the control has not yet been evaluated against the product.

Controls

Control GOVERN 1.1

Legal and regulatory requirements involving AI are understood, managed, and documented.

SCF Controls: CPL-01, GOV-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 1.2

The characteristics of trustworthy AI are integrated into organizational policies, processes, procedures, and practices.

SCF Controls: AAT-01, GOV-01, OPS-01.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 1.3

Processes, procedures, and practices are in place to determine the needed level of risk management activities based on the organization's risk tolerance.

SCF Controls: OPS-01.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 1.4

The risk management process and its outcomes are established through transparent policies, procedures, and other controls based on organizational risk priorities.

SCF Controls: AAT-07, RSK-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 1.5

Ongoing monitoring and periodic review of the risk management process and its outcomes are planned and organizational roles and responsibilities clearly defined, including determining the frequency of periodic review.

SCF Controls: AAT-08, HRS-03

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 1.6

Mechanisms are in place to inventory AI systems and are resourced according to organizational risk priorities.

SCF Controls: AAT-02, AST-02

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides a centralized inventory of AI models through the Model Gallery in Harbor Registry. Each model gallery has a unique name and is organized within Harbor projects with configured user access controls. Model revisions are stored as OCI artifacts with immutable manifests, and each revision is assigned a unique digest, giving a verifiable record of every model version in the inventory.

OCI metadata describes model contents and dependencies, supporting identification of compatible platforms and giving visibility into what each model requires. The vcf pais models list command retrieves and displays all models available in a model gallery, giving operators a direct inventory view. Model Runtime tracks which models are actively deployed to inference endpoints, distinguishing between stored and deployed models.

For third-party AI components, PAIS validates running AI workloads based on NVIDIA GPU Cloud (NGC) containers, a curated catalog of GPU-optimized, pre-tested containers. In disconnected environments, a Harbor registry stores container images from the NGC catalog, acting as a controlled intermediary that makes third-party AI components visible and trackable within the organization's infrastructure.

VMware vDefend (Contributes)

VMware vDefend's Security Services Platform (SSP) maintains an inventory of network-visible workloads organized into a hierarchical asset collection spanning regions, zones, environments, applications, and application tiers. The Publish Inventory Assets tab in SSP displays infrastructure asset details including asset names, asset types, IP addresses, and total VM counts for assets sourced from imported CSV files, discovered services, or published infrastructure assets, providing accurate, granular records of network-visible assets.

Platform Inventory monitoring supports ongoing inventory accuracy through several mechanisms. Administrators can filter infrastructure assets by Name and Status, identifying assets that are timed out or out of sync. The inventory hierarchy view shows regions, zones, and VMs so administrators can verify correct definition. SSP Inventory displays assigned tags for each infrastructure asset, and environment rows can be expanded to view tag assignments and confirm hierarchy accuracy. The platform also collects and displays VM tags with their respective scopes, supporting classification and tracking. Published applications and their application types are verifiable in the Inventory > Applications interface.

SSP identifies and publishes application assets, classifying them as either critical or regular applications. Application Asset Collections represent groups of workloads delivering enterprise software functions. Security Intelligence within SSP displays application associations for compute entities, indicating the tiers within each application to which a given compute entity belongs. Administering compute entity classifications in Security Intelligence requires appropriate privileges, supporting access control over classification changes. Consistent naming conventions are recommended when preparing inventory data for import, and the Asset Prefix setting enables standardized naming during segmentation planning and monitoring.

In disconnected mode, usage files include an asset_id field providing a unique asset identifier, supporting asset tracking in environments without continuous connectivity.

vDefend's inventory scope is limited to network-visible workloads and does not replace enterprise-wide TAASD inventory tools, authorized software catalogs with business justification records, or data asset registries required for full AST-02 compliance.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) contributes to asset inventory requirements by providing built-in visibility into the database instances, infrastructure resources, and users it manages. This visibility supports tracking and accountability for the subset of technology assets that DSM governs. Organization-wide TAASD inventory across all asset categories requires organizational processes and additional tooling beyond DSM itself.

DSM tracks the association between deployed databases and the infrastructure policies under which they were provisioned. Administrators can review databases and their configuration details through the Databases tab in the Infrastructure Policies view of the DSM Administrator UI, which displays each policy alongside the databases bound to it. Because infrastructure policies define the compute, storage, and networking resources allocated to databases, including allowed VM classes, this view gives administrators a structured picture of how platform resources are being consumed. The About the DSM UI Views section confirms the Infrastructure Policies view lists all policies available in the environment and the databases associated with each.

DSM administrators can define and configure data service policies for the data services DSM manages, supporting compliance with organizational standards across tenants. Each policy can be tailored to specific requirements and assigned to one or more organizations to enforce consistent configurations. For tenant-facing self-service, the DSM Consumption Operator creates infrastructure policies as cluster-scoped resources in the consumption cluster and requires an allowedInfrastructurePolicies list that defines which policies a tenant cluster may use, giving administrators a clear inventory of the policies in scope per cluster. Tenant users create and consume databases under a Supervisor namespace through the VCF Automation Cloud Consumption Interface (CCI), and the DSM portal exposes a Consumption Operator endpoint that provides access to deployment artifacts.

For database-level visibility, the Database Health view in the DSM UI collects metric data for each database instance and cross-references alerts with resolution paths. DSM deploys a Telegraf agent on the Provider Appliance and a local Telegraf agent on each workload cluster to gather VM-level metrics such as CPU, memory, and disk usage, plus engine-specific metrics. DSM can forward metrics to an external Metrics target server, supporting integration with external monitoring or inventory aggregation systems.

For auditors, DSM's inventory-relevant information is accessible through the Administrator UI and is available to personnel with the Admin role. The Databases tab, Infrastructure Policies view, Database Health view, and Consumption Operator policy inventory collectively provide the granularity needed to account for DSM-managed database assets and the policies governing them. The documented DSM user roles (vSphere administrator, VCF Automation organization administrator, VCF Automation project user) clarify which personnel have visibility into which inventory views.

DSM provides resource tracking and visibility within its own management scope. DSM does not perform organization-wide asset discovery and does not inventory physical infrastructure, non-database applications, or data assets outside its management domain. A TAASD inventory program spanning all asset categories requires a CMDB or dedicated asset management solution that can aggregate DSM resource data alongside information from other platform components.

Not applicable for this control: VCF Protection and Recovery, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 1.7

Processes and procedures are in place for decommissioning and phasing out AI systems safely and in a manner that does not increase risks or decrease the organization's trustworthiness.

SCF Controls: AAT-01, AST-30

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 2.1

Roles and responsibilities and lines of communication related to mapping, measuring, and managing AI risks are documented and are clear to individuals and teams throughout the organization.

SCF Controls: AAT-08

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) supports differentiated roles and responsibilities across the AI workload lifecycle through platform-level access controls layered on the VCF identity model. Platform administrators manage infrastructure setup: they deploy infrastructure, configure GPU-accelerated workload domains, and add the PAIS supervisor service to the supervisor before any AI workloads can be deployed. This role controls the foundational infrastructure on which AI operations depend.

Organization administrators control namespace imports and grant permissions for AI catalog items. The organization administrator token is required for API calls that import namespaces into VCF Automation organizations, centralizing oversight of which AI capabilities are exposed. Organization administrators must grant permissions before users can request AI catalog items, providing a gatekeeping function over AI deployments.

DevOps engineers are scoped to project and namespace membership for provisioning GPU-accelerated VMware Kubernetes Service (VKS) clusters and related infrastructure. Data scientists and MLOps engineers provision models and retrieval-augmented generation (RAG) applications through PAIS. The MLOps role carries specific validation responsibilities: MLOps engineers validate ML models against the security, privacy, and technical requirements of the organization before uploading them to the model gallery, including testing inference functionality and evaluating performance and safety against business use cases.

Access to PAIS is authorized through OIDC groups specified in the `authorizedGroups` configuration, tying role enforcement to VCF SSO. Namespace-scoped permissions confine each role's actions to assigned boundaries within the platform.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 3.2

Policies and procedures are in place to define and differentiate roles and responsibilities for human-AI configurations and oversight of AI systems.

SCF Controls: AAT-08

Aggregate Coverage: Contributes

VCF (Contributes)

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Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 4.1

Organizational policies and practices are in place to foster a critical thinking and safety-first mindset in the design, development, deployment, and uses of AI systems to minimize potential negative impacts. Continued on next page

SCF Controls: AAT-01, GOV-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 4.2

Organizational teams document the risks and potential impacts of the AI technology they design, develop, deploy, evaluate, and use, and they communicate about the impacts more broadly.

SCF Controls: AAT-07.2, AAT-09

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 4.3

Organizational practices are in place to enable AI testing, identification of incidents, and information sharing.

SCF Controls: AAT-10

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure that supports AI Test, Evaluation, Validation and Verification (TEVV) activities throughout the model lifecycle. Deep learning VMs provisioned from PAIS images provide dedicated testing environments for AI prototyping, fine-tuning, validation, and inference workloads. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so TEVV activities occur in conditions comparable to the deployment environment.

The validate-then-publish workflow creates a formal gate for AI deployment. MLOps engineers validate models for inference functionality using the Triton Inference Server and evaluate performance and safety against business use cases, including examination of inference requests for malicious behavior and functional testing. Models must pass this validation before being uploaded to the Model Gallery for production use. Model integrity is verified through hash code validation, and each revision is assigned a unique digest stored as an OCI artifact with an immutable manifest, giving cryptographic verification that the tested model is the same artifact that gets deployed.

The platform also supports TEVV through its monitoring stack. VCF Operations monitors GPU consumption at cluster, host system, and host properties levels. Each DL VM includes a pre-installed DCGM Exporter with Prometheus collection at 5-second scrape intervals and Grafana visualization, supplying performance telemetry during testing and evaluation. The software stack for running AI applications on top of NVIDIA GPUs is validated in advance, and NVIDIA NGC containers are pre-tested and ready to run, reducing variables during TEVV by providing a known-good baseline.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 5.1

Organizational policies and practices are in place to collect, consider, prioritize, and integrate feedback from those external to the team that developed or deployed the AI system regarding the potential individual and societal impacts related to AI risks.

SCF Controls: AAT-11

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 6.1

Policies and procedures are in place that address AI risks associated with third-party entities, including risks of infringement of a third-party's intellectual property or other rights.

SCF Controls: AAT-01, AAT-12

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GOVERN 6.2

Contingency processes are in place to handle failures or incidents in third-party data or AI systems deemed to be high-risk.

SCF Controls: AAT-01, AAT-10, AAT-11.4

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.1-001

Align GAI development and use with applicable laws and regulations, including those related to data privacy, copyright and intellectual property law.

SCF Controls: CPL-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.2-001

Establish transparency policies and processes for documenting the origin and history of training data and generated data for GAI applications to advance digital content transparency, while balancing the proprietary nature of training approaches.

SCF Controls: AAT-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.2-002

Establish policies to evaluate risk-relevant capabilities of GAI and robustness of safety measures, both prior to deployment and on an ongoing basis, through internal and external evaluations.

SCF Controls: AAT-01, GOV-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.3-002

Establish minimum thresholds for performance or assurance criteria and review as part of deployment approval (“go/no-go”) policies, procedures, and processes, with reviewed processes and approval thresholds reflecting measurement of GAI capabilities and risks.

SCF Controls: GOV-05, GOV-05.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.3-003

Establish a test plan and response policy, before developing highly capable models, to periodically evaluate whether the model may misuse CBRN information or capabilities and/or offensive cyber capabilities.

SCF Controls: AAT-10

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.3-004

Obtain input from stakeholder communities to identify unacceptable use, in accordance with activities in the AI RMF Map function.

SCF Controls: GOV-01.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.3-005

Maintain an updated hierarchy of identified and expected GAI risks connected to contexts of GAI model advancement and use, potentially including specialized risk levels for GAI systems that address issues such as model collapse and algorithmic monoculture.

SCF Controls: AAT-01, AAT-02.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.3-006

Reevaluate organizational risk tolerances to account for unacceptable negative risk (such as where significant negative impacts are imminent, severe harms are actually occurring, or large-scale risks could occur); and broad GAI negative risks, including: Immature safety or risk cultures related to AI and GAI design, development and deployment, public information integrity risks, including impacts on democratic processes, unknown long-term performance characteristics of GAI.

SCF Controls: AAT-10.4

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.3-007

Devise a plan to halt development or deployment of a GAI system that poses unacceptable negative risk.

SCF Controls: AAT-18.1

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides AI-specific mechanisms that support an organization's risk response process for AI systems. MLOps engineers validate ML models onboarded to PAIS against the security, privacy, and technical requirements of the organization, providing a structured gate for remediation when an existing model is identified as risky and a replacement must be vetted before publication. When remediation requires substituting a model, operators can deploy completion or embedding model endpoints and use A/B testing to compare the behavior of candidate models against the incumbent in real-world settings, supporting empirical evaluation of the replacement before full rollout. Embedding model selection is recommended to be based on empirical evaluation using the organization's own data rather than relying solely on benchmarks.

Health and performance metrics for PAIS components running in a VCF Automation namespace can be visualized in observability platforms such as Grafana and VCF Operations. MLOps engineers and application developers can monitor the health, quality, and behavior of AI applications through these observability surfaces to detect anomalies that warrant a risk response. PAIS observability is configured through a Prometheus runtime with a specified storage class and a configurable metrics retention period, with a default of 14 days, giving risk responders historical context for investigation. Metrics collection is activated by enabling the `spec.observability.prometheusRuntime` section of the `PAISConfiguration` CR.

Organizational processes for prioritizing AI risks based on assessments, coordinating remediation across stakeholders, and tracking risk closure remain organizational responsibilities outside the scope of PAIS infrastructure capabilities. General platform resilience inherited from VCF is covered under VCF Coverage and is not a PAIS-specific contribution to this control.

VMware vDefend (Contributes)

VMware vDefend contributes to prioritizing, responding to, and remediating security risks affecting AI workloads running on VCF by providing severity-based threat prioritization, multiple response mechanisms, and policy-based remediation at the network layer.

Prioritization. Security Intelligence categorizes detected suspicious traffic events by impact type (Critical, High, Medium, and Low), helping security teams focus first on the most urgent risks to AI workloads. The Security Services Platform Campaign Impact Score measures the overall severity of correlated threat campaigns on a scale from 0 through 100, with scores calculated using the confidence level of each constituent detection and higher scores indicating higher-priority campaigns for triage. Malware Prevention's Process Analysis Report classifies risk estimates and flags artifacts as High risk when they represent a critical risk requiring priority attention. vDefend IDS/IPS follows a prioritized event-handling approach under memory pressure, dropping lower-priority severity data (Low and Informational) first and retaining Critical and High severity events, so that the most significant detections are preserved for triage and response.

Response. When security risks to AI workloads are identified, vDefend provides multiple response options calibrated to the situation. Administrators can deploy microsegmentation rules to isolate affected AI workloads, adjust IDS/IPS profiles to address newly identified threat patterns, or modify Gateway Firewall policies to restrict traffic at the perimeter. The Security Intelligence threat visualization canvas supports rapid threat containment by identifying compromised assets and guiding microsegmentation policy updates. Intelligent Assist analyzes detected anomalies and recommends remediation policies; remediation rules are added using the naming convention prefix 'ai-' and are not enabled by default, requiring an analyst or administrator to review and manually enable them after due diligence and verification. When Intelligent Assist determines that a given anomaly cannot be remediated, it indicates this automatically, allowing analysts to focus on actionable findings. Security Intelligence retains the IDS policies, rules, and security profiles created by Intelligent Assist as part of remediation operations, providing an audit trail of response actions taken.

Remediation. The Security Intelligence recommendation engine generates firewall policy recommendations based on observed traffic patterns between virtual machines. These recommendations can be published as firewall policies applied to environment category pairs, enabling organizations to close segmentation gaps identified during risk assessments. Administrators can designate high-priority applications and groups through prioritized collections, focusing Security Intelligence's flow analysis on the AI workloads most critical to protect.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.4-001

Establish policies and mechanisms to prevent GAI systems from generating CSAM, NCII or content that violates the law.

SCF Controls: GOV-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.4-002

Establish transparent acceptable use policies for GAI that address illegal use or applications of GAI.

SCF Controls: GOV-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.5-001

Define organizational responsibilities for periodic review of content provenance and incident monitoring for GAI systems.

SCF Controls: AAT-08, AAT-12.1, HRS-03

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.5-002

Establish organizational policies and procedures for after action reviews of GAI system incident response and incident disclosures, to identify gaps; Update incident response and incident disclosure processes as required.

SCF Controls: AAT-01, GOV-02, OPS-01.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.5-003

Maintain a document retention policy to keep history for test, evaluation, validation, and verification (TEVV), and digital content transparency methods for GAI.

SCF Controls: AAT-01, AAT-10, DCH-18

Aggregate Coverage: Contributes

VCF (Contributes)

VCF provides configurable data retention controls within VCF Operations that allow administrators to align operational data retention periods with organizational requirements. These controls do not address the full scope of media and data retention governance, but they give administrators granular tuning over how long different categories of platform-generated data are kept.

VCF Operations exposes data retention settings through Administration > Global Settings > Data Retention. Administrators can configure retention periods independently for several data categories. Metrics data supports a retention window between 1 month and 13 months. Entities and configuration data can be retained for a minimum of 1 month and a maximum of 3 months. Organization reports support a maximum data retention period of 1,440 days. At the normal retention tier, VCF Operations retains data at 5-minute granularity for 6 months, then rolls up the seventh month into 1-hour aggregations. A real-time retention tier provides 5-minute granularity, a standard tier retains data for 1 month, and a long-term tier retains data for 13 months. Different data retention policies can be configured and controlled for each category according to organizational requirements. Log data is governed separately through VCF Log Management, where administrators define log partitions that each carry their own retention period in days, with logs removed from disk automatically once a partition's period elapses and the option to archive partitions to external NFS or S3 storage for longer-term preservation.

VCF Operations for Networks provides additional granular retention controls. The default data retention period is 1 month in the advanced license edition. Flow data retention is fixed at a 1-month window within the platform. For organizations that need to retain flow data beyond one month, the Streaming Databus provides a mechanism to export flows to an external subscriber via HTTPS/HTTP for extended retention. Miscellaneous data retention can be configured with up to 100 GB of additional disk space.

VCF Operations also provides backup capabilities that support regulatory and operational retention needs. Administrators can take regular backups of both custom and out-of-the-box content. When reducing an entity metrics retention period in VCF Operations for Networks, the recommendation is to back up log files first to prevent data loss, which supports the principle of preserving data before modifying retention windows.

The Consumption layer of VCF, comprising VMware Kubernetes Service (VKS), vSphere Supervisor, and vSphere namespaces, adds Kubernetes-native persistent storage lifecycle controls relevant to data retention for containerized workloads. Persistent volumes in Supervisor namespaces store application data outside the lifecycle of individual pods and workloads, so data remains available when workloads restart or scale. The central mechanism governing retention behavior is the PersistentVolume reclaim policy, which controls what happens to the underlying storage and its data when the associated PersistentVolumeClaim is deleted. The Retain policy preserves the volume and its data in a Released state for manual administrator reclamation, leaving the data intact for inspection, archival, or transfer before final disposal. The Delete policy causes the storage asset to be removed when the claim is released. Retain is the default for manually provisioned PersistentVolumes. StorageClass definitions carry a ReclaimPolicy field that sets the default behavior for all dynamically provisioned volumes within a given storage class, allowing cluster administrators to enforce organization-wide retention defaults at the storage class level rather than per-volume. For StatefulSet workloads, Kubernetes provides a persistentVolumeClaimRetentionPolicy setting with a whenDeleted field that controls whether PVCs and their backing volumes are retained or deleted when the StatefulSet is removed; scaling a StatefulSet down does not delete the associated volumes by default. Organizations subject to retention obligations should configure StorageClasses with the Retain policy so that workload termination does not destroy data that must be kept for a defined period.

Storage Policy Based Management (SPBM) provides a unified control plane across VKS storage, enabling storage policies to be associated with persistent volumes that capture the storage characteristics required by a given workload or retention regime. VKS clusters surface a storage compliance status per volume: a Compliant status indicates that the datastore backing the volume satisfies the requirements of the associated policy; an Out of Date status indicates the policy has been updated but the new requirements have not yet been communicated to the datastore, requiring the policy to be reapplied to bring the volume back into compliance. This policy-driven model allows administrators to codify storage requirements and verify adherence continuously.

For Harbor Registry deployments on VKS clusters, administrators should note that configuring a StorageClass or PersistentVolume with a Delete reclaim policy results in all Harbor data being deleted when the Harbor package is removed. Image artifact retention requirements should drive reclaim policy selection for Harbor-backing storage. Platform teams should also manage data protection storage locations in VCF Automation to align with company-specific data residency and infrastructure standards.

These retention controls apply to the operational, monitoring, and configuration data that VCF Operations collects and manages, and to the persistent volume lifecycle for containerized workloads on VKS. They do not extend to application data stored within virtual machines outside the Kubernetes layer, user-generated content on datastores not managed by Kubernetes, or media outside the VCF platform. Organizations must establish broader data retention policies, classification schemes, and lifecycle management processes that address all media and data types subject to statutory, regulatory, and contractual obligations. VCF's retention settings are one component of a larger retention program that requires organizational governance, legal review of applicable obligations, and potentially additional tools for archival and disposition of data beyond what VCF Operations and VKS storage lifecycle controls manage.

VMware vDefend (Contributes)

VMware vDefend provides configurable data retention capabilities for the security telemetry and flow data it collects, contributing to an organization's ability to retain security-relevant data in accordance with statutory, regulatory, and contractual obligations.

Security Services Platform, which underpins Network Detection and Response (NDR) and related analytics services, offers control over flow data retention. Flow data is retained for 30 days by default via the flow-data-retention setting. If storage capacity is insufficient to handle flow volume from workloads, some flow metadata may be lost, flow ingestion may be paused, or data retention may be reduced dynamically to stay within available storage. Administrators can monitor retention health through the `ssp_flow_storage.avg_flow_storage_ratio` metric, which reports the ratio of current retention days to predicted full days for flow storage, and through the `ssp_analytics_storage.avg_predicted_tabl_retention_days` metric, which tracks predicted retention for the correlated flow visualization table (default 30 days). Security Services Platform persistent storage collects data even when the License Hub is offline, providing resilience against brief service interruptions that might otherwise affect data continuity.

Network Detection and Response retains detection events and campaign data for 12 months, providing a fixed retention window for threat detection artifacts.

For organizations using cloud-connected vDefend features, Security Services Platform maintains defined retention periods for data collected under active service subscriptions: IDS events and other network events, VM identifiers, rules and policies, conversation history, and security profiles are retained for one year during an active subscription, and for 90 days following subscription expiration.

Administrators should note that when the Malware Prevention Service feature is deleted, all previously gathered data analytics are permanently removed. Organizations with compliance-driven retention requirements should account for this before decommissioning that service.

These retention controls allow organizations to align vDefend's security data storage with their compliance requirements. The broader scope of media and data retention across the full infrastructure stack, including operational data, log archives, backup snapshots, and storage-level data protection, is handled by other VCF components such as VCF Operations, VCF Log Management, and vSAN Data Protection.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) provides configurable retention mechanisms that allow administrators to align disaster recovery and cyber recovery data with retention obligations. PNR snapshot schedules define a retention policy that determines how long each snapshot is retained, with hourly, daily, weekly, and monthly cadences and a maximum retention duration of three years. Administrators can also select predefined retention templates that combine recovery point objective (RPO) and snapshot retention parameters for quick configuration of protection groups.

For vSphere Replication, the retention policy determines the expiration of point-in-time copies, and storage consumption drops after all point-in-time copies referring to the original disk are expired. Administrators configure the number of retention slots and the spacing between recovery points, so older instances expire automatically once the configured limit is exceeded.

Cyber Recovery snapshot schedules in PNR follow the same retention model, defining a retention policy that controls how long each snapshot is retained. Administrators should account for storage capacity when configuring longer retention schedules, since extended retention periods increase storage consumption on the underlying datastores.

These retention controls allow organizations to align disaster recovery and cyber recovery data retention with applicable statutory, regulatory, and contractual obligations. PNR provides the technical mechanisms; determining the specific retention durations that satisfy each obligation is an organizational policy decision.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) provides configurable backup retention controls that support an organization's data retention requirements for its managed databases. These controls do not define retention obligations, which come from the organization's legal and compliance functions, but they provide the technical means to implement those obligations for PostgreSQL, MySQL, SQL Server, and MinIO data.

Retention periods are configured through Data Service Policies in the DSM administration interface. Each policy supports a backup-retention-period setting that defines the duration for which backups are kept. When multiple policies apply to a namespace, DSM aggregates the retention constraints by selecting the more restrictive minimum and maximum retention day values, producing a resultant retention window that reflects all applicable policies. Data service policies can also be tailored to specific requirements and enforced across different tenants, giving administrators a consistent mechanism for applying retention standards in multi-tenant deployments.

For deleted databases, DSM retains automated backups according to the configured retention period and exposes an Archives tab in the administration interface where administrators can access and adjust the retention period of retained backups. This allows organizations to maintain access to backup data even after a database instance has been removed, supporting retention obligations that outlast the operational lifetime of a specific database. When a database is deleted, automated backups continue to be retained until the retention policy is met.

DSM also supports point-in-time restore within the backup retention period, enabling recovery to any point within the retained window. On-demand backups behave differently from automated backups: they are permanent and do not expire automatically. On-demand backups remain in the backup storage location until an administrator manually deletes them, which means they can consume storage quota indefinitely and require separate tracking for retention compliance.

Data service policies also define backup storage locations (S3-compatible targets) alongside retention duration. If a data service policy that is in use is deleted, the data services associated with that policy become non-compliant, alerting administrators to a configuration gap.

Meeting this control fully requires the organization to determine the applicable retention periods, communicate those requirements to DSM administrators, and configure policies accordingly. DSM does not automatically derive retention obligations from regulatory or contractual sources. Its retention controls also cover only the database data it manages and do not address retention of all media and data types that may fall within an organization's broader retention obligations.

Not applicable for this control: VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of this product.

Control GV-1.6-001

Enumerate organizational GAI systems for incorporation into AI system inventory and adjust AI system inventory requirements to account for GAI risks.

SCF Controls: AAT-02, AST-02

Aggregate Coverage: Contributes

VMware Private AI Services (PAIS) (Contributes)

VMware Private AI Services (PAIS) provides a centralized inventory of AI models through the Model Gallery in Harbor Registry. Each model gallery has a unique name and is organized within Harbor projects with configured user access controls. Model revisions are stored as OCI artifacts with immutable manifests, and each revision is assigned a unique digest, giving a verifiable record of every model version in the inventory.

OCI metadata describes model contents and dependencies, supporting identification of compatible platforms and giving visibility into what each model requires. The `vcf pais` models list command retrieves and displays all models available in a model gallery, giving operators a direct inventory view. Model Runtime tracks which models are actively deployed to inference endpoints, distinguishing between stored and deployed models.

For third-party AI components, PAIS validates running AI workloads based on NVIDIA GPU Cloud (NGC) containers, a curated catalog of GPU-optimized, pre-tested containers. In disconnected environments, a Harbor registry stores container images from the NGC catalog, acting as a controlled intermediary that makes third-party AI components visible and trackable within the organization's infrastructure.

VMware vDefend (Contributes)

VMware vDefend's Security Services Platform (SSP) maintains an inventory of network-visible workloads organized into a hierarchical asset collection spanning regions, zones, environments, applications, and application tiers. The Publish Inventory Assets tab in SSP displays infrastructure asset details including asset names, asset types, IP addresses, and total VM counts for assets sourced from imported CSV files, discovered services, or published infrastructure assets, providing accurate, granular records of network-visible assets.

Platform Inventory monitoring supports ongoing inventory accuracy through several mechanisms. Administrators can filter infrastructure assets by Name and Status, identifying assets that are timed out or out of sync. The inventory hierarchy view shows regions, zones, and VMs so administrators can verify correct definition. SSP Inventory displays assigned tags for each infrastructure asset, and environment rows can be expanded to view tag assignments and confirm hierarchy accuracy. The platform also collects and displays VM tags with their respective scopes, supporting classification and tracking. Published applications and their application types are verifiable in the Inventory > Applications interface.

SSP identifies and publishes application assets, classifying them as either critical or regular applications. Application Asset Collections represent groups of workloads delivering enterprise software functions. Security Intelligence within SSP displays application associations for compute entities, indicating the tiers within each application to which a given compute entity belongs. Administering compute entity classifications in Security Intelligence requires appropriate privileges, supporting access control over classification changes. Consistent naming conventions are recommended when preparing inventory data for import, and the Asset Prefix setting enables standardized naming during segmentation planning and monitoring.

In disconnected mode, usage files include an `asset_id` field providing a unique asset identifier, supporting asset tracking in environments without continuous connectivity.

vDefend's inventory scope is limited to network-visible workloads and does not replace enterprise-wide TAASD inventory tools, authorized software catalogs with business justification records, or data asset registries required for full AST-02 compliance.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) contributes to asset inventory requirements by providing built-in visibility into the database instances, infrastructure resources, and users it manages. This visibility supports tracking and accountability for the subset of technology assets that DSM governs. Organization-wide TAASD inventory across all asset categories requires organizational processes and additional tooling beyond DSM itself.

DSM tracks the association between deployed databases and the infrastructure policies under which they were provisioned. Administrators can review databases and their configuration details through the Databases tab in the Infrastructure Policies view of the DSM Administrator UI, which displays each policy alongside the databases bound to it. Because infrastructure policies define the compute, storage, and networking resources allocated to databases, including allowed VM classes, this view gives administrators a structured picture of how platform resources are being consumed. The About the DSM UI Views section confirms the Infrastructure Policies view lists all policies available in the environment and the databases associated with each.

DSM administrators can define and configure data service policies for the data services DSM manages, supporting compliance with organizational standards across tenants. Each policy can be tailored to specific requirements and assigned to one or more organizations to enforce consistent configurations. For tenant-facing self-service, the DSM Consumption Operator creates infrastructure policies as cluster-scoped resources in the consumption cluster and requires an allowedInfrastructurePolicies list that defines which policies a tenant cluster may use, giving administrators a clear inventory of the policies in scope per cluster. Tenant users create and consume databases under a Supervisor namespace through the VCF Automation Cloud Consumption Interface (CCI), and the DSM portal exposes a Consumption Operator endpoint that provides access to deployment artifacts.

For database-level visibility, the Database Health view in the DSM UI collects metric data for each database instance and cross-references alerts with resolution paths. DSM deploys a Telegraf agent on the Provider Appliance and a local Telegraf agent on each workload cluster to gather VM-level metrics such as CPU, memory, and disk usage, plus engine-specific metrics. DSM can forward metrics to an external Metrics target server, supporting integration with external monitoring or inventory aggregation systems.

For auditors, DSM's inventory-relevant information is accessible through the Administrator UI and is available to personnel with the Admin role. The Databases tab, Infrastructure Policies view, Database Health view, and Consumption Operator policy inventory collectively provide the granularity needed to account for DSM-managed database assets and the policies governing them. The documented DSM user roles (vSphere administrator, VCF Automation organization administrator, VCF Automation project user) clarify which personnel have visibility into which inventory views.

DSM provides resource tracking and visibility within its own management scope. DSM does not perform organization-wide asset discovery and does not inventory physical infrastructure, non-database applications, or data assets outside its management domain. A TAASD inventory program spanning all asset categories requires a CMDB or dedicated asset management solution that can aggregate DSM resource data alongside information from other platform components.

Not applicable for this control: VCF Protection and Recovery, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.6-002

Define any inventory exemptions in organizational policies for GAI systems embedded into application software.

SCF Controls: AAT-02, AST-02

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-1.7-001

Protocols are put in place to ensure GAI systems are able to be deactivated when necessary.

SCF Controls: AAT-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-2.1-001

Establish organizational roles, policies, and procedures for communicating GAI incidents and performance to AI Actors and downstream stakeholders (including those potentially impacted), via community or official resources (e.g., AI incident database, AVID, CVE, NVD, or OECD AI incident monitor).

SCF Controls: AAT-01, AAT-08

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-2.1-002

Establish procedures to engage teams for GAI system incident response with diverse composition and responsibilities based on the particular incident type.

SCF Controls: AAT-01, AAT-08

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-2.1-003

Establish processes to verify the AI Actors conducting GAI incident response tasks demonstrate and maintain the appropriate skills and training.

SCF Controls: AAT-05

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-2.1-004

When systems may raise national security risks, involve national security professionals in mapping, measuring, and managing those risks.

SCF Controls: AAT-01, AAT-09.1, AAT-16.9, GOV-06

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-2.1-005

Create mechanisms to provide protections for whistleblowers who report, based on reasonable belief, when the organization violates relevant laws or poses a specific and empirically well-substantiated negative risk to public safety (or has already caused harm).

SCF Controls: HRS-15

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-3.2-001

Policies are in place to bolster oversight of GAI systems with independent evaluations or assessments of GAI models or systems where the type and robustness of evaluations are proportional to the identified risks.

SCF Controls: AAT-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-3.2-003

Define acceptable use policies for GAI interfaces, modalities, and human-AI configurations (i.e., for chatbots and decision-making tasks), including criteria for the kinds of queries GAI applications should refuse to respond to.

SCF Controls: AAT-01, TDA-21

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-3.2-004

Establish policies for user feedback mechanisms for GAI systems which include thorough instructions and any mechanisms for recourse.

SCF Controls: AAT-01, AAT-11.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-3.2-005

Engage in threat modeling to anticipate potential risks from GAI systems.

SCF Controls: TDA-06.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-4.1-001

Establish policies and procedures that address continual improvement processes for GAI risk measurement. Address general risks associated with a lack of explainability and transparency in GAI systems by using ample documentation and techniques such as: application of gradient-based attributions, occlusion/term reduction, counterfactual prompts and prompt engineering, and analysis of embeddings; Assess and update risk measurement approaches at regular cadences.

SCF Controls: AAT-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-4.1-002

Establish policies, procedures, and processes detailing risk measurement in context of use with standardized measurement protocols and structured public feedback exercises such as AI red-teaming or independent external evaluations.

SCF Controls: AAT-01, AAT-02.1, AAT-07.2, AAT-15, AAT-15.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-4.1-003

Establish policies, procedures, and processes for oversight functions (e.g., senior leadership, legal, compliance, including internal evaluation) across the GAI lifecycle, from problem formulation and supply chains to system decommission.

SCF Controls: AAT-01, AAT-11

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-4.2-001

Establish terms of use and terms of service for GAI systems.

SCF Controls: AAT-11, HRS-05.4

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-4.2-002

Include relevant AI Actors in the GAI system risk identification process.

SCF Controls: AAT-02.1, AAT-09, AAT-11, AAT-11.3, RSK-03

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-4.2-003

Verify that downstream GAI system impacts (such as the use of third-party plugins) are included in the impact documentation process.

SCF Controls: AAT-07.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-4.3-002

Establish organizational practices to identify the minimum set of criteria necessary for GAI system incident reporting such as: System ID (auto-generated most likely), Title, Reporter, System/Source, Data Reported, Date of Incident, Description, Impact(s), Stakeholder(s) Impacted.

SCF Controls: AAT-01, AAT-16.8, AAT-16.9, MON-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-4.3-003

Verify information sharing and feedback mechanisms among individuals and organizations regarding any negative impact from GAI systems.

SCF Controls: AAT-10, AAT-11

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-5.1-001

Allocate time and resources for outreach, feedback, and recourse processes in GAI system development.

SCF Controls: AAT-11, AAT-11.1, AAT-11.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-5.1-002

Document interactions with GAI systems to users prior to interactive activities, particularly in contexts involving more significant risks.

SCF Controls: AAT-07.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.1-001

Categorize different types of GAI content with associated third-party rights (e.g., copyright, intellectual property, data privacy).

SCF Controls: AAT-12, AAT-12.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.1-002

Conduct joint educational activities and events in collaboration with third parties to promote best practices for managing GAI risks.

SCF Controls: SAT-02

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.1-003

Develop and validate approaches for measuring the success of content provenance management efforts with third parties (e.g., incidents detected and response times).

SCF Controls: AAT-10.17, AAT-12.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.1-004

Draft and maintain well-defined contracts and service level agreements (SLAs) that specify content ownership, usage rights, quality standards, security requirements, and content provenance expectations for GAI systems.

SCF Controls: AAT-12.1, IAO-03.2, TPM-05

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.1-005

Implement a use-cased based supplier risk assessment framework to evaluate and monitor third-party entities' performance and adherence to content provenance standards and technologies to detect anomalies and unauthorized changes; services acquisition and value chain risk management; and legal compliance.

SCF Controls: TPM-04.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.1-006

Include clauses in contracts which allow an organization to evaluate third-party GAI processes and standards.

SCF Controls: TPM-04.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.1-007

Inventory all third-party entities with access to organizational content and establish approved GAI technology and service provider lists.

SCF Controls: TPM-01.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.1-008

Maintain records of changes to content made by third parties to promote content provenance, including sources, timestamps, metadata.

SCF Controls: AAT-12.1, AAT-12.4

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides explicit data source identification and tracking for AI workloads. The Data Indexing and Retrieval service offers a Knowledge Bases interface where users create and manage knowledge bases, and a knowledge base can be linked to up to ten data sources. Each data source is created with a name, description, type, and the required properties for retrieving content from its original location, and a single data source can be linked to multiple knowledge bases. The data source type is fixed at creation. Supported data source types include Microsoft SharePoint sites, Amazon S3 compatible stores, and other enterprise content management systems. Each data source stores the credentials required for retrieval, such as service account key file information when applicable, which creates a documented connection between the AI system and its data origins.

During knowledge base indexing, PAIS tracks document-level detail including crawled documents, new documents found, deleted documents, existing unmodified documents, existing modified documents requiring embedding updates, and skipped documents. It captures size metrics including total megabytes downloaded, new chunks indexed corresponding to inference server invocations for embeddings, and final index size after removal of deleted documents. It records timing metrics for download, embedding generation, and the overall index operation. Document counts returned by retrieval queries are tracked, and counts of documents per media type such as PDF and PPTX are maintained for each knowledge base. Automatic collection and indexing of updates from linked data sources over time keeps these records current.

For runtime visibility, PAIS supports LLM trace collection through an OpenTelemetry Collector by configuring the `spec.observability.llmTraces` section of the `PAISConfiguration` custom resource, with traces transmitted over HTTP/protobuf or gRPC. PAIS metrics, including knowledge base activity, can be visualized in third-party observability platforms such as Grafana. The Data Indexing and Retrieval component also implements a Model Context Protocol (MCP) server that exports a set of search tools for each knowledge base, which gives downstream AI applications a documented programmatic surface tied back to specific data sources. PAIS AI catalog items, including AI Workstation, DSM Database, Triton Inferencing Server, and AI Kubernetes Cluster, are added to the VCF Automation catalog by running the PAIS Quickstart, with required images uploaded in advance to a Harbor registry in disconnected environments.

Organizational decisions about which data sources are approved for AI use, how data lineage is documented outside the platform, and how training data provenance is governed remain the responsibility of the organization. PAIS supplies the technical registration and tracking infrastructure that the organization configures and operates.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.1-009

Update and integrate due diligence processes for GAI acquisition and procurement vendor assessments to include intellectual property, data privacy, security, and other risks. For example, update processes to: Address solutions that may rely on embedded GAI technologies; Address ongoing monitoring, assessments, and alerting, dynamic risk assessments, and real-time reporting tools for monitoring third-party GAI risks; Consider policy adjustments across GAI modeling libraries, tools and APIs, fine-tuned models, and embedded tools; Assess GAI vendors, open-source or proprietary GAI tools, or GAI service providers against incident or vulnerability databases.

SCF Controls: AAT-01, TPM-01, TPM-04.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.1-010

Update GAI acceptable use policies to address proprietary and open-source GAI technologies and data, and contractors, consultants, and other third-party personnel.

SCF Controls: HRS-05.1, HRS-05.3, TPM-05

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.2-001

Document GAI risks associated with system value chain to identify over-reliance on third-party data and to identify fallbacks.

SCF Controls: AAT-25, AAT-25.1

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) defines a structured, multi-stage value chain for creating and deploying AI workloads, with distinct stakeholder roles at each stage. The platform separates responsibilities across VI administrators, organization administrators, MLOps engineers, DevOps engineers, and AI application developers, with each role scoped to a specific phase of the deployment lifecycle.

Infrastructure provisioning begins with VI administrators configuring GPU-accelerated workload domains, namespaces, and VMware Kubernetes Service (VKS) Supervisors in VMware vCenter and VMware ESX. Organization administrators set up the VCF Automation organization. PAIS must be activated in organization namespaces by an organization administrator or MLOps engineer after namespace creation, establishing the foundation before any AI workloads can be deployed.

Model acquisition is registry-driven. A Harbor administrator creates a dedicated Harbor project for PAIS in the private Harbor registry, into which PAIS artifacts are uploaded as part of the installation process. OCI-compliant registries from other vendors are also supported. The Harbor project's user access controls restrict who can push or pull artifacts, creating an auditable record of which images are available and how they arrived. PAIS deployments can run in connected environments with internet access or in disconnected (air-gapped) environments, with disconnected operation requiring the Harbor registry to be pre-populated with the AI catalog images in advance.

Compute provisioning is driven by VCF Automation catalog items created through the Private AI Services Quickstart. The catalog items include AI Workstation, DSM Database, Triton Inferencing Server, and AI Kubernetes Cluster, and are provisioned by users in selected namespaces. Catalog item visibility is verified per project so that PAIS catalog items are available only to members of the selected project. A machine retrieving information about PAIS must have access to VCF Automation, the VCF Consumption CLI, and kubectl, which scopes the provisioning interfaces to the responsible stakeholder.

PAIS provides a UI and API for AI application developers to interact with deployed models and services. Data scientists and MLOps engineers compose applications by using PAIS Knowledge Bases linked to external data sources such as Microsoft SharePoint sites and Amazon S3 compatible stores. A Knowledge Base data source is created with a specific type that cannot be changed after creation, and one data source can be linked to multiple Knowledge Bases. DevOps engineers visualize PAIS component health and performance metrics in observability platforms such as Grafana and VCF Operations, providing accountability for each stage of the value chain.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.2-002

Document incidents involving third-party GAI data and systems, including open- data and open-source software.

SCF Controls: CFG-04.1, TPM-04

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.2-003

Establish incident response plans for third-party GAI technologies: Align incident response plans with impacts enumerated in MAP 5.1; Communicate third-party GAI incident response plans to all relevant AI Actors; Define ownership of GAI incident response functions; Rehearse third-party GAI incident response plans at a regular cadence; Improve incident response plans based on retrospective learning; Review incident response plans for alignment with relevant breach reporting, data protection, data privacy, or other laws.

SCF Controls: IRO-04, IRO-06, TPM-11

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.2-004

Establish policies and procedures for continuous monitoring of third-party GAI systems in deployment.

SCF Controls: MON-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.2-005

Establish policies and procedures that address GAI data redundancy, including model weights and other system artifacts.

SCF Controls: AAT-01, AAT-09, TDA-01.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.2-006

Establish policies and procedures to test and manage risks related to rollover and fallback technologies for GAI systems, acknowledging that rollover and fallback may include manual processing.

SCF Controls: AAT-25.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control GV-6.2-007

Review vendor contracts and avoid arbitrary or capricious termination of critical GAI technologies or vendor services and non-standard terms that may amplify or defer liability in unexpected ways and/or contribute to unauthorized data collection by vendors or third-parties (e.g., secondary data use). Consider: Clear assignment of liability and responsibility for incidents, GAI system changes over time (e.g., fine-tuning, drift, decay); Request: Notification and disclosure for serious incidents arising from third-party data and systems; Service Level Agreements (SLAs) in vendor contracts that address incident response, response times, and availability of critical support.

SCF Controls: TPM-01, TPM-05

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MANAGE 1.3

Responses to the AI risks deemed high priority, as identified by the MAP function, are developed, planned, and documented. Risk response options can include mitigating, transferring, avoiding, or accepting.

SCF Controls: AAT-07, AAT-09

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MANAGE 2.2

Mechanisms are in place and applied to sustain the value of deployed AI systems.

SCF Controls: AAT-01.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MANAGE 2.3

Procedures are followed to respond to and recover from a previously unknown risk when it is identified.

SCF Controls: AAT-17.3

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to responding to and recovering from previously unknown AI-related risks through vSphere HA (automatic VM restart on host failure), DRS (workload rebalancing), vSAN fault domains (storage resilience), and VM snapshots (point-in-time recovery). If AI infrastructure is affected by a novel threat, these mechanisms support rapid recovery. VCF Operations lifecycle management supports emergency patching of the platform.

The VCF platform's Consumption layer, delivered through VMware Kubernetes Service (VKS) and the Supervisor, extends these capabilities to containerized AI workloads running on Kubernetes and provides observability, containment, and recovery mechanisms relevant to novel AI risks.

Observability and Detection. Supervisor integrates with the Continuous Integration and Delivery with Harness supervisor service, which includes a Dynatrace AI engine that automatically detects anomalies, identifies root causes, and surfaces intelligent alerts for workloads including AI workloads. The Kubernetes API server audit log records API server activity for VKS clusters, including changes to workload configurations and access patterns. Pod Security Admission in warn or audit mode generates warnings when pods running in a namespace violate the configured security level, flagging unrestricted capabilities, allowPrivilegeEscalation, or non-standard seccomp profiles, which supports identifying workloads that may pose elevated risk. VKS Cluster Management Policy Insights provide centralized visibility into the governance health of the VKS cluster estate and generate actionable data for troubleshooting policy enforcement issues, including Health alerts for infrastructure-level failures such as an unhealthy policy operator or a failed Gatekeeper installation, Sync signals indicating that policy definitions failed to propagate to target clusters, and Incompatibility warnings when a cluster contains an external Gatekeeper installation not managed by cluster management. MachineHealthCheck monitors node health conditions classified by severity and can trigger automated remediation when nodes hosting AI workloads become unhealthy.

Containment and Isolation. When a risk is identified, Kubernetes NetworkPolicy objects enforced by Antrea at the pod level can be applied or updated to isolate the affected namespace or workload from the rest of the cluster. Kubernetes RBAC restricts access to impacted services and, when properly scoped, helps limit privilege escalation. Pod Security Admission enforcement can be tightened on specific namespaces to block workloads that do not meet the required security posture, limiting the scope of a risk involving a compromised or misbehaving AI component. The Kubernetes documentation recommends adopting a zero trust architecture that minimizes attack surfaces, including for internal threats, which supports limiting the lateral spread of a newly identified risk.

Recovery. VKS supports deployment rollback and revision history, allowing operators to revert AI workloads to a prior known-good version. The MachineDeployment remediation.maxInFlight parameter controls how many in-flight remediations of unhealthy machines occur simultaneously, expressed as an integer or percentage, limiting disruption during large-scale node replacements. Velero provides workload backup and restore for VKS clusters, enabling restoration of AI workloads to a pre-incident state. The declarative validation feature includes a safety mechanism for downgrade and rollback scenarios, allowing operators to disable the feature and fall back to prior validation behavior during recovery.

Supply Chain Integrity. AI and machine learning workloads on VKS clusters require Ubuntu VKr as the operating environment, providing a known software baseline that can be validated and replaced when a vulnerability is identified in the base image. Harbor, an open-source cloud-native private container registry that ships with Supervisor, can be deployed as a Supervisor Service and used as the private registry for VKS clusters. Harbor Registry instances run on the Supervisor with images organized by vSphere Namespace, and all VKS clusters on the same Supervisor automatically trust Harbor and can pull container images from it. In air-gapped environments, Harbor Supervisor Service must be configured as the private registry before upgrading VKS from a private registry, supporting controlled image provenance. Container images and other artifacts can be scanned for known vulnerabilities as part of supply chain security. The Kubernetes Image Policy admission controller

provides an additional enforcement layer by evaluating container images against a remote service before allowing access; the service returns an ImageReview response based on image reference, annotations, and namespace context, enabling operators to block images associated with a newly identified threat across VKS clusters. The Kubernetes Security Response Committee tracks and announces CVEs with security-related announcements via the kube-announce channel, giving operators an authoritative feed for newly identified platform-level risks. VKS cluster management enables governance of the operational consistency and security posture of the VKS cluster estate through policy-driven automation, helping maintain a controlled baseline from which deviations due to a novel risk can be detected.

Identifying novel AI-specific threats and developing response procedures are organizational responsibilities. Incident response plans must cover AI workload categories, personnel must be trained to recognize AI-specific threat indicators, and communication workflows must exist for escalating newly identified risks. VCF and the Consumption layer provide the infrastructure controls for containment and recovery, but the identification and classification of a novel AI risk depends on organizational judgment and monitoring practices built on top of these capabilities.

VMware Private AI Services (PAIS) contributes to responding to and recovering from previously unknown AI-related risks through AI-service-layer mechanisms that operate above the underlying VCF platform's infrastructure resilience.

VCF provides infrastructure-level resilience that supports incident response: vSphere High Availability restarts AI workload VMs on healthy hosts, Distributed Resource Scheduler rebalances workloads when host-level anomalies occur, vMotion migrates running AI workloads without downtime, and snapshot-based recovery enables point-in-time restoration of affected VMs. These capabilities allow rapid response at the infrastructure layer when an AI-related risk is identified.

At the AI service layer, PAIS provides several response mechanisms specific to AI workloads. Organization administrators or MLOps engineers activate PAIS in organization namespaces, and the same roles can deactivate or restrict access when a previously unknown risk is identified. Access to PAIS is gated through OIDC integration over HTTPS with the OIDC provider; restricting OIDC group membership or scope at the identity provider provides an identity-layer cutoff for PAIS service access. MLOps engineers validate ML models onboarded to PAIS against organizational security, privacy, and technical requirements, which helps identify problematic models before deployment and gives operators a basis for re-validation when a new risk surfaces.

For model storage and recovery, PAIS uses a Harbor registry, or another OCI-compliant registry, as the model storage backend, requiring authentication and certificate-based trust for model retrieval. Operators can remove a problematic model or restore a prior model version from the registry without redeploying the underlying infrastructure. PAIS Knowledge Bases support linked data sources including SharePoint sites and S3-compatible stores, with a maximum of ten data sources per knowledge base; an administrator can unlink a data source that is later identified as a risk source. PAIS observability is configured with a Prometheus runtime and a metricsRetention parameter (14 days by default), retaining metrics that support investigation and post-incident analysis when a risk is identified.

VMware vDefend (Contributes)

VMware vDefend contributes to responding to previously unknown AI and Autonomous Technologies-related risks through behavioral detection and automated response mechanisms at the network layer.

Security Intelligence uses a proprietary machine-learning algorithm to identify anomalous compute entity behavior by analyzing traffic flow patterns across the east-west distributed firewall environment. The machine learning runs as a scheduled process that makes inferred infrastructure classifications to establish behavioral baselines. Detected suspicious traffic events are categorized by impact type (Critical, High, Medium, and Low) to support prioritized response activities. The Security Intelligence Campaign view tracks the total number of detections and threats identified under Malicious Activity, providing a running record of security events that helps assess how effectively controls are responding to novel activity. The threat visualization canvas is a tool for rapid threat containment, identifying compromised assets, and updating microsegmentation policy in response to detected events.

The Network Detection and Response (NDR) component extends detection coverage by applying a combination of IDS, malware analysis, and network behavior analytics through the NDR Sensor to detect suspicious activity, lateral movement, and advanced threats. The NDR Sensor exports data to the vDefend Advanced Threat Prevention Cloud Service to enhance

detection capabilities for emerging and previously unknown threats. When the NDR component processes a custom IDS detection without a specified tactic, it defaults to an 'Undetermined' classification rather than discarding the event, flagging it for analyst review. The NDR component also dynamically manages the IDS signature set by deactivating signatures that are consistently false-positive prone, reducing noise and maintaining detection quality.

Intelligent Assist is an AI assistant feature accessible through the NSX Manager UI or Security Services Platform UI that provides analysis support for IDS/IPS events, detections, and campaigns. When threats are identified, Intelligent Assist can recommend remediation policies to support containment of newly identified risks. Remediation rules are added to the Intelligent Assist Remediation policy using the naming convention prefix 'ai-' and are not enabled by default; they require manual review and verification by the analyst or administrator before activation. Not all network anomalies can be remediated automatically; Intelligent Assist determines when remediation is not possible and flags the limitation for the operator. Applied remediation policies can be verified by navigating to Security > IDS/IPS & Malware Prevention > Emergency Threat Rules.

Knowledge-based signatures in vDefend IDS/IPS are limited to attacks that are already known. The behavioral analysis and ML-based anomaly detection in Security Intelligence, combined with the NDR Sensor's multi-method analytics, address this gap by providing visibility into network-layer behavior that falls outside known signature patterns.

Recovery activities beyond network-level containment and policy remediation, such as restoring affected AI systems, validating their integrity, or conducting post-incident recovery planning, fall outside vDefend's scope and require organizational incident response and recovery processes.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) contributes to responding to and recovering from previously unknown AI-related risks through its failover, isolated recovery, and integrated security analysis capabilities.

If an AI workload is affected by a novel threat, PNR's recovery plans can orchestrate failover to a protected site where the workload can resume operations. Recovery plan testing validates that all virtual machines will recover correctly, reducing the risk that an actual disaster recovery event fails to restore critical workloads.

The Cyber Recovery component of PNR provides an isolated clean room environment where AI workloads can be restored and analyzed separately from potentially compromised production infrastructure. Within the clean room, CrowdStrike integration performs behavioral analysis and machine learning-based detection to identify both known and unknown threats in real time, continuously monitoring endpoint activities during recovery validation. CrowdStrike ExPRT (Exploitability and Predictive Risk Technology) ratings assess vulnerability risk based on exploitability likelihood and predicted potential impact, and vulnerability risk scores on a 0-to-10 scale enable prioritization of remediation before workloads are promoted from the clean room to production. Carbon Black integration displays vulnerability analysis results including risk score details and environmental impact information to support remediation of discovered issues.

Replication scheduling with configurable recovery point objectives allows organizations to retain multiple recovery points, making it possible to restore AI workloads to a state that predates a threat materialization.

Identifying novel AI-related risks and developing response procedures are organizational responsibilities. PNR supplies the technical recovery and analysis mechanisms that organizations can incorporate into their incident response processes.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) contributes to responding to and recovering from previously unknown AI-related risks by providing backup, restore, high availability, and alerting capabilities for the databases that AI workloads depend on. If AI training data or model artifacts stored in DSM-managed databases are compromised by a novel threat, point-in-time recovery allows restoration to a state before the threat materialized, and automated backups with configurable retention provide multiple recovery points. Database replicas with automatic promotion support continued operation when a primary database is affected. For PostgreSQL workloads, multi-site deployment supports cross-site disaster recovery, including controlled redirection of a secondary instance to a new primary after promotion, with explicit acknowledgment of data-loss risk when demoting a previously blocked primary.

DSM emits a structured catalog of global alerts that help operators detect novel issues affecting database services. Alert types include DATABASE_ENGINE_HEALTH_ALERT for database engine outages, read-only mode, or reduced availability; SYSTEM_DISK_HEALTH_ALERT for system disk health; data-disk warning thresholds; and infrastructure policy alerts such as INFRASTRUCTURE_POLICY_ORPHAN_VM_FOUND, INFRASTRUCTURE_POLICY_VC_ALARMS, INFRASTRUCTURE_POLICY_CLUSTER_COMPLIANT, INVALID_SUPERVISOR_INFRASTRUCTURE_POLICY_STATUS, and NetworkOverlap Found. Alerts for database operation failures can be activated or deactivated, and webhooks can be configured to deliver CRITICAL or WARNING alert level information to external response tooling when database parameter thresholds are crossed. Administrators can collect support bundles for the affected cluster to assist in root-cause analysis and recovery.

Identifying novel AI-related risks and developing response and recovery procedures are organizational responsibilities. DSM supplies the technical mechanisms that those response actions act on.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer provides several application-layer mechanisms that organizations can deploy to respond to previously unknown Artificial Intelligence (AI) and Autonomous Technologies (AAT)-related risks as they are identified.

The Bot Management capability addresses AI-driven bot traffic through a three-step pipeline of detection, classification, and action. Avi Bot Management detects and manages bot traffic, distinguishing between good bots (search engine crawlers, website health monitors, vulnerability scanners) and bad bots (scrapers, spam, click fraud, bot impersonators, botnets), and enforces HTTP security policy actions such as closing connections for dangerous classifications. Bot Management assigns a confidence level to each classification decision of Bot, Human, or Undetermined, allowing policy actions to be tuned to the certainty of the detection. Avi Virtual Service Analytics App Learning classifies application traffic through trusted IP groups, supporting bot categories including Bad Bot, Dangerous Bot, Good Bot, Human, Unknown, and User-defined Bot. WAF Policies attach a Positive Security group to apply learned data and define hit and miss actions; both Application Insights and WAF Positive Security initially gather data through decoupled Learning functionality, enabling the WAF to adapt its enforcement posture as new threat patterns emerge and observed baseline behavior shifts.

The Live Security Threat Intelligence service, available through the Avi Cloud Console, delivers security threat feeds for various attack vectors in real time to protect applications from evolving threats. This is an opt-in service that is disabled by default on the Avi Load Balancer Controller and must be explicitly enabled. The IP Reputation service is used as a source for bot detection and classification, providing protection from bad IPs such as botnets, phishing sources, and spam. Virtual Service Network Security Policy can match on IP Reputation to identify or categorize IP addresses based on associated threats. The User Agent Database Sync capability can be enabled to receive User-Agent updates for threat detection, allowing detection logic to adapt as new bot user-agent patterns are identified. The Service Engine Bot Management feature relies on IP Reputation and L7 GeoDB lookup for threat detection and mitigation.

For Kubernetes-native environments, the Avi Kubernetes Operator (AKO) L7Rule custom resource definition supports a botPolicyRef field to reference a bot policy for bot detection and mitigation, extending Bot Management capabilities to Kubernetes ingress traffic.

Avi Analytics detects anomalies defined as traffic outside the statistical norm for a given time period and assigns an anomaly penalty to flag potential risk. This visibility into abnormal traffic patterns can support early identification of novel AI-driven threats in progress.

Control MANAGE 2.4

Mechanisms are in place and applied, and responsibilities are assigned and understood, to supersede, disengage, or deactivate AI systems that demonstrate performance or outcomes inconsistent with intended use.

SCF Controls: AAT-15.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MANAGE 3.1

AI risks and benefits from third-party resources are regularly monitored, and risk controls are applied and documented.

SCF Controls: AAT-02, AAT-04.4

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides a centralized inventory of AI models through the Model Gallery in Harbor Registry. Each model gallery has a unique name and is organized within Harbor projects with configured user access controls. Model revisions are stored as OCI artifacts with immutable manifests, and each revision is assigned a unique digest, giving a verifiable record of every model version in the inventory.

OCI metadata describes model contents and dependencies, supporting identification of compatible platforms and giving visibility into what each model requires. The vcf pais models list command retrieves and displays all models available in a model gallery, giving operators a direct inventory view. Model Runtime tracks which models are actively deployed to inference endpoints, distinguishing between stored and deployed models.

For third-party AI components, PAIS validates running AI workloads based on NVIDIA GPU Cloud (NGC) containers, a curated catalog of GPU-optimized, pre-tested containers. In disconnected environments, a Harbor registry stores container images from the NGC catalog, acting as a controlled intermediary that makes third-party AI components visible and trackable within the organization's infrastructure.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MANAGE 3.2

Pre-trained models which are used for development are monitored as part of AI system regular monitoring and maintenance.

SCF Controls: AAT-10.13

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to post-deployment monitoring of AI workloads through VCF Operations, which provides continuous health, performance, and capacity monitoring for the infrastructure hosting those workloads. VCF Operations dashboards surface CPU, memory, storage, and network metrics for VMs running AI workloads. VCF Log Management aggregates and analyzes log data from VMware vCenter, VMware ESX, and NSX components. vCenter alarms provide threshold-based alerting for infrastructure conditions that could affect AI workload availability.

For AI workloads running as containerized applications on VMware Kubernetes Service (VKS) clusters, the platform extends infrastructure-level observability to the Kubernetes layer. VKS clusters provisioned with the automated-monitoring label automatically receive monitoring configuration on each new node, integrating cluster telemetry with VCF Operations. The Prometheus standard package, installable on VKS clusters through the Supervisor Management Proxy Service, deploys kube-state-metrics, node-exporter, pushgateway, Prometheus, and Alertmanager as components. The prometheus-kube-state-metrics component monitors node status and capacity, ReplicaSet compliance, pod status, job and CronJob status, and resource requests and limits, providing continuous metric collection that can be directed at AI inference services or training job pods. Telegraf is available as a standard package that deploys as a Deployment component to scrape metrics from kube-state-metrics and node-exporter and forward them to external observability backends. Monitoring and alerting for Kubernetes deployments is also available through the Argo CD Supervisor Service using Grafana Operator and Prometheus.

Kubernetes-native health probes and status tracking support workload condition detection at the pod level. Liveness and readiness probes can be configured on AI workload containers to detect unresponsive or overloaded model serving instances; a readiness probe removes the pod from service routing when it cannot handle requests, and a liveness probe triggers a container restart when the application becomes unresponsive. The Kubernetes Deployment controller continuously monitors running application instances and automatically replaces instances when nodes fail or are removed from service, providing a self-healing mechanism that addresses machine failure or maintenance. Kubernetes DeploymentStatus exposes observedGeneration and conditions fields with type, status, lastTransitionTime, and reason values, allowing operators to detect rollout failures, stuck deployments, or configuration drift from the expected specification. VKS cluster management proactively monitors the cluster estate to verify that assigned policies are enforced correctly, and generates Policy Insights when configuration conflicts or environmental errors are detected.

For organizations using Harness CI/CD on VKS, Dynatrace integration provides full-stack observability including application performance monitoring, infrastructure monitoring, and digital experience monitoring. The Dynatrace operator automates deployment, scaling, and maintenance of the monitoring stack within VKS clusters, and automatically correlates application deployment service metrics with underlying VKS infrastructure including nodes, pods, and the network. The Dynatrace AI engine detects anomalies, identifies root causes, and provides intelligent alerts. Harness SRM monitors deployed services post-deployment to calculate a real-time health score based on transaction success and latency.

The VCF Automation Monitor tab provides deployment health information drawn from VCF Operations, and deployment cards in the self-service catalog surface deployment state, last request status, and operational history for workload deployments. AI-specific behavior monitoring (model accuracy drift, output quality) requires application-level tooling outside VCF's scope.

VMware Private AI Services (PAIS) provides AI-specific monitoring mechanisms that support continuous post-deployment observation of AI workloads on VCF.

At the PAIS layer, observability is activated by updating the PAISConfiguration custom resource with parameters that enable metrics collection and forward large language model traces to an OpenTelemetry Collector. PAIS deploys a Prometheus

server pod for metrics collection that must be ready and running for observability to function, configured with a chosen storage class and a configurable metrics retention period. PAIS controller pod metrics are integrated into observability platforms through Telegraf configuration in the Supervisor instance, and the telegraf-user-config ConfigMap in the vmware-system-monitoring namespace manages user-configured monitoring endpoints for PAIS. The PAIS Supervisor Service exposes service-level metrics through `pais.vmware.com` that VI administrators access to monitor the health of the service.

DevOps engineers visualize PAIS health and performance metrics in observability platforms such as Grafana, where dashboards report pod restart frequency, vGPU licensing status, and idle or underutilized GPU devices in passthrough mode. VI administrators visualize PAIS controller pod operation metrics in VCF Operations using the same metrics pipeline. MLOps engineers and application developers track the health, quality, and behavior of AI applications and the models used in agents through Grafana, supporting continuous observation of inference behavior. Model Endpoint deployment status can be verified using the `kubectl get modelendpoints` command, which returns the Ready status for each deployed endpoint, and the `vcf` cluster command-line interface provides cluster provisioning monitoring, registration, and kubeconfig retrieval for VKS clusters that host AI workloads. PAIS deployment leases can be configured to bound how long a deployment can exist, supporting lifecycle oversight of catalog-deployed AI workloads.

Enabling these observability features, defining retention and alerting policy, and acting on monitoring signals are organizational process responsibilities. PAIS provides the monitoring infrastructure and integration points, but continuous post-deployment monitoring as a practice requires the organization to assign operational ownership, review observability data on an ongoing basis, and respond to anomalies in AI workload behavior.

VMware vDefend (Contributes)

VMware vDefend (vDefend) contributes to proactive and continuous monitoring of deployed AI workloads running on VCF through multiple layers of ongoing network-level security monitoring.

Security Intelligence continuously monitors for new flows between environments and alerts administrators to unauthorized or unexpected traffic, enabling ongoing policy adjustments as workloads evolve. It also monitors infrastructure services to detect newly deployed servers. A classification cron job runs automatically at 2:00 AM local time and repeats every 24 hours, evaluating traffic flows from the preceding 30 days to classify compute entities providing network infrastructure services. When the Recommendations Monitoring feature is enabled, Security Intelligence checks every hour for changes in the scope of monitored entities, allowing policy recommendations to remain current as AI workloads change. The Security Intelligence Flow Insights Dashboard supports detection and display of MITRE ATT&CK framework tactics and techniques, showing counts of detected suspicious events per tactic and technique during a selected time period.

Segmentation Monitoring provides dedicated views for environment-level traffic flows, accessible through the Monitor & Plan interface. Administrators can monitor traffic between configured environment pairs, view flows related to detected infrastructure servers within specified time intervals, and export all unique flows to a CSV file for offline analysis or compliance evidence. The Environments monitoring window provides assessment of current security posture and enables improvement actions for environment pairs. When no protection policy is configured for an environment pair, Segmentation Monitoring offers a Run Recommendations option to generate policy recommendations, closing monitoring gaps around AI workloads.

The Security Services Platform (SSP) collects metrics to monitor the readiness and performance of vDefend IDS/IPS in the environment. IDS/IPS engine statistics accessible per host include uptime, alert count, suppressed alerts, alert queue overflow status, engine ID, and last reload timestamp. IDS/IPS monitoring dashboards can be scoped to custom projects, displaying events triggered for that project only. For bare metal deployments, Bare Metal Agents stream metrics and detailed host-level telemetry to the SSP. The NDR Sensor periodically checks the status of activated features on the SSP every 15 minutes and updates its configuration accordingly.

Intelligent Assist provides AI-assisted analysis within the NSX Manager UI and SSP UI, displaying on the IDS/IPS, Detections, and Campaigns pages to help administrators interpret monitoring data.

Malware Prevention Service provides deployment status monitoring at the cluster level and per-node level, displaying deployment errors and individual transport node status. Network Detection and Response (NDR) applies behavioral analysis to detect suspicious activity targeting monitored workloads and correlates events into campaigns for investigation.

Monitoring AI system functionality at the application layer, including model performance, output quality, and behavioral drift, is outside vDefend's domain.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MANAGE 4.1

Post-deployment AI system monitoring plans are implemented, including mechanisms for capturing and evaluating input from users and other relevant AI actors, appeal and override, decommissioning, incident response, recovery, and change management.

SCF Controls: AAT-10.1, AAT-16.5, AAT-16.8

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MANAGE 4.2

Measurable activities for continual improvements are integrated into AI system updates and include regular engagement with interested parties, including relevant AI actors.

SCF Controls: AAT-07.3

Aggregate Coverage: Contributes

VCF (Contributes)

Continuous improvement of Artificial Intelligence and Autonomous Technologies capabilities is an organizational activity carried out by the teams that build, evaluate, and operate AI workloads. VMware Private AI Services (PAIS) is the AI infrastructure layer of VCF and supplies the platform on which those workloads run; it does not perform the improvement cycle itself. The decision to retrain a model, change a data source, adjust a prompt, or retire a capability is made by the data science, MLOps, and product teams that own the workload, governed by their own internal processes.

PAIS supplies platform mechanisms that workload teams can incorporate into their own improvement cycles. A central model gallery is recommended for managing ML models with consistency across deployment targets. Knowledge bases can be created with linked data sources including Microsoft SharePoint sites and Amazon S3 compatible stores, allowing teams to update the context available to model responses without rebuilding the underlying model. Private AI Services Agent Builder maintains a list of approved tools for use in AI agents when connecting to Model Context Protocol (MCP) servers, giving workload owners a mechanism to add or restrict capabilities over time. PAIS can be installed or updated to later versions independently of the core Supervisor Service, allowing platform-level improvements to ship without a full core update. Controller pod metrics are integrated into observability platforms through Telegraf configuration in the Supervisor instance, so operations teams can feed platform telemetry into their evaluation loops. These are tools that workload teams use; PAIS does not own the improvement decision.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MANAGE 4.3

Incidents and errors are communicated to relevant AI actors, including affected communities. Processes for tracking, responding to, and recovering from incidents and errors are followed and documented.

SCF Controls: AAT-11.4

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MAP 1.1

Intended purposes, potentially beneficial uses, contextspecific laws, norms and expectations, and prospective settings in which the AI system will be deployed are understood and documented.

Considerations include: the specific set or types of users along with their expectations; potential positive and negative impacts of system uses to individuals, communities, organizations, society, and the planet; assumptions and related limitations about AI system purposes, uses, and risks across the development or product AI lifecycle; and related TEV and system metrics.

SCF Controls: AAT-03

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MAP 1.2

Interdisciplinary AI actors, competencies, skills, and capacities for establishing context reflect demographic diversity and broad domain and user experience expertise, and their participation is documented. Opportunities for interdisciplinary collaboration are prioritized.

SCF Controls: AAT-07

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MAP 2.1

The specific tasks and methods used to implement the tasks that the AI system will support are defined (e.g., classifiers, generative models, recommenders).

SCF Controls: AAT-14.1

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) defines the AI task types the platform supports through a set of catalog items, model and knowledge-base constructs, and namespace activation steps that translate organizational AI scoping decisions into platform configuration. The PAIS catalog in VCF Automation, populated through the Private AI Services Quickstart, exposes a defined set of AI workload patterns to users: AI Workstation, DSM Database, Triton Inferencing Server, and AI Kubernetes Cluster. An organization administrator or MLOps engineer must activate PAIS in a namespace before any of these AI workloads can be requested, and catalog item permissions control which task types are visible to which users. AI workloads can run on GPU devices in vGPU or passthrough mode or in CPU-only configurations, allowing the task definitions to match the underlying hardware policy.

For generative AI tasks, PAIS provides Model Endpoints for completion and embedding models, Agent Builder for assembling agents on top of those endpoints, and Knowledge Bases that link to external data sources such as Microsoft SharePoint sites and Amazon S3 compatible stores so that retrieval-augmented generation patterns are scoped to approved corpora. Agent Builder also maintains a list of approved tools when connecting to MCP servers, giving organizations a mechanism to constrain the specialized capabilities an AI agent can invoke. Deployment connectivity mode is itself a defined attribute: PAIS can be installed in connected or disconnected air-gapped environments, with a Harbor registry, or another OCI-compliant registry, holding the model gallery in both modes.

Because PPTDF for this control is Process only, the organization remains responsible for identifying the AI use cases the environment will support, approving task categories, and documenting the operational scope. PAIS supplies the platform mechanisms that enforce those decisions but does not perform the scoping itself.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MAP 2.2

Information about the AI system's knowledge limits and how system output may be utilized and overseen by humans is documented. Documentation provides sufficient information to assist relevant AI actors when making decisions and taking subsequent actions.

SCF Controls: AAT-14.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MAP 2.3

Scientific integrity and TEVV considerations are identified and documented, including those related to experimental design, data collection and selection (e.g., availability, representativeness, suitability), system trustworthiness, and construct validation.

SCF Controls: AAT-10, AAT-12.2

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure that supports AI Test, Evaluation, Validation and Verification (TEVV) activities throughout the model lifecycle. Deep learning VMs provisioned from PAIS images provide dedicated testing environments for AI prototyping, fine-tuning, validation, and inference workloads. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so TEVV activities occur in conditions comparable to the deployment environment.

The validate-then-publish workflow creates a formal gate for AI deployment. MLOps engineers validate models for inference functionality using the Triton Inference Server and evaluate performance and safety against business use cases, including examination of inference requests for malicious behavior and functional testing. Models must pass this validation before being uploaded to the Model Gallery for production use. Model integrity is verified through hash code validation, and each revision is assigned a unique digest stored as an OCI artifact with an immutable manifest, giving cryptographic verification that the tested model is the same artifact that gets deployed.

The platform also supports TEVV through its monitoring stack. VCF Operations monitors GPU consumption at cluster, host system, and host properties levels. Each DL VM includes a pre-installed DCGM Exporter with Prometheus collection at 5-second scrape intervals and Grafana visualization, supplying performance telemetry during testing and evaluation. The software stack for running AI applications on top of NVIDIA GPUs is validated in advance, and NVIDIA NGC containers are pre-tested and ready to run, reducing variables during TEVV by providing a known-good baseline.

VMware vDefend (Contributes)

VMware vDefend contributes to protecting the integrity of source data used by AI workloads on VCF through network-level security controls that restrict access to data stores and pipelines and detect malicious activity targeting data repositories.

Distributed Firewall microsegmentation enforces least-privilege network access at the vNIC level, limiting which workloads and users can reach databases, object stores, and data pipelines used for AI training and operation. Attacks that attempt to contaminate or corrupt AI source data require network access to the stores where training data resides; microsegmentation restricts that access to authorized workloads and identities. Security Intelligence provides visibility into traffic flows between workloads, generates firewall rule recommendations, and validates that security policies are enforced. Security Intelligence administrators can monitor for traffic leakage between environments, establish exceptions for authorized communication paths, and implement lockdown protection rules to control unauthorized inter-environment traffic. Gateway Firewall provides an additional enforcement point at the perimeter of network segments containing sensitive data repositories.

vDefend IDS/IPS inspects traffic to and from data repositories for known exploit patterns, command-and-control traffic, lateral movement, and data exfiltration signatures. The IDS/IPS filestore capability stores matched files to disk when a signature rule fires, supporting forensic analysis and post-incident review of suspected data integrity compromises. IDS/IPS signature-based detection also uses file magic inspection, identifying file types by their binary structure via libmagic, to detect misrepresented or malformed files in transit to data repositories.

Malware Prevention intercepts and analyzes files transferred over the network to data repositories. Profiles can be configured to send files to the cloud for detailed analysis or to process them locally without cloud connectivity. Verdict decisions classify files as Benign, Trusted, Highly Trusted, or Malicious. Security Services Platform Malware Prevention collects file analysis data including file hash submissions, assigns an Analyst UUID to each cloud submission for tracking, and records in-memory script buffer inspection details with associated analysis reports. The Security Services Platform also provides backup and restore

mechanisms covering hardware malfunctions, software malfunctions, data corruption, and malware incidents, providing a recovery path if SSP itself is compromised.

vDefend contributes to this control because its microsegmentation, threat detection, and file inspection capabilities reduce the attack surface and detection gap for data poisoning attacks targeting AI workloads. Protecting data integrity at the application layer, including content validation, provenance tracking, and checksums within data stores, is outside vDefend's domain and requires organizational processes and additional tooling.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MAP 3.4

Processes for operator and practitioner proficiency with AI system performance and trustworthiness – and relevant technical standards and certifications – are defined, assessed, and documented.

SCF Controls: AAT-10.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MAP 4.1

Approaches for mapping AI technology and legal risks of its components – including the use of third-party data or software – are in place, followed, and documented, as are risks of infringement of a third party’s intellectual property or other rights.

SCF Controls: AAT-01, AAT-02.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MAP 5.1

Likelihood and magnitude of each identified impact (both potentially beneficial and harmful) based on expected use, past uses of AI systems in similar contexts, public incident reports, feedback from those external to the team that developed or deployed the AI system, or other data are identified and documented.

SCF Controls: AAT-07.1, AAT-07.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MAP 5.2

Practices and personnel for supporting regular engagement with relevant AI actors and integrating feedback about positive, negative, and unanticipated impacts are in place and documented.

SCF Controls: AAT-11

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 1.1

Approaches and metrics for measurement of AI risks enumerated during the MAP function are selected for implementation starting with the most significant AI risks. The risks or trustworthiness characteristics that will not – or cannot – be measured are properly documented.

SCF Controls: AAT-07, AAT-16.4

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 1.3

Internal experts who did not serve as front-line developers for the system and/or independent assessors are involved in regular assessments and updates. Domain experts, users, AI actors external to the team that developed or deployed the AI system, and affected communities are consulted in support of assessments as necessary per organizational risk tolerance.

SCF Controls: AAT-11.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 2.2

Evaluations involving human subjects meet applicable requirements (including human subject protection) and are representative of the relevant population.

SCF Controls: AAT-17.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 2.3

AI system performance or assurance criteria are measured qualitatively or quantitatively and demonstrated for conditions similar to deployment setting(s). Measures are documented.

SCF Controls: AAT-20.2

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to AI implementation documentation by providing documented compute and hardware resource requirements, event log collection mechanisms, and infrastructure configuration references that support the documentation of AI deployment environments. VCF's compatibility guides document supported hardware, VCF Operations tracks infrastructure configuration, and VCF Log Management documents the event collection architecture.

Documenting AI model architecture, training methodology, and decision logic is outside VCF's scope.

VMware Private AI Services (PAIS), formerly Private AI Services, provides documentation and platform features that address several of the listed implementation documentation requirements. For computational and hardware resource requirements (item 6), the platform documents the system architecture of PAIS deployments, including GPU-accelerated workload domain requirements, vGPU profile configurations, VM class definitions for AI workloads, and the NVIDIA GPU and driver compatibility requirements. VI administrators configure vGPU-based VM classes that define the GPU resource allocations available to AI workloads. The PAIS catalog items in VCF Automation, including AI Workstation, DSM Database, Triton Inferencing Server, and AI Kubernetes Cluster, document the components users can provision in selected namespaces, and the deployment Overview tab summarizes the installed software and the access points for the application, services, and workstation VM. PAIS setup is documented for both connected and disconnected (air-gapped) deployment models, with a Harbor registry in the Supervisor or another OCI-compliant registry used to store validated ML models.

For event logging mechanisms (item 8), the platform documents multiple logging and telemetry capabilities. Deep Learning VMs record cloud-init script logs in /var/log/dl.log and GPU driver installation logs in /var/log/gpu-install.log, and they provide JupyterLab instances and console logs for verifying component installation and operation. The vLLM inference server logging level is configurable through the VLLM_LOGGING_LEVEL environment variable. PAIS Controller pod metrics are exposed at a Prometheus endpoint on the pais-controller-manager service and are integrated into observability platforms through Telegraf configuration in the Supervisor instance, with VCF Operations as the documented consuming platform. Deep Learning VMs also expose vGPU performance metrics through Prometheus. PAIS supports LLM traces collection in an OpenTelemetry Collector by configuring the spec.observability.llmTraces section of the PAISConfiguration custom resource. Knowledge Base operations track crawled documents, new documents found, deleted documents, modified documents requiring embeddings updates, and skipped documents, and record timing metrics for download, embedding generation, and the overall index operation.

For human oversight measures (item 5) and for interpreting outputs (items 4 and 5), MLOps engineers analyze LLM traces and Prometheus metrics to inspect agent execution flow across components and knowledge base indexing. The PAIS Agent Builder maintains a list of approved tools for use in AI agents and requires examination and approval of MCP server tools and capabilities before they are used by agents. Knowledge Bases are created with linked data sources such as Microsoft SharePoint sites and Amazon S3-compatible stores, which gives implementers a documented data lineage for retrieval-augmented generation outputs. Access to PAIS is controlled through OIDC authorized groups, where user accounts must be members of specified groups to consume the service, and personal access tokens generated from user account settings provide programmatic access to Knowledge Bases and data sources.

Requirements 1 (provider contact details), 2 (characteristics, capabilities, and limitations of the AI system), 3 (errata from initial conformity assessment), and 7 (projected useable lifetime) are documentation obligations carried by model vendors and the deploying organization rather than by the infrastructure platform. PAIS supplies registry and catalog mechanisms that organizations can attach revision and provenance metadata to, but the substantive documentation content for those items is outside the platform's scope.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 2.5

The AI system to be deployed is demonstrated to be valid and reliable. Limitations of the generalizability beyond the conditions under which the technology was developed are documented.

SCF Controls: AAT-10.3

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) supports demonstrating model validity, reliability, and intended operation through its validate-then-publish workflow on the Model Gallery. MLOps engineers validate ML models against the security, privacy, and technical requirements of the organization before onboarding them to the gallery, which provides the verification step the control calls for. Before validation can take place, Deep Learning VMs require issuer certificates to be installed to the trust store via the update-ca-certificates command, establishing the trusted channel between the validation workstation and the model storage backend.

The Model Gallery is implemented on a private Harbor registry that PAIS integrates with as the model storage backend, requiring authentication and certificate-based trust for model retrieval. A Harbor administrator must create a dedicated PAIS project in the private Harbor registry as part of the deployment workflow, and OCI-compliant registries from other vendors are supported for organizations that already operate a model store. Project-level access controls in Harbor restrict who can upload, retrieve, and publish models, which keeps the gate between validation and production aligned with approved designs.

To support ongoing demonstration that deployed models continue to operate as intended, PAIS metrics are integrated with observability platforms such as Grafana and VCF Operations. Observability is configured during PAIS activation in VCF Automation and must be verified as active in the PAIS instance namespace, and PAIS controller pod metrics are forwarded through Telegraf configuration in the Supervisor instance. The vcf pais CLI plug-in distributes the artifact mirroring tool used to move validated content into disconnected environments, helping keep production deployments aligned with the content that was validated upstream.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 2.6

The AI system is evaluated regularly for safety risks – as identified in the MAP function. The AI system to be deployed is demonstrated to be safe, its residual negative risk does not exceed the risk tolerance, and it can fail safely, particularly if made to operate beyond its knowledge limits. Safety metrics reflect system reliability and robustness, real-time monitoring, and response times for AI system failures.

SCF Controls: AAT-16

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to monitoring deployed AI workloads through VCF Operations, which provides continuous visibility into the health, performance, and capacity of the infrastructure those workloads run on. VCF Operations dashboards display real-time and historical metrics for compute, storage, and networking resources consumed by AI VMs. VCF Log Management aggregates log data across the stack. VMware vCenter alarms trigger on configurable thresholds for CPU, memory, disk, and network conditions.

For AI and autonomous technology workloads deployed as containerized applications on VMware Kubernetes Service (VKS) clusters, the platform provides a layered observability stack. The Prometheus Standard Package, available for deployment on VKS clusters, collects metrics from configured targets at defined intervals, evaluates alerting rules, and forwards alerts to Alertmanager. The package includes prometheus-server, which handles scraping, rule processing, and alerting, as well as prometheus-kube-state-metrics for tracking node status, replica-set compliance, pod and job status, and resource requests and limits across namespaces. For batch inference or training jobs that cannot be scraped directly, Prometheus Pushgateway accepts pushed metrics. Grafana Operator pairs with Prometheus to provide dashboards and configurable alert thresholds, enabling operators to define visual representations of AI workload metrics and receive notifications when behavior deviates from expected patterns.

Monitoring agents for Kubernetes device plugin resources, including GPUs commonly consumed by AI inference and training workloads, can be deployed as DaemonSets on worker nodes to track resource utilization and associate metrics with specific containers. The Kubernetes Node Problem Detector can be extended with custom monitoring scripts written in any language, which allows AI-specific health checks to be added without modifying core Kubernetes components. Kubernetes also provides built-in auditing capabilities for monitoring and recording cluster activities.

Istio, available as a Standard Package for VKS clusters, provides mechanisms to secure, connect, and monitor services. For AI inference services that expose APIs, Istio supplies request rate, latency, and error rate metrics without requiring application-level instrumentation.

Deployments using the Harness CI/CD Supervisor Service can integrate Dynatrace, which automatically instruments applications and infrastructure upon deployment and collects metrics, logs, and traces. The Dynatrace AI engine detects anomalies, identifies root causes, and generates intelligent alerts. The Dynatrace operator automates the deployment, scaling, and maintenance of the monitoring stack within the VKS cluster. Harness Service Reliability Management monitors deployed services post-deployment to calculate a real-time health score based on transaction success and latency.

VKS cluster management proactively monitors the VKS cluster estate to confirm assigned policies are enforced correctly, and generates Policy Insights when configuration conflicts or environmental errors are detected.

Monitoring AI model behavior at the application level, including model drift, inference accuracy, and data distribution shifts, requires organization-defined monitoring logic implemented using these platform tools. VCF and its Kubernetes consumption layer provide the collection, storage, and alerting infrastructure, but the organization must configure the signals to collect, define thresholds for AI-specific behavioral anomalies, and establish processes for reviewing and responding to monitoring data.

VMware Private AI Services (PAIS) provides production monitoring for deployed AI workloads through metrics collection, telemetry, and integration with observability platforms. The Private AI Services controller exposes a Prometheus metrics endpoint on the pais-controller-manager service, and Private AI Services deployments include a Prometheus server pod that collects metrics for observability platforms. DevOps engineers visualize health and performance metrics for Private AI Services components running in a VCF Automation namespace using observability platforms such as Grafana.

Observability is activated by updating the PAISConfiguration custom resource with parameters that enable metrics collection for observability platforms and LLM traces for the OpenTelemetry (OTel) Collector. When LLM traces collection is activated, MLOps engineers can analyze traces from agents and knowledge base indexing in the OpenTelemetry Collector, supporting observation of model behavior and application health in agentic workflows. Private AI Services supports sending traces to an OpenTelemetry collector over HTTP/protobuf or gRPC.

Operators can monitor Private AI Services deployments in Grafana for pod restart frequency, vGPU licensing status, and idle or underutilized GPU devices in passthrough mode. Private AI Services controller pod metrics are integrated into observability platforms through Telegraf configuration in the Supervisor instance, which allows VCF Operations to ingest PAIS controller metrics alongside the broader infrastructure telemetry it already collects for GPU-enabled workload domains. Deployment health is verified through Kubernetes pod state of the API, ingress, rex worker, and Prometheus server pods.

MLOps engineers and application developers can perform monitoring and diagnostics of models used in agents through observability platforms like Grafana to track the health, quality, and behavior of AI applications.

VMware vDefend (Contributes)

VMware vDefend contributes to monitoring the functionality and behavior of deployed AI workloads running on VCF by providing continuous network-level behavioral monitoring, threat detection, and AI-assisted security analysis around those workloads.

Security Intelligence, running on the Security Services Platform, provides continuous network-level behavioral monitoring that detects new traffic flows, unauthorized communications, and anomalous activity around deployed systems, including AI workloads. It continuously monitors for new flows between environments and alerts administrators to unauthorized or unexpected traffic, supporting ongoing policy adjustments. The Segmentation Monitoring Dashboard provides dedicated views for monitoring environment-level traffic flows; when monitoring identifies an environment pair without a configured protection policy, it offers a Run Recommendations option to generate policy recommendations based on observed traffic. Security Intelligence provides Infrastructure, Environment, and Application Monitoring capabilities. The application monitoring view within the Segmentation Monitoring Dashboard surfaces metrics including unprotected flows and applications with unprotected rule hits, providing targeted visibility into workloads, including AI systems, that lack active firewall policy coverage. The Policy Recommendations monitoring capability checks for changes in monitored entity scope every hour when enabled. Security Intelligence also runs a scheduled daily classification job, automatically categorizing compute entities based on traffic flows observed over the prior 30 days.

Intelligent Assist is an AI-assisted analysis feature accessible through the NSX Manager UI and Security Services Platform UI. It provides contextual analysis support on the IDS/IPS, Detections, and Campaigns pages, enabling administrators to investigate security events affecting deployed workloads with AI-driven guidance.

IDS/IPS provides ongoing threat detection through monitoring dashboards scoped to both distributed and gateway deployments. Project-scoped dashboards display events triggered for specific projects, allowing targeted monitoring of AI workload deployments. Network Detection and Response (NDR) applies behavioral analysis to identify suspicious activity targeting monitored workloads. The NDR Sensor periodically checks the status of activated features on the Security Services Platform every 15 minutes and updates its configuration accordingly.

Monitoring AI system functionality at the application layer, including model performance, output quality, and behavioral drift, is outside vDefend's domain and requires additional tooling.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 2.7

AI system security and resilience – as identified in the MAP function – are evaluated and documented.

SCF Controls: AAT-10.5

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to evaluating the security and resilience of AI workloads through VCF Operations compliance dashboards that assess infrastructure security posture, configuration drift detection, and health monitoring across the compute, storage, and networking layers that AI workloads depend on. VCF Log Management aggregates event data that supports security assessment. The Security Configuration Guide and DISA STIG provide hardening baselines that can be validated against the infrastructure hosting AI workloads.

The Consumption layer of the VCF platform, delivered through VMware Kubernetes Service (VKS) and vSphere Supervisor, provides pre-deployment evaluation mechanisms for AI and machine learning workloads. DISA STIGs are available for both the Supervisor control plane and VKS cluster VKr images, supplying a structured body of security requirements against which AI workload infrastructure can be evaluated before production deployment. AI and machine learning workloads on Supervisor run on Ubuntu-based Kubernetes release images (VKrs), which are the designated runtime for such workloads in VKS clusters. Cluster readiness assessment is built into the VKS Controller, which reports status conditions covering cluster phase and component health; administrators can query these conditions using `kubectl` to verify that the infrastructure meets defined readiness criteria before AI workloads are scheduled.

Pod Security Admission is enforced on VKS clusters and provides a pre-execution gate for workload pods. Starting with VKr v1.25, the `kubernetes.security.podSecurityStandard` configuration variable selects Baseline, Restricted, or Privileged profiles to constrain what workload pods can execute. VKS Cluster Management adds a fleet-wide policy governance layer powered by OPA Gatekeeper, with out-of-box security policy templates that mirror industry-standard Pod Security Standards profiles; policies applied at the organization or project level are automatically inherited by all child resources, providing scalable governance across the VKS cluster estate. The VKS Cluster Management Policy Insights capability surfaces health alerts for infrastructure-level failures that would prevent policy enforcement, such as an unhealthy policy operator or a failed Gatekeeper installation, giving administrators visibility into the operational state of the security control.

The Kubernetes security documentation available through Supervisor includes guidance for pre-deployment security evaluation: threat modeling to identify trust boundaries and risks, cloud-native security lifecycle checklists organized around the develop phase, and recommendations for advanced security assessment techniques such as fuzzing and security chaos engineering where appropriate. The Kubernetes Security Response Committee publishes CVE feeds and tracks vulnerabilities in the Kubernetes platform, acknowledging reports within three working days, which gives teams visibility into known security issues in the AI and machine learning runtime infrastructure. For high-trust environments where AI workloads handle sensitive data, Supervisor supports deploying workloads on confidential virtual machines, adding hardware-level memory isolation as an additional element of the security profile that can be evaluated during a pre-deployment review.

The broader AI TEVV program, including evaluation criteria, result interpretation, and model-level testing for safety, fairness, and adversarial robustness, is an organizational responsibility. VCF provides the observable, policy-governed runtime environment that supports those activities but does not perform AI-specific evaluation itself.

VMware Private AI Services (PAIS) supports pre-deployment security and resilience evaluation of AI workloads through PAIS-native mechanisms in the model lifecycle workflow. MLOps engineers validate ML models against the organization's security, privacy, and technical requirements before onboarding them to the PAIS model gallery, with deep learning VMs and the PAIS model runtime available as the validation environment for functional and behavioral testing.

PAIS observability integrates with platforms such as Grafana and VCF Operations so that MLOps engineers, application developers, and VI administrators have ongoing visibility into model and component health, quality, and behavior in deployed AI applications. Metrics collection is activated through the `PAISConfiguration` custom resource, with a Prometheus runtime, a configurable storage class, a configurable retention period, and optional LLM traces forwarded through the OpenTelemetry

(OTel) Collector. This telemetry provides the data inputs for continued security and resilience review of models in production, including tracking of anomalous model behavior over time.

PAIS deployment leases bound how long an AI workload deployment can exist, providing a control for time-limited evaluation environments where candidate models can be exercised before broader rollout. PAIS supports both connected and disconnected (air-gapped) deployment models, allowing organizations to perform security evaluation of AI workloads in isolated environments. A Harbor or other OCI-compliant registry is used as the artifact source for models and container images, giving organizations a controlled location from which validated artifacts are retrieved.

Resilience evaluation for AI workloads depends on the underlying VCF platform, which provides vSphere High Availability for automatic VM restart on host failure, Distributed Resource Scheduler for workload balancing, and vMotion for live migration of GPU-accelerated VMs. These capabilities are VCF platform mechanisms; see VCF Coverage for the underlying resilience infrastructure.

VMware vDefend (Contributes)

VMware vDefend contributes to evaluating the security and resilience of AI and autonomous technology workloads by providing continuous network-level security assessment capabilities that operate during and after deployment.

The vDefend distributed firewall includes a dedicated Security Assessment workflow that generates Security Segmentation Reports and Blast Radius Reports. These reports identify at-risk workloads, communication chains, and traffic flows, providing structured input for evaluating whether network isolation is correctly applied to AI workload segments and identifying gaps in protection coverage.

Security Intelligence offers traffic flow analysis, a quantitative Security Segmentation Score, and environment-level monitoring that can inform resilience assessments. The Security Segmentation Score calculates posture by evaluating Distributed Firewall configuration and traffic rules; score values from 0 to 50 indicate low security, 50 to 80 indicate medium security, and values above 80 indicate a highly secure environment. The score breakdown report details total environments and environment pairs, protection rule actions, and highlights environment pairs lacking inter-environment policies. Security Intelligence generates policy recommendations by analyzing VM traffic flows over specified time periods and boundary definitions, and monitors inter-environment traffic to detect leakage and support lockdown rule implementation. The Security Intelligence dashboard displays environments, environment pairs, and segmentation state to help administrators understand current security posture.

The Security Services Platform collects and surfaces IDS/IPS readiness and performance metrics. Administrators can view intrusion details including severity, CVE identifiers, CVSS scores, and affected VMs through vDefend IDS/IPS to inform mitigation planning. Intelligent Assist assists with IDS policies, rules, and security profiles during remediation operations. The vDefend Threat Intelligence Service Portal supplies application-level threat intelligence that can inform distributed and gateway firewall policies targeting lateral and perimeter network threats.

The broader TEVV program for AI security and resilience, including defining evaluation criteria, establishing test plans, and interpreting results against organizational risk thresholds, is an organizational responsibility outside vDefend's scope.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 2.8

Risks associated with transparency and accountability – as identified in the MAP function – are examined and documented.

SCF Controls: AAT-10.6

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 2.9

The AI model is explained, validated, and documented, and AI system output is interpreted within its context – as identified in the MAP function – to inform responsible use and governance.

SCF Controls: AAT-10, AAT-20.1

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure that supports AI Test, Evaluation, Validation and Verification (TEVV) activities throughout the model lifecycle. Deep learning VMs provisioned from PAIS images provide dedicated testing environments for AI prototyping, fine-tuning, validation, and inference workloads. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so TEVV activities occur in conditions comparable to the deployment environment.

The validate-then-publish workflow creates a formal gate for AI deployment. MLOps engineers validate models for inference functionality using the Triton Inference Server and evaluate performance and safety against business use cases, including examination of inference requests for malicious behavior and functional testing. Models must pass this validation before being uploaded to the Model Gallery for production use. Model integrity is verified through hash code validation, and each revision is assigned a unique digest stored as an OCI artifact with an immutable manifest, giving cryptographic verification that the tested model is the same artifact that gets deployed.

The platform also supports TEVV through its monitoring stack. VCF Operations monitors GPU consumption at cluster, host system, and host properties levels. Each DL VM includes a pre-installed DCGM Exporter with Prometheus collection at 5-second scrape intervals and Grafana visualization, supplying performance telemetry during testing and evaluation. The software stack for running AI applications on top of NVIDIA GPUs is validated in advance, and NVIDIA NGC containers are pre-tested and ready to run, reducing variables during TEVV by providing a known-good baseline.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 2.10

Privacy risk of the AI system – as identified in the MAP function – is examined and documented.

SCF Controls: AAT-10.7

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 2.11

| Fairness and bias – as identified in the MAP function – are evaluated and results are documented.

SCF Controls: AAT-10.8

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 2.12

Environmental impact and sustainability of AI model training and management activities – as identified in the MAP function – are assessed and documented.

SCF Controls: AAT-17.2

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides PAIS-native mechanisms that help organizations assess the environmental footprint of AI workloads. Metrics collection for PAIS is activated by uncommenting the `spec.observability.prometheusRuntime` section of the `PAISConfiguration` custom resource, after which DevOps and MLOps engineers can visualize PAIS health and performance metrics for components running in a VCF Automation namespace through observability platforms such as Grafana. The same observability pipeline lets MLOps engineers and application developers monitor models used in agents, giving operators visibility into the operational characteristics of individual AI workloads across deep learning VMs, VKS clusters, and RAG applications. This telemetry is the raw operational data that organizations use to quantify the energy and resource footprint of AI activity.

PAIS also supports resource efficiency through deployment leases. A lease controls how long a PAIS deployment can exist, so that AI workloads provisioned from VCF Automation catalog items are reclaimed at the end of a defined period rather than continuing to consume GPU and compute resources indefinitely. Disconnected (air-gapped) and connected deployment modes both rely on the same observability and lease controls, so the same sustainability-relevant data and lifecycle controls apply regardless of connectivity model.

Broader sustainability and green-metric monitoring at the cluster and host level is a VCF platform capability and is covered under VCF Coverage. The organizational activity of setting sustainability targets, analyzing collected metrics, and documenting environmental impact is a process obligation outside PAIS itself, which is why the rating is Contributes rather than Meets.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 2.13

Effectiveness of the employed TEVV metrics and processes in the MEASURE function are evaluated and documented.

SCF Controls: AAT-10

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 3.2

Risk tracking approaches are considered for settings where AI risks are difficult to assess using currently available measurement techniques or where metrics are not yet available.

SCF Controls: AAT-07

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 3.3

Feedback processes for end users and impacted communities to report problems and appeal system outcomes are established and integrated into AI system evaluation metrics.

SCF Controls: AAT-11.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MEASURE 4.2

Measurement results regarding AI system trustworthiness in deployment context(s) and across the AI lifecycle are informed by input from domain experts and relevant AI actors to validate whether the system is performing consistently as intended. Results are documented.

SCF Controls: AAT-10.1

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure for evaluating AI systems for trustworthy behavior, specifically addressing the security dimension of trustworthiness assessment. The validate-then-publish workflow requires MLOps engineers to validate ML models against the security, privacy, and technical requirements of the organization before uploading them to the Model Gallery. This validation includes evaluating models for inference functionality using the Triton Inference Server and examining inference requests for malicious behavior, directly addressing the security evaluation component of trustworthiness.

Deep learning VMs provisioned from PAIS images are isolated evaluation environments where data scientists and MLOps engineers can assess model behavior before production deployment. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so trustworthiness evaluation occurs under conditions comparable to the deployment environment. Model integrity is verified through hash code validation and unique content digests on each revision, stored as OCI artifacts with immutable manifests, providing cryptographic verification that the evaluated model is the same artifact that reaches production.

PAIS does not address the anonymization or disaggregation aspects of this control. Data anonymization, de-identification, and disaggregation of captured and stored data are application-layer data governance activities that fall outside the scope of infrastructure-level platform controls.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-1.3-001

Document trade-offs, decision processes, and relevant measurement and feedback results for risks that do not surpass organizational risk tolerance, for example, in the context of model release: Consider different approaches for model release, for example, leveraging a staged release approach. Consider release approaches in the context of the model and its projected use cases. Mitigate, transfer, or avoid risks that surpass organizational risk tolerances.

SCF Controls: AAT-10.4, AAT-15.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-1.3-002

Monitor the robustness and effectiveness of risk controls and mitigation plans (e.g., via red-teaming, field testing, participatory engagements, performance assessments, user feedback mechanisms).

SCF Controls: AAT-16.2

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to measuring the effectiveness of security controls for AI workloads through VCF Operations compliance dashboards that report on configuration conformance, hardening posture, and drift from baselines across the infrastructure hosting those workloads. VCF Operations generates health scores and compliance reports that can be reviewed to assess whether infrastructure-level controls are functioning. VCF Operations also monitors Kubernetes cluster health metrics, including node status and kubelet status, when the Supervisor is integrated as a data source, extending this infrastructure-level visibility to containerized workload environments.

At the Consumption layer, VMware Kubernetes Service (VKS) and the vSphere Supervisor provide technical mechanisms that support ongoing assessment of control effectiveness across containerized workloads. Pod Security Admission (PSA) supports three enforcement modes, enforce, warn, and audit, applied per namespace. When a namespace is configured with the `pod-security.kubernetes.io/audit` or `pod-security.kubernetes.io/warn` labels, violations are recorded in the Kubernetes audit log or surfaced as warnings without blocking workload deployment. This allows organizations to evaluate the compliance impact of a security policy before moving to full enforcement, and the `pod_security_evaluations_total` metric tracks the number of policy evaluations occurring across namespaces, giving administrators a quantitative view of PSA activity.

Kubernetes audit logging records all API server activity, including authentication outcomes, authorization decisions, and resource access patterns. The audit policy controls which events are captured and at what level of detail, and the resulting log stream can be delivered to a webhook backend or written to log files for integration with external monitoring and analysis platforms. Antrea NetworkPolicyStats, enabled via the NetworkPolicyStats feature flag, collects per-policy enforcement statistics including packet and flow counts for traffic matched by each network policy, allowing administrators to verify that policies are actively enforcing network boundaries and to identify policies with no traffic matches.

VKS Policy Insights provides centralized visibility into the governance health of the VKS cluster estate and generates actionable data for troubleshooting and remediating policy enforcement issues. It surfaces Health alerts when infrastructure-level conditions prevent policy enforcement, such as an unhealthy policy operator or a failed Gatekeeper installation, and raises Incompatibility insights when a pre-existing Gatekeeper installation is detected outside of cluster management. Custom VKS Cluster Management Security Policies allow administrators to fine-tune security context parameters to match specific organizational requirements. The Prometheus standard package for VKS clusters measures API server availability as the ratio of successful requests to total requests over 30-day windows, with separate tracking for read and write verb categories, and also captures request duration and error rate metrics. The Telegraf standard package complements this by scraping metrics from service endpoints such as `kube-state-metrics` and `node-exporter`.

MachineHealthCheck, a Cluster API resource configurable on VKS clusters, defines unhealthyConditions as an array of node condition types, statuses, and timeout thresholds, enabling declarative detection of nodes that have fallen outside expected health parameters. When a defined condition is met, the controller marks the node for remediation, creating a record of detected infrastructure failures against which administrators can assess the state of resilience controls. VKS cluster machine health conditions, including `KubernetesReleaseCompatible`, indicate when the Kubernetes release in use by a cluster is no longer compatible with the Supervisor, supporting assessment of upgrade readiness and compatibility posture. VCF's Network Inventory provides visibility into Kubernetes node health by surfacing standard node conditions including `MemoryPressure`, `PIDPressure`, `DiskPressure`, and the `Ready` condition, allowing administrators to assess node-level health as part of broader infrastructure reviews. The health status of VKS infrastructure components can also be observed directly by inspecting the running state and restart counts of system pods, including `etcd`, `kube-apiserver`, `antrea-agent`, `antrea-controller`, and `coredns`, running in the Supervisor cluster.

Harbor Registry, available as a Supervisor Service and as a standard package on VKS clusters, provides vulnerability scanning, content trust, and policy-based access control for container images. Harbor Registry instances on the Supervisor are organized by vSphere Namespace and are automatically trusted by all VKS clusters on the same Supervisor, creating a consistent point of control for supply chain assessment. Security Technical Implementation Guides (STIGs) are available for Supervisor components and VKS Kubernetes releases, providing hardening benchmarks against which administrators can assess deployed cluster configuration and identify gaps. GitOps tooling, such as the Argo CD Supervisor Service, combined with infrastructure as code in version control, creates an audit trail of every authorized change to cluster configuration and supports retrospective review of configuration drift.

Measuring the effectiveness of AI-specific controls such as model accuracy, bias detection, and output filtering is outside VCF's scope. The organization must define assessment cadences, review audit log data and vulnerability reports, track findings to resolution, and evaluate potential impacts on affected communities.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.2-001

Compare GAI system outputs against pre-defined organization risk tolerance, guidelines, and principles, and review and test AI-generated content against these guidelines.

SCF Controls: AAT-10.4

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.2-002

Document training data sources to trace the origin and provenance of AI-generated content.

SCF Controls: AAT-12.1, AAT-12.2

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides explicit data source identification and tracking for AI workloads. The Data Indexing and Retrieval service offers a Knowledge Bases interface where users create and manage knowledge bases, and a knowledge base can be linked to up to ten data sources. Each data source is created with a name, description, type, and the required properties for retrieving content from its original location, and a single data source can be linked to multiple knowledge bases. The data source type is fixed at creation. Supported data source types include Microsoft SharePoint sites, Amazon S3 compatible stores, and other enterprise content management systems. Each data source stores the credentials required for retrieval, such as service account key file information when applicable, which creates a documented connection between the AI system and its data origins.

During knowledge base indexing, PAIS tracks document-level detail including crawled documents, new documents found, deleted documents, existing unmodified documents, existing modified documents requiring embedding updates, and skipped documents. It captures size metrics including total megabytes downloaded, new chunks indexed corresponding to inference server invocations for embeddings, and final index size after removal of deleted documents. It records timing metrics for download, embedding generation, and the overall index operation. Document counts returned by retrieval queries are tracked, and counts of documents per media type such as PDF and PPTX are maintained for each knowledge base. Automatic collection and indexing of updates from linked data sources over time keeps these records current.

For runtime visibility, PAIS supports LLM trace collection through an OpenTelemetry Collector by configuring the `spec.observability.llmTraces` section of the `PAISConfiguration` custom resource, with traces transmitted over HTTP/protobuf or gRPC. PAIS metrics, including knowledge base activity, can be visualized in third-party observability platforms such as Grafana. The Data Indexing and Retrieval component also implements a Model Context Protocol (MCP) server that exports a set of search tools for each knowledge base, which gives downstream AI applications a documented programmatic surface tied back to specific data sources. PAIS AI catalog items, including AI Workstation, DSM Database, Triton Inferencing Server, and AI Kubernetes Cluster, are added to the VCF Automation catalog by running the PAIS Quickstart, with required images uploaded in advance to a Harbor registry in disconnected environments.

Organizational decisions about which data sources are approved for AI use, how data lineage is documented outside the platform, and how training data provenance is governed remain the responsibility of the organization. PAIS supplies the technical registration and tracking infrastructure that the organization configures and operates.

VMware vDefend (Contributes)

VMware vDefend contributes to protecting the integrity of source data used by AI workloads on VCF through network-level security controls that restrict access to data stores and pipelines and detect malicious activity targeting data repositories.

Distributed Firewall microsegmentation enforces least-privilege network access at the vNIC level, limiting which workloads and users can reach databases, object stores, and data pipelines used for AI training and operation. Attacks that attempt to contaminate or corrupt AI source data require network access to the stores where training data resides; microsegmentation restricts that access to authorized workloads and identities. Security Intelligence provides visibility into traffic flows between workloads, generates firewall rule recommendations, and validates that security policies are enforced. Security Intelligence administrators can monitor for traffic leakage between environments, establish exceptions for authorized communication paths, and implement lockdown protection rules to control unauthorized inter-environment traffic. Gateway Firewall provides an additional enforcement point at the perimeter of network segments containing sensitive data repositories.

vDefend IDS/IPS inspects traffic to and from data repositories for known exploit patterns, command-and-control traffic, lateral movement, and data exfiltration signatures. The IDS/IPS filestore capability stores matched files to disk when a signature rule

fires, supporting forensic analysis and post-incident review of suspected data integrity compromises. IDS/IPS signature-based detection also uses file magic inspection, identifying file types by their binary structure via libmagic, to detect misrepresented or malformed files in transit to data repositories.

Malware Prevention intercepts and analyzes files transferred over the network to data repositories. Profiles can be configured to send files to the cloud for detailed analysis or to process them locally without cloud connectivity. Verdict decisions classify files as Benign, Trusted, Highly Trusted, or Malicious. Security Services Platform Malware Prevention collects file analysis data including file hash submissions, assigns an Analyst UUID to each cloud submission for tracking, and records in-memory script buffer inspection details with associated analysis reports. The Security Services Platform also provides backup and restore mechanisms covering hardware malfunctions, software malfunctions, data corruption, and malware incidents, providing a recovery path if SSP itself is compromised.

vDefend contributes to this control because its microsegmentation, threat detection, and file inspection capabilities reduce the attack surface and detection gap for data poisoning attacks targeting AI workloads. Protecting data integrity at the application layer, including content validation, provenance tracking, and checksums within data stores, is outside vDefend's domain and requires organizational processes and additional tooling.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.2-003

Evaluate feedback loops between GAI system content provenance and human reviewers, and update where needed. Implement real-time monitoring systems to affirm that content provenance protocols remain effective.

SCF Controls: AAT-12.1, AAT-12.2, AAT-16.4

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides explicit data source identification and tracking for AI workloads. The Data Indexing and Retrieval service offers a Knowledge Bases interface where users create and manage knowledge bases, and a knowledge base can be linked to up to ten data sources. Each data source is created with a name, description, type, and the required properties for retrieving content from its original location, and a single data source can be linked to multiple knowledge bases. The data source type is fixed at creation. Supported data source types include Microsoft SharePoint sites, Amazon S3 compatible stores, and other enterprise content management systems. Each data source stores the credentials required for retrieval, such as service account key file information when applicable, which creates a documented connection between the AI system and its data origins.

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For runtime visibility, PAIS supports LLM trace collection through an OpenTelemetry Collector by configuring the `spec.observability.llmTraces` section of the `PAISConfiguration` custom resource, with traces transmitted over HTTP/protobuf or gRPC. PAIS metrics, including knowledge base activity, can be visualized in third-party observability platforms such as Grafana. The Data Indexing and Retrieval component also implements a Model Context Protocol (MCP) server that exports a set of search tools for each knowledge base, which gives downstream AI applications a documented programmatic surface tied back to specific data sources. PAIS AI catalog items, including AI Workstation, DSM Database, Triton Inferencing Server, and AI Kubernetes Cluster, are added to the VCF Automation catalog by running the PAIS Quickstart, with required images uploaded in advance to a Harbor registry in disconnected environments.

Organizational decisions about which data sources are approved for AI use, how data lineage is documented outside the platform, and how training data provenance is governed remain the responsibility of the organization. PAIS supplies the technical registration and tracking infrastructure that the organization configures and operates.

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vDefend IDS/IPS inspects traffic to and from data repositories for known exploit patterns, command-and-control traffic, lateral movement, and data exfiltration signatures. The IDS/IPS filestore capability stores matched files to disk when a signature rule fires, supporting forensic analysis and post-incident review of suspected data integrity compromises. IDS/IPS signature-based detection also uses file magic inspection, identifying file types by their binary structure via libmagic, to detect misrepresented or malformed files in transit to data repositories.

Malware Prevention intercepts and analyzes files transferred over the network to data repositories. Profiles can be configured to send files to the cloud for detailed analysis or to process them locally without cloud connectivity. Verdict decisions classify files as Benign, Trusted, Highly Trusted, or Malicious. Security Services Platform Malware Prevention collects file analysis data including file hash submissions, assigns an Analyst UUID to each cloud submission for tracking, and records in-memory script buffer inspection details with associated analysis reports. The Security Services Platform also provides backup and restore mechanisms covering hardware malfunctions, software malfunctions, data corruption, and malware incidents, providing a recovery path if SSP itself is compromised.

vDefend contributes to this control because its microsegmentation, threat detection, and file inspection capabilities reduce the attack surface and detection gap for data poisoning attacks targeting AI workloads. Protecting data integrity at the application layer, including content validation, provenance tracking, and checksums within data stores, is outside vDefend's domain and requires organizational processes and additional tooling.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.2-004

Evaluate GAI content and data for representational biases and employ techniques such as re-sampling, re-ranking, or adversarial training to mitigate biases in the generated content.

SCF Controls: AAT-10.8

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.2-005

Engage in due diligence to analyze GAI output for harmful content, potential misinformation, and CBRN-related or NCII content.

SCF Controls: AAT-27

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure-level mechanisms that support output filtering for AI-generated content, but the specific content policies and filtering criteria are the responsibility of the deploying organization.

The retrieval-augmented generation (RAG) architecture available through PAIS helps reduce false or fabricated content. RAG workloads combine a large language model with an external knowledge base to ground responses in organizational data rather than relying solely on the model's training data. Data scientists can create knowledge bases linked to data sources hosted in external content management systems for production use cases, including Microsoft SharePoint On-Premises deployments, with a single knowledge base linked to up to ten data sources. PAIS knowledge base operations track document counts including crawled documents, newly found documents, deleted documents, and existing modified documents requiring embeddings update, so that the grounding corpus stays current and traceable.

PAIS extends grounding through Model Context Protocol (MCP) integration. PAIS can connect to MCP servers to integrate real-time data and specialized capabilities with generative AI applications, helping models respond from current authoritative sources rather than improvising answers from training data alone.

For model lifecycle, PAIS uses a Harbor registry, or another OCI-compliant registry, as the model gallery from which models are pulled into the platform. This supports a curated set of approved models rather than ad hoc model use. MLOps engineers and application developers can monitor model and agent behavior through observability platforms such as Grafana to track the health, quality, and behavior of AI applications, which gives operators a feedback path for identifying when a deployed model is producing problematic output.

PAIS does not include a built-in content classification or filtering API that automatically screens outputs against predefined content categories. The platform supplies several mechanisms that organizations can use to implement output filtering: RAG grounding, MCP-based real-time data integration, curated model deployment, and runtime observability. The specific content policies, acceptance thresholds, and filtering logic are defined by the deploying organization to match its regulatory and business requirements.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.2-006

Use feedback from internal and external AI Actors, users, individuals, and communities, to assess impact of AI-generated content.

SCF Controls: AAT-11.1, AAT-11.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.2-007

Use real-time auditing tools where they can be demonstrated to aid in the tracking and validation of the lineage and authenticity of AI-generated data.

SCF Controls: AAT-10

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure that supports AI Test, Evaluation, Validation and Verification (TEVV) activities throughout the model lifecycle. Deep learning VMs provisioned from PAIS images provide dedicated testing environments for AI prototyping, fine-tuning, validation, and inference workloads. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so TEVV activities occur in conditions comparable to the deployment environment.

The validate-then-publish workflow creates a formal gate for AI deployment. MLOps engineers validate models for inference functionality using the Triton Inference Server and evaluate performance and safety against business use cases, including examination of inference requests for malicious behavior and functional testing. Models must pass this validation before being uploaded to the Model Gallery for production use. Model integrity is verified through hash code validation, and each revision is assigned a unique digest stored as an OCI artifact with an immutable manifest, giving cryptographic verification that the tested model is the same artifact that gets deployed.

The platform also supports TEVV through its monitoring stack. VCF Operations monitors GPU consumption at cluster, host system, and host properties levels. Each DL VM includes a pre-installed DCGM Exporter with Prometheus collection at 5-second scrape intervals and Grafana visualization, supplying performance telemetry during testing and evaluation. The software stack for running AI applications on top of NVIDIA GPUs is validated in advance, and NVIDIA NGC containers are pre-tested and ready to run, reducing variables during TEVV by providing a known-good baseline.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.2-008

Use structured feedback mechanisms to solicit and capture user input about AI- generated content to detect subtle shifts in quality or alignment with community and societal values.

SCF Controls: AAT-11.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.3-001

Develop and update GAI system incident response and recovery plans and procedures to address the following: Review and maintenance of policies and procedures to account for newly encountered uses; Review and maintenance of policies and procedures for detection of unanticipated uses; Verify response and recovery plans account for the GAI system value chain; Verify response and recovery plans are updated for and include necessary details to communicate with downstream GAI system Actors: Points-of-Contact (POC), Contact information, notification format.

SCF Controls: AAT-01, BCD-01, IRO-04

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.4-001

Establish and maintain communication plans to inform AI stakeholders as part of the deactivation or disengagement process of a specific GAI system (including for open-source models) or context of use, including reasons, workarounds, user access removal, alternative processes, contact information, etc.

SCF Controls: AAT-11

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.4-002

Establish and maintain procedures for escalating GAI system incidents to the organizational risk management authority when specific criteria for deactivation or disengagement is met for a particular context of use or for the GAI system as a whole.

SCF Controls: AAT-15.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.4-003

Establish and maintain procedures for the remediation of issues which trigger incident response processes for the use of a GAI system, and provide stakeholders timelines associated with the remediation plan.

SCF Controls: AAT-18.1

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides AI-specific mechanisms that support an organization's risk response process for AI systems. MLOps engineers validate ML models onboarded to PAIS against the security, privacy, and technical requirements of the organization, providing a structured gate for remediation when an existing model is identified as risky and a replacement must be vetted before publication. When remediation requires substituting a model, operators can deploy completion or embedding model endpoints and use A/B testing to compare the behavior of candidate models against the incumbent in real-world settings, supporting empirical evaluation of the replacement before full rollout. Embedding model selection is recommended to be based on empirical evaluation using the organization's own data rather than relying solely on benchmarks.

Health and performance metrics for PAIS components running in a VCF Automation namespace can be visualized in observability platforms such as Grafana and VCF Operations. MLOps engineers and application developers can monitor the health, quality, and behavior of AI applications through these observability surfaces to detect anomalies that warrant a risk response. PAIS observability is configured through a Prometheus runtime with a specified storage class and a configurable metrics retention period, with a default of 14 days, giving risk responders historical context for investigation. Metrics collection is activated by enabling the `spec.observability.prometheusRuntime` section of the `PAISConfiguration` CR.

Organizational processes for prioritizing AI risks based on assessments, coordinating remediation across stakeholders, and tracking risk closure remain organizational responsibilities outside the scope of PAIS infrastructure capabilities. General platform resilience inherited from VCF is covered under VCF Coverage and is not a PAIS-specific contribution to this control.

VMware vDefend (Contributes)

VMware vDefend contributes to prioritizing, responding to, and remediating security risks affecting AI workloads running on VCF by providing severity-based threat prioritization, multiple response mechanisms, and policy-based remediation at the network layer.

Prioritization. Security Intelligence categorizes detected suspicious traffic events by impact type (Critical, High, Medium, and Low), helping security teams focus first on the most urgent risks to AI workloads. The Security Services Platform Campaign Impact Score measures the overall severity of correlated threat campaigns on a scale from 0 through 100, with scores calculated using the confidence level of each constituent detection and higher scores indicating higher-priority campaigns for triage. Malware Prevention's Process Analysis Report classifies risk estimates and flags artifacts as High risk when they represent a critical risk requiring priority attention. vDefend IDS/IPS follows a prioritized event-handling approach under memory pressure, dropping lower-priority severity data (Low and Informational) first and retaining Critical and High severity events, so that the most significant detections are preserved for triage and response.

Response. When security risks to AI workloads are identified, vDefend provides multiple response options calibrated to the situation. Administrators can deploy microsegmentation rules to isolate affected AI workloads, adjust IDS/IPS profiles to address newly identified threat patterns, or modify Gateway Firewall policies to restrict traffic at the perimeter. The Security Intelligence threat visualization canvas supports rapid threat containment by identifying compromised assets and guiding microsegmentation policy updates. Intelligent Assist analyzes detected anomalies and recommends remediation policies; remediation rules are added using the naming convention prefix 'ai-' and are not enabled by default, requiring an analyst or administrator to review and manually enable them after due diligence and verification. When Intelligent Assist determines that a given anomaly cannot be remediated, it indicates this automatically, allowing analysts to focus on actionable findings.

Security Intelligence retains the IDS policies, rules, and security profiles created by Intelligent Assist as part of remediation operations, providing an audit trail of response actions taken.

Remediation. The Security Intelligence recommendation engine generates firewall policy recommendations based on observed traffic patterns between virtual machines. These recommendations can be published as firewall policies applied to environment category pairs, enabling organizations to close segmentation gaps identified during risk assessments. Administrators can designate high-priority applications and groups through prioritized collections, focusing Security Intelligence's flow analysis on the AI workloads most critical to protect.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-2.4-004

Establish and regularly review specific criteria that warrants the deactivation of GAI systems in accordance with set risk tolerances and appetites.

SCF Controls: AAT-15.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-3.1-001

Apply organizational risk tolerances and controls (e.g., acquisition and procurement processes; assessing personnel credentials and qualifications, performing background checks; filtering GAI input and outputs, grounding, fine tuning, retrieval-augmented generation) to third-party GAI resources: Apply organizational risk tolerance to the utilization of third-party datasets and other GAI resources; Apply organizational risk tolerances to fine-tuned third-party models; Apply organizational risk tolerance to existing third-party models adapted to a new domain; Reassess risk measurements after fine-tuning third-party GAI models.

SCF Controls: AAT-02.3

Aggregate Coverage: Contributes

VCF (Contributes)

VCF provides the security and resilience infrastructure that protects AI and autonomous technology workloads running on the platform. VCF's layered security model includes VMware vCenter RBAC with granular roles and permissions that control who can manage AI workload VMs, VCF SSO for centralized authentication, ESX Secure Boot and TPM for host integrity, VM encryption and vSAN encryption at rest for data protection, and VDS/NSX network segmentation for workload isolation. VCF Operations provides continuous monitoring with compliance dashboards that assess the security posture of the infrastructure hosting AI workloads. VCF Operations also monitors Kubernetes-specific health metrics for Supervisor-managed clusters, including node status and kubelet status. vSphere HA and DRS provide compute resilience, and vSAN fault domains provide storage resilience. VKS clusters support MachineHealthCheck, a Kubernetes Cluster API resource that monitors defined node health conditions and triggers automatic machine remediation when worker nodes become unhealthy, providing a cluster-native resilience mechanism for AI workload infrastructure.

The Consumption layer of VCF, delivered through VMware Kubernetes Service (VKS) and the vSphere Supervisor, extends these protections to containerized AI and machine learning workloads. VKS requires the Ubuntu VKr node operating system for AI/ML workloads, establishing a defined and consistent runtime configuration to which security policy can be applied. Pod Security Admission (PSA) is enforced at the namespace level and supports three modes (warn, audit, and enforce) combined with three profiles (privileged, baseline, and restricted). The restricted profile disallows privileged containers, limits Linux capabilities, requires non-root execution contexts, and restricts volume types. The podSecurityStandard variable in VKS cluster configuration sets the cluster-wide default pod security policy, with per-namespace exemptions defined where workload requirements justify them. VKS cluster management provides out-of-the-box security policy templates for the baseline and restricted profiles, letting operators match the level of restriction to each workload's assessed risk without manual policy authoring.

VKS clusters enforce Kubernetes NetworkPolicy objects that restrict ingress and egress traffic at the pod and namespace level, allowing AI/ML workloads to be deployed in dedicated namespaces with rules that limit communication to authorized services. The Antrea AntreaPolicy feature gate enables fine-grained network policy enforcement through the Antrea CNI. The Istio CNI plugin can be enabled on VKS clusters to manage pod network interfaces and apply service mesh policies without requiring privileged containers, supporting AI inference pipelines that need encrypted inter-service communication and fine-grained traffic routing.

Kubernetes Deployments and ReplicaSets apply an immutable infrastructure pattern in which containers are replaced rather than modified in place, making unauthorized persistent changes to AI/ML workloads visible and reducing the risk of persistent compromise. More broadly, Kubernetes workload resources including Deployments, StatefulSets, and DaemonSets provide self-healing capabilities that automatically replace failed containers and reschedule workloads to available nodes when machines become unavailable. Liveness probes detect and remedy broken application states by restarting containers that fail health checks, supporting continuity of AI/ML workload operation without operator intervention. Harbor, available as a Supervisor Service and deployable as a standard package on VKS clusters, provides a private container image registry scoped by vSphere Namespace. Integrating workload deployments with a governed Harbor registry restricts AI/ML containers to

images sourced from a controlled registry, reducing supply chain risk. In air-gapped environments, Harbor Supervisor Service serves as the private registry from which VKS images and upgrade packages are pulled. Content libraries in the Supervisor environment support creation of VM images with current patches and required security settings following organizational security policies.

VCF Automation Organizations provide centralized access control and establish secure boundaries between organizational units consuming VKS clusters, supporting delegation of cluster management with enforced separation. VKS cluster management policies enable governance of the operational consistency and security posture of the cluster estate through policy-driven automation, covering both baseline and restricted security configurations.

Kubernetes documentation in the Consumption layer addresses cloud-native security practices applicable to AI/ML workloads, including threat modeling during the Develop lifecycle phase to identify trust boundaries and risks, and applying cryptographic protections for audit logs in high-assurance environments. Confidential virtual machines are available for workloads requiring additional hardware-level isolation.

AI-specific risk assessment and the alignment of protections to AI threat models are organizational responsibilities. VCF provides the infrastructure and platform mechanisms that those assessments direct, including the ability to calibrate pod security profile stringency, network segmentation scope, and image provenance controls to the assessed risk of each AI workload. Fully satisfying this control requires organizational processes to define acceptable risk thresholds and deliberately select the appropriate policy configuration based on each workload's risk assessment.

This control asks whether the organization ensures AI and autonomous technologies include reasonable security, compliance, and resilience protections commensurate with assessed risks and threats. The control is categorized as Process under NIST CSF Govern, which evaluates organizational risk assessment and governance practice rather than platform capability. VMware Private AI Services (PAIS) supplies technical mechanisms that the organization can configure when planning AI workload protections, including MLOps-driven validation of ML models against organizational security, privacy, and technical requirements before onboarding, examination and approval workflows for Model Context Protocol (MCP) server tools and capabilities before agents use them, CA trust bundle configuration for secure communication with external services and databases, Harbor and OCI-compliant registry integration with certificate-based trust for model retrieval, observability through Prometheus and Grafana for monitoring AI model health, quality, and behavior, and support for air-gapped deployment models for environments with strict isolation requirements. Determining which protections are commensurate with the organization's assessed risks and threats, and governing AI initiatives under an enterprise risk program, are organizational activities that PAIS does not perform.

VMware vDefend (Contributes)

VMware vDefend contributes to ensuring AI workloads running on VCF include reasonable cybersecurity protections commensurate with assessed risks and threats.

The vDefend distributed firewall and microsegmentation capabilities enforce network-level protections that can be scaled proportionally to each AI workload's risk profile. Security Intelligence monitors traffic flows, generates firewall rule recommendations, and validates security policies for the Distributed Firewall, providing Infrastructure, Environment, and Application Monitoring across the deployment. Security Intelligence generates policy recommendations for environment pairs based on observed inter-VM traffic patterns, and these recommendations can be published as firewall policies applied to environment category pairs. The Recommendation engine selects inter-environment traffic that hits protection rules or default category rules, producing granular application-level rules that allow only authorized traffic between environments. Administrators can identify all inter-environment flows and secure them with specific policies before enforcing strict block rules between environments.

Security Services Platform Environment Protection evaluates whether workloads are protected by environment-specific segmentation policies such as Dev, Prod, and Test. Security Intelligence provides metrics for monitoring firewall rule effectiveness and proactive threat detection. Comparing the ratio of detected versus prevented events provides a means to assess how effectively security controls are protecting infrastructure, giving administrators insight into whether current

protections are proportionate to AI workload risk. Security Intelligence also collects IDS policies, rules, and security profiles created by Intelligent Assist as part of remediation operations, enabling policy adjustments in response to detected threats.

IDS/IPS inspects traffic to and from AI workloads for exploit attempts, command-and-control traffic, lateral movement, and exfiltration patterns. Malware Prevention analyzes files entering the environment for malicious content. vDefend rules can enforce isolation between projects, denying all inter-project communication except for required infrastructure protocols such as DHCP, DNS, NTP, and ICMP.

vDefend contributes to this control because its Distributed Firewall, IDS/IPS, Malware Prevention, and Security Intelligence capabilities provide network-layer protections that can be scaled to each AI workload's assessed risk level. However, ensuring adequate protections across all dimensions of AI risk, including application security, data governance, model integrity, and operational resilience, extends beyond the network security domain that vDefend addresses.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) contributes to the security and resilience protections for AI and autonomous technology workloads through replication, failover orchestration, and isolated recovery capabilities. When AI workloads run on VCF infrastructure, PNR protects them with configurable replication intervals, automated recovery plans, and tested failover procedures. Protection and Recovery organizes virtual machines into protection groups as the basis for cross-site replication and failover, and recovery plans can be tested in quarantined test networks that have no lasting effects on either the protected site or the recovery site.

Cyber Recovery extends protection with vulnerability analysis and clean room recovery capabilities. The Integrated Security and Vulnerability Analysis with CrowdStrike assigns vulnerability risk scores on a scale from 0 (no risk) to 10.0 (maximum risk), assessing the risk associated with vulnerabilities based on exploitability likelihood and predicted potential impact on virtual machines in the recovery environment. Cyber Recovery implements a "Quarantined + Analysis" isolation level that restricts virtual machine connectivity to integrated security and vulnerability servers, providing a controlled environment for analyzing workloads before returning them to production. Periodic security scans of virtual machines in the clean room region support compliance validation prior to production restoration.

The broader requirements of this control, including AI-specific risk assessment, threat identification, and alignment of protections to assessed risk levels, are organizational responsibilities outside PNR's scope.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) contributes to the security and resilience protections for AI workloads that store data in DSM-managed databases. DSM provides role-based access control with Admin and User roles that restrict who can administer the environment and who can manage databases, and namespace-scoped restrictions that allow a Cloud Administrator to control which InfrastructurePolicies and BackupLocations are available in each user namespace. Infrastructure policies act as guardrails that restrict the quality and quantity of resources DSM users can consume from vSphere clusters, covering CPU, memory, storage policies, IP pool, VM classes, and related resources, and the Infrastructure Policies view in the DSM administrator UI gives administrators a single place to review available policies and the databases associated with them.

Data service policies can be defined for MySQL, PostgreSQL, and SQL Server and configured to enforce consistent security and compliance posture across tenants. Each policy can require, allow, or disallow automated backups, and when backups are required at least one backup location must be configured with a trusted root certificate. Resilience for database workloads is supported through high-availability topologies; for workloads that need both data protection and high read throughput, a three-node HA cluster can be deployed with multiple read replicas, where the primary node synchronously replicates to failover nodes and asynchronously replicates to read replicas. DSM generates infrastructure policy compliance alerts, including `INFRASTRUCTURE_POLICY_CLUSTER_COMPLIANT` and `INVALID_SUPERVISOR_INFRASTRUCTURE_POLICY_STATUS`, that monitor cluster configuration and Supervisor infrastructure policy state so administrators can detect drift from defined policy.

The broader requirements of this control, including AI-specific risk assessment and alignment of protections to assessed risk levels, are organizational responsibilities outside DSM's scope.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer (Avi) provides application-layer security mechanisms that organizations can use to protect AI and autonomous technology workloads running behind Avi virtual services. These capabilities address the security and resilience dimensions of this control at the application delivery layer.

Avi's Web Application Firewall (WAF) supports a positive security model through WAF Positive Security rules, which define allowed application behavior as an allowlist of expected request patterns. WAF Positive Security Groups can be created and configured within a WAF Policy to define allowed request patterns and enforce positive security models. Avi WAF Policies attach a Positive Security group to apply learned data for security and to define hit and miss actions. Administrators can enable Application Insights to analyze known-good traffic and generate a Positive Security group for ongoing enforcement. For AI API endpoints where expected request structures can be defined, this provides a mechanism to enforce consistent application-layer access controls.

The Avi Cloud Console Application Rules Service allows organizations to protect their applications from vulnerabilities by enabling the service on their Avi Controllers, providing a cloud-delivered mechanism for rule-based protection of AI application endpoints.

Avi's Bot Management capability detects and classifies automated traffic, including bots and autonomous agents, and applies enforcement actions based on classification results. The Avi Cloud Console BOT Management Service can allow, deny, or rate-limit bad bots, and supports User Agent Database Sync to receive updated user agent signatures for threat detection. Bot detection policies are created using the `configure botdetectionpolicy` CLI command or through HTTP security policy rules. WAF HTTP Security policies support Bot Management as a match condition, allowing administrators to select specific bot classifications and assign enforcement actions such as allow, deny, or rate limiting. The BOT Management Service requires the IP Reputation service to be enabled for full coverage of bad bot traffic. A `System-BotDetectionPolicy` can be bound to virtual services to enforce bot detection rules across AI workload endpoints. For AI workloads that interact with automated clients, this gives administrators visibility and control over which automated systems are permitted access.

In Kubernetes deployments using the Avi Kubernetes Operator (AKO), the `L7Rule` custom resource definition supports a `botPolicyRef` parameter to reference a bot policy for bot detection and mitigation on individual ingress routes. Virtual services can also reference a `SecurityPolicy` to perform DDoS attack mitigation on traffic flowing through the virtual service, giving platform teams a Kubernetes-native path to apply Avi security capabilities to AI workloads running in clusters.

Avi Service Engines automatically defend against DDoS attacks by implementing TCP SYN cookies, idle connection reaping, and protocol validation. The Service Engine data-plane architecture isolates attacks on the data plane from management interfaces, so a volumetric attack against a backend AI service does not affect Controller management. Service Engines detect and mitigate a wide range of Layer 4-7 network attacks to protect application traffic.

Avi's Network Security Policy supports creation of rules that match on IP address and apply IP Groups to control access, providing a mechanism to restrict which sources can reach AI application endpoints.

Avi supports role-based access control for administrative access to the Controller. Administrators can restrict administrative access by source IP address range and use RBAC to limit access levels, scoping the administrative attack surface for Avi infrastructure that fronts AI workloads.

Control MG-3.1-002

Test GAI system value chain risks (e.g., data poisoning, malware, other software and hardware vulnerabilities; labor practices; data privacy and localization compliance; geopolitical alignment).

SCF Controls: AAT-10

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure that supports AI Test, Evaluation, Validation and Verification (TEVV) activities throughout the model lifecycle. Deep learning VMs provisioned from PAIS images provide dedicated testing environments for AI prototyping, fine-tuning, validation, and inference workloads. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so TEVV activities occur in conditions comparable to the deployment environment.

The validate-then-publish workflow creates a formal gate for AI deployment. MLOps engineers validate models for inference functionality using the Triton Inference Server and evaluate performance and safety against business use cases, including examination of inference requests for malicious behavior and functional testing. Models must pass this validation before being uploaded to the Model Gallery for production use. Model integrity is verified through hash code validation, and each revision is assigned a unique digest stored as an OCI artifact with an immutable manifest, giving cryptographic verification that the tested model is the same artifact that gets deployed.

The platform also supports TEVV through its monitoring stack. VCF Operations monitors GPU consumption at cluster, host system, and host properties levels. Each DL VM includes a pre-installed DCGM Exporter with Prometheus collection at 5-second scrape intervals and Grafana visualization, supplying performance telemetry during testing and evaluation. The software stack for running AI applications on top of NVIDIA GPUs is validated in advance, and NVIDIA NGC containers are pre-tested and ready to run, reducing variables during TEVV by providing a known-good baseline.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-3.1-003

Re-assess model risks after fine-tuning or retrieval-augmented generation implementation and for any third-party GAI models deployed for applications and/or use cases that were not evaluated in initial testing.

SCF Controls: AAT-10.1, AAT-11.2

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure for evaluating AI systems for trustworthy behavior, specifically addressing the security dimension of trustworthiness assessment. The validate-then-publish workflow requires MLOps engineers to validate ML models against the security, privacy, and technical requirements of the organization before uploading them to the Model Gallery. This validation includes evaluating models for inference functionality using the Triton Inference Server and examining inference requests for malicious behavior, directly addressing the security evaluation component of trustworthiness.

Deep learning VMs provisioned from PAIS images are isolated evaluation environments where data scientists and MLOps engineers can assess model behavior before production deployment. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so trustworthiness evaluation occurs under conditions comparable to the deployment environment. Model integrity is verified through hash code validation and unique content digests on each revision, stored as OCI artifacts with immutable manifests, providing cryptographic verification that the evaluated model is the same artifact that reaches production.

PAIS does not address the anonymization or disaggregation aspects of this control. Data anonymization, de-identification, and disaggregation of captured and stored data are application-layer data governance activities that fall outside the scope of infrastructure-level platform controls.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-3.1-004

Take reasonable measures to review training data for CBRN information, and intellectual property, and where appropriate, remove it. Implement reasonable measures to prevent, flag, or take other action in response to outputs that reproduce particular training data (e.g., plagiarized, trademarked, patented, licensed content or trade secret material).

SCF Controls: AAT-12, AAT-26

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-3.1-005

Review various transparency artifacts (e.g., system cards and model cards) for third-party models.

SCF Controls: AAT-20.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-3.2-003

Document sources and types of training data and their origins, potential biases present in the data related to the GAI application and its content provenance, architecture, training process of the pre-trained model including information on hyperparameters, training duration, and any fine-tuning or retrieval-augmented generation processes applied.

SCF Controls: AAT-10.8

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-3.2-004

Evaluate user reported problematic content and integrate feedback into system updates.

SCF Controls: AAT-11.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-3.2-005

Implement content filters to prevent the generation of inappropriate, harmful, false, illegal, or violent content related to the GAI application, including for CSAM and NCII. These filters can be rule-based or leverage additional machine learning models to flag problematic inputs and outputs.

SCF Controls: AAT-27

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure-level mechanisms that support output filtering for AI-generated content, but the specific content policies and filtering criteria are the responsibility of the deploying organization.

The retrieval-augmented generation (RAG) architecture available through PAIS helps reduce false or fabricated content. RAG workloads combine a large language model with an external knowledge base to ground responses in organizational data rather than relying solely on the model's training data. Data scientists can create knowledge bases linked to data sources hosted in external content management systems for production use cases, including Microsoft SharePoint On-Premises deployments, with a single knowledge base linked to up to ten data sources. PAIS knowledge base operations track document counts including crawled documents, newly found documents, deleted documents, and existing modified documents requiring embeddings update, so that the grounding corpus stays current and traceable.

PAIS extends grounding through Model Context Protocol (MCP) integration. PAIS can connect to MCP servers to integrate real-time data and specialized capabilities with generative AI applications, helping models respond from current authoritative sources rather than improvising answers from training data alone.

For model lifecycle, PAIS uses a Harbor registry, or another OCI-compliant registry, as the model gallery from which models are pulled into the platform. This supports a curated set of approved models rather than ad hoc model use. MLOps engineers and application developers can monitor model and agent behavior through observability platforms such as Grafana to track the health, quality, and behavior of AI applications, which gives operators a feedback path for identifying when a deployed model is producing problematic output.

PAIS does not include a built-in content classification or filtering API that automatically screens outputs against predefined content categories. The platform supplies several mechanisms that organizations can use to implement output filtering: RAG grounding, MCP-based real-time data integration, curated model deployment, and runtime observability. The specific content policies, acceptance thresholds, and filtering logic are defined by the deploying organization to match its regulatory and business requirements.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-3.2-006

Implement real-time monitoring processes for analyzing generated content performance and trustworthiness characteristics related to content provenance to identify deviations from the desired standards and trigger alerts for human intervention.

SCF Controls: MON-01.2, MON-05.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-3.2-007

Leverage feedback and recommendations from organizational boards or committees related to the deployment of GAI applications and content provenance when using third-party pre-trained models.

SCF Controls: AAT-11.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-3.2-008

Use human moderation systems where appropriate to review generated content in accordance with human-AI configuration policies established in the Govern function, aligned with socio-cultural norms in the context of use, and for settings where AI models are demonstrated to perform poorly.

SCF Controls: AAT-27.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-3.2-009

Use organizational risk tolerance to evaluate acceptable risks and performance metrics and decommission or retrain pre-trained models that perform outside of defined limits.

SCF Controls: AAT-10.4

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-4.1-001

Collaborate with external researchers, industry experts, and community representatives to maintain awareness of emerging best practices and technologies in measuring and managing identified risks.

SCF Controls: AAT-16.5

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-4.1-002

Establish, maintain, and evaluate effectiveness of organizational processes and procedures for post-deployment monitoring of GAI systems, particularly for potential confabulation, CBRN, or cyber risks.

SCF Controls: AAT-10.11

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-4.1-003

Evaluate the use of sentiment analysis to gauge user sentiment regarding GAI content performance and impact, and work in collaboration with AI Actors experienced in user research and experience.

SCF Controls: AAT-01, AAT-10

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-4.1-005

Share transparency reports with internal and external stakeholders that detail steps taken to update the GAI system to enhance transparency and accountability.

SCF Controls: AAT-10.6

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-4.1-006

Track dataset modifications for provenance by monitoring data deletions, rectification requests, and other changes that may impact the verifiability of content origins.

SCF Controls: AAT-12.1, AAT-12.2

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides explicit data source identification and tracking for AI workloads. The Data Indexing and Retrieval service offers a Knowledge Bases interface where users create and manage knowledge bases, and a knowledge base can be linked to up to ten data sources. Each data source is created with a name, description, type, and the required properties for retrieving content from its original location, and a single data source can be linked to multiple knowledge bases. The data source type is fixed at creation. Supported data source types include Microsoft SharePoint sites, Amazon S3 compatible stores, and other enterprise content management systems. Each data source stores the credentials required for retrieval, such as service account key file information when applicable, which creates a documented connection between the AI system and its data origins.

During knowledge base indexing, PAIS tracks document-level detail including crawled documents, new documents found, deleted documents, existing unmodified documents, existing modified documents requiring embedding updates, and skipped documents. It captures size metrics including total megabytes downloaded, new chunks indexed corresponding to inference server invocations for embeddings, and final index size after removal of deleted documents. It records timing metrics for download, embedding generation, and the overall index operation. Document counts returned by retrieval queries are tracked, and counts of documents per media type such as PDF and PPTX are maintained for each knowledge base. Automatic collection and indexing of updates from linked data sources over time keeps these records current.

For runtime visibility, PAIS supports LLM trace collection through an OpenTelemetry Collector by configuring the `spec.observability.llmTraces` section of the `PAISConfiguration` custom resource, with traces transmitted over HTTP/protobuf or gRPC. PAIS metrics, including knowledge base activity, can be visualized in third-party observability platforms such as Grafana. The Data Indexing and Retrieval component also implements a Model Context Protocol (MCP) server that exports a set of search tools for each knowledge base, which gives downstream AI applications a documented programmatic surface tied back to specific data sources. PAIS AI catalog items, including AI Workstation, DSM Database, Triton Inferencing Server, and AI Kubernetes Cluster, are added to the VCF Automation catalog by running the PAIS Quickstart, with required images uploaded in advance to a Harbor registry in disconnected environments.

Organizational decisions about which data sources are approved for AI use, how data lineage is documented outside the platform, and how training data provenance is governed remain the responsibility of the organization. PAIS supplies the technical registration and tracking infrastructure that the organization configures and operates.

VMware vDefend (Contributes)

VMware vDefend contributes to protecting the integrity of source data used by AI workloads on VCF through network-level security controls that restrict access to data stores and pipelines and detect malicious activity targeting data repositories.

Distributed Firewall microsegmentation enforces least-privilege network access at the vNIC level, limiting which workloads and users can reach databases, object stores, and data pipelines used for AI training and operation. Attacks that attempt to contaminate or corrupt AI source data require network access to the stores where training data resides; microsegmentation restricts that access to authorized workloads and identities. Security Intelligence provides visibility into traffic flows between workloads, generates firewall rule recommendations, and validates that security policies are enforced. Security Intelligence administrators can monitor for traffic leakage between environments, establish exceptions for authorized communication paths, and implement lockdown protection rules to control unauthorized inter-environment traffic. Gateway Firewall provides an additional enforcement point at the perimeter of network segments containing sensitive data repositories.

vDefend IDS/IPS inspects traffic to and from data repositories for known exploit patterns, command-and-control traffic, lateral movement, and data exfiltration signatures. The IDS/IPS filestore capability stores matched files to disk when a signature rule fires, supporting forensic analysis and post-incident review of suspected data integrity compromises. IDS/IPS signature-based detection also uses file magic inspection, identifying file types by their binary structure via libmagic, to detect misrepresented or malformed files in transit to data repositories.

Malware Prevention intercepts and analyzes files transferred over the network to data repositories. Profiles can be configured to send files to the cloud for detailed analysis or to process them locally without cloud connectivity. Verdict decisions classify files as Benign, Trusted, Highly Trusted, or Malicious. Security Services Platform Malware Prevention collects file analysis data including file hash submissions, assigns an Analyst UUID to each cloud submission for tracking, and records in-memory script buffer inspection details with associated analysis reports. The Security Services Platform also provides backup and restore mechanisms covering hardware malfunctions, software malfunctions, data corruption, and malware incidents, providing a recovery path if SSP itself is compromised.

vDefend contributes to this control because its microsegmentation, threat detection, and file inspection capabilities reduce the attack surface and detection gap for data poisoning attacks targeting AI workloads. Protecting data integrity at the application layer, including content validation, provenance tracking, and checksums within data stores, is outside vDefend's domain and requires organizational processes and additional tooling.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-4.1-007

Verify that AI Actors responsible for monitoring reported issues can effectively evaluate GAI system performance including the application of content provenance data tracking techniques, and promptly escalate issues for response.

SCF Controls: AAT-10.13

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to post-deployment monitoring of AI workloads through VCF Operations, which provides continuous health, performance, and capacity monitoring for the infrastructure hosting those workloads. VCF Operations dashboards surface CPU, memory, storage, and network metrics for VMs running AI workloads. VCF Log Management aggregates and analyzes log data from VMware vCenter, VMware ESX, and NSX components. vCenter alarms provide threshold-based alerting for infrastructure conditions that could affect AI workload availability.

For AI workloads running as containerized applications on VMware Kubernetes Service (VKS) clusters, the platform extends infrastructure-level observability to the Kubernetes layer. VKS clusters provisioned with the automated-monitoring label automatically receive monitoring configuration on each new node, integrating cluster telemetry with VCF Operations. The Prometheus standard package, installable on VKS clusters through the Supervisor Management Proxy Service, deploys kube-state-metrics, node-exporter, pushgateway, Prometheus, and Alertmanager as components. The prometheus-kube-state-metrics component monitors node status and capacity, ReplicaSet compliance, pod status, job and CronJob status, and resource requests and limits, providing continuous metric collection that can be directed at AI inference services or training job pods. Telegraf is available as a standard package that deploys as a Deployment component to scrape metrics from kube-state-metrics and node-exporter and forward them to external observability backends. Monitoring and alerting for Kubernetes deployments is also available through the Argo CD Supervisor Service using Grafana Operator and Prometheus.

Kubernetes-native health probes and status tracking support workload condition detection at the pod level. Liveness and readiness probes can be configured on AI workload containers to detect unresponsive or overloaded model serving instances; a readiness probe removes the pod from service routing when it cannot handle requests, and a liveness probe triggers a container restart when the application becomes unresponsive. The Kubernetes Deployment controller continuously monitors running application instances and automatically replaces instances when nodes fail or are removed from service, providing a self-healing mechanism that addresses machine failure or maintenance. Kubernetes DeploymentStatus exposes observedGeneration and conditions fields with type, status, lastTransitionTime, and reason values, allowing operators to detect rollout failures, stuck deployments, or configuration drift from the expected specification. VKS cluster management proactively monitors the cluster estate to verify that assigned policies are enforced correctly, and generates Policy Insights when configuration conflicts or environmental errors are detected.

For organizations using Harness CI/CD on VKS, Dynatrace integration provides full-stack observability including application performance monitoring, infrastructure monitoring, and digital experience monitoring. The Dynatrace operator automates deployment, scaling, and maintenance of the monitoring stack within VKS clusters, and automatically correlates application deployment service metrics with underlying VKS infrastructure including nodes, pods, and the network. The Dynatrace AI engine detects anomalies, identifies root causes, and provides intelligent alerts. Harness SRM monitors deployed services post-deployment to calculate a real-time health score based on transaction success and latency.

The VCF Automation Monitor tab provides deployment health information drawn from VCF Operations, and deployment cards in the self-service catalog surface deployment state, last request status, and operational history for workload deployments. AI-specific behavior monitoring (model accuracy drift, output quality) requires application-level tooling outside VCF's scope.

VMware Private AI Services (PAIS) provides AI-specific monitoring mechanisms that support continuous post-deployment observation of AI workloads on VCF.

At the PAIS layer, observability is activated by updating the PAISConfiguration custom resource with parameters that enable metrics collection and forward large language model traces to an OpenTelemetry Collector. PAIS deploys a Prometheus server pod for metrics collection that must be ready and running for observability to function, configured with a chosen storage class and a configurable metrics retention period. PAIS controller pod metrics are integrated into observability platforms through Telegraf configuration in the Supervisor instance, and the telegraf-user-config ConfigMap in the vmware-system-monitoring namespace manages user-configured monitoring endpoints for PAIS. The PAIS Supervisor Service exposes service-level metrics through pais.vmware.com that VI administrators access to monitor the health of the service.

DevOps engineers visualize PAIS health and performance metrics in observability platforms such as Grafana, where dashboards report pod restart frequency, vGPU licensing status, and idle or underutilized GPU devices in passthrough mode. VI administrators visualize PAIS controller pod operation metrics in VCF Operations using the same metrics pipeline. MLOps engineers and application developers track the health, quality, and behavior of AI applications and the models used in agents through Grafana, supporting continuous observation of inference behavior. Model Endpoint deployment status can be verified using the `kubectl get modelendpoints` command, which returns the Ready status for each deployed endpoint, and the `vcf` cluster command-line interface provides cluster provisioning monitoring, registration, and kubeconfig retrieval for VKS clusters that host AI workloads. PAIS deployment leases can be configured to bound how long a deployment can exist, supporting lifecycle oversight of catalog-deployed AI workloads.

Enabling these observability features, defining retention and alerting policy, and acting on monitoring signals are organizational process responsibilities. PAIS provides the monitoring infrastructure and integration points, but continuous post-deployment monitoring as a practice requires the organization to assign operational ownership, review observability data on an ongoing basis, and respond to anomalies in AI workload behavior.

VMware vDefend (Contributes)

VMware vDefend (vDefend) contributes to proactive and continuous monitoring of deployed AI workloads running on VCF through multiple layers of ongoing network-level security monitoring.

Security Intelligence continuously monitors for new flows between environments and alerts administrators to unauthorized or unexpected traffic, enabling ongoing policy adjustments as workloads evolve. It also monitors infrastructure services to detect newly deployed servers. A classification cron job runs automatically at 2:00 AM local time and repeats every 24 hours, evaluating traffic flows from the preceding 30 days to classify compute entities providing network infrastructure services. When the Recommendations Monitoring feature is enabled, Security Intelligence checks every hour for changes in the scope of monitored entities, allowing policy recommendations to remain current as AI workloads change. The Security Intelligence Flow Insights Dashboard supports detection and display of MITRE ATT&CK framework tactics and techniques, showing counts of detected suspicious events per tactic and technique during a selected time period.

Segmentation Monitoring provides dedicated views for environment-level traffic flows, accessible through the Monitor & Plan interface. Administrators can monitor traffic between configured environment pairs, view flows related to detected infrastructure servers within specified time intervals, and export all unique flows to a CSV file for offline analysis or compliance evidence. The Environments monitoring window provides assessment of current security posture and enables improvement actions for environment pairs. When no protection policy is configured for an environment pair, Segmentation Monitoring offers a Run Recommendations option to generate policy recommendations, closing monitoring gaps around AI workloads.

The Security Services Platform (SSP) collects metrics to monitor the readiness and performance of vDefend IDS/IPS in the environment. IDS/IPS engine statistics accessible per host include uptime, alert count, suppressed alerts, alert queue overflow status, engine ID, and last reload timestamp. IDS/IPS monitoring dashboards can be scoped to custom projects, displaying events triggered for that project only. For bare metal deployments, Bare Metal Agents stream metrics and detailed host-level telemetry to the SSP. The NDR Sensor periodically checks the status of activated features on the SSP every 15 minutes and updates its configuration accordingly.

Intelligent Assist provides AI-assisted analysis within the NSX Manager UI and SSP UI, displaying on the IDS/IPS, Detections, and Campaigns pages to help administrators interpret monitoring data.

Malware Prevention Service provides deployment status monitoring at the cluster level and per-node level, displaying deployment errors and individual transport node status. Network Detection and Response (NDR) applies behavioral analysis to detect suspicious activity targeting monitored workloads and correlates events into campaigns for investigation.

Monitoring AI system functionality at the application layer, including model performance, output quality, and behavioral drift, is outside vDefend's domain.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-4.2-001

Conduct regular monitoring of GAI systems and publish reports detailing the performance, feedback received, and improvements made.

SCF Controls: AAT-07.3

Aggregate Coverage: Contributes

VCF (Contributes)

Continuous improvement of Artificial Intelligence and Autonomous Technologies capabilities is an organizational activity carried out by the teams that build, evaluate, and operate AI workloads. VMware Private AI Services (PAIS) is the AI infrastructure layer of VCF and supplies the platform on which those workloads run; it does not perform the improvement cycle itself. The decision to retrain a model, change a data source, adjust a prompt, or retire a capability is made by the data science, MLOps, and product teams that own the workload, governed by their own internal processes.

PAIS supplies platform mechanisms that workload teams can incorporate into their own improvement cycles. A central model gallery is recommended for managing ML models with consistency across deployment targets. Knowledge bases can be created with linked data sources including Microsoft SharePoint sites and Amazon S3 compatible stores, allowing teams to update the context available to model responses without rebuilding the underlying model. Private AI Services Agent Builder maintains a list of approved tools for use in AI agents when connecting to Model Context Protocol (MCP) servers, giving workload owners a mechanism to add or restrict capabilities over time. PAIS can be installed or updated to later versions independently of the core Supervisor Service, allowing platform-level improvements to ship without a full core update. Controller pod metrics are integrated into observability platforms through Telegraf configuration in the Supervisor instance, so operations teams can feed platform telemetry into their evaluation loops. These are tools that workload teams use; PAIS does not own the improvement decision.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-4.2-002

Practice and follow incident response plans for addressing the generation of inappropriate or harmful content and adapt processes based on findings to prevent future occurrences. Conduct post-mortem analyses of incidents with relevant AI Actors, to understand the root causes and implement preventive measures.

SCF Controls: IRO-04, IRO-13

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-4.3-001

Conduct after-action assessments for GAI system incidents to verify incident response and recovery processes are followed and effective, including to follow procedures for communicating incidents to relevant AI Actors and where applicable, relevant legal and regulatory bodies.

SCF Controls: IRO-10, IRO-13

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-4.3-002

Establish and maintain policies and procedures to record and track GAI system reported errors, near-misses, and negative impacts.

SCF Controls: AAT-17

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MG-4.3-003

Report GAI incidents in compliance with legal and regulatory requirements (e.g., HIPAA breach reporting, e.g., OCR (2023) or NHTSA (2022) autonomous vehicle crash reporting requirements.

SCF Controls: CPL-01, IRO-10.2

Aggregate Coverage: Contributes

VCF (Contributes)

VCF provides detection, monitoring, alerting, notification delivery, and investigation capabilities through VCF Operations that support timely identification and reporting of incidents involving sensitive or regulated data. While the organizational process of formally reporting such incidents to regulators, affected individuals, or other stakeholders falls outside the platform's scope, VCF supplies the technical foundation that enables an organization to detect incidents quickly, gather evidence, and route information to the people and systems responsible for response.

VCF Operations event auditing provides visibility into platform interaction changes across the VCF infrastructure and surfaces suspicious access events and policy violations. This capability allows security teams to identify unauthorized or anomalous activity that may indicate a data incident. VCF Operations also provides audit tracking that identifies who made changes to the data center during a specific time range, which is important for establishing the scope and timeline of an incident involving sensitive data.

VCF Operations compliance management continuously monitors infrastructure by evaluating collected data against defined policies. When a compliance alert is triggered on VMware vCenter instances, VMware ESX hosts, virtual machines, distributed port groups, or distributed switches, compliance violations are recorded for investigation. This automated detection of deviations from compliance baselines helps organizations identify potential data-related incidents as they occur rather than discovering them after the fact. Compliance management also highlights risk areas and provides actionable remediation recommendations, supporting the triage process that precedes formal incident reporting.

VCF Operations generates alerts based on configurable thresholds and event definitions. When an alert fires, VCF Operations can deliver notifications through multiple outbound channels: SNMP traps to network management platforms, webhook calls to REST-enabled systems (in JSON or XML format), a ServiceNow plug-in that automatically creates incidents in the ServiceNow ticketing system, a Slack plug-in that forwards alerts to Slack channels with links to alert details, and email. Notification rules let administrators filter which alerts are forwarded and to which endpoints, so that events potentially involving data exposure reach the right teams without delay.

All VCF components support syslog forwarding to external log aggregation and SIEM platforms. vCenter syslog forwarding can be configured through the vCenter Management API, and ESX hosts send syslog feeds through Log Collections configuration. VCF Log Management ingests and indexes these events from across the stack, providing real-time analytics to search and filter logs instantly for troubleshooting or auditing. Users can collect and analyze log feeds, view log-related metrics, and dynamically extract fields from log messages based on customized queries. This centralized log management gives security operations teams the ability to correlate events across the environment.

Accurate log correlation depends on synchronized clocks across all components. The vSphere documentation notes that incorrect time settings in vCenter and ESX hosts make it difficult to inspect and correlate log files to detect attacks and render auditing inaccurate; NTP should be configured on all vCenter instances and ESX hosts so that log timestamps are reliable for incident investigation and reporting.

The Diagnostic Findings feature provides a consolidated view of known product issues, VMSA-based security exposures, and industry best-practice recommendations across the VCF environment. Diagnostic Findings are alerts without notifications that result from a scan of the platform against known issues, logs, and properties. This provides an additional signal that may be relevant during incident triage, helping teams determine whether an observed event is related to a known vulnerability.

For incident tracking and investigation, VCF Operations provides capabilities at multiple levels. The Troubleshooting Incident Page in VCF Network Operations displays information related to incidents, including status, root cause metrics, closing

remarks, and timestamps for when incidents were created. During investigation, analysts can flag metrics and mark root causes, creating a documented trail that supports the incident reporting process. Incident status can be tracked as resolved or unresolved within the platform. VCF Operations also supports scheduled automated reports at regular intervals to track current resources and identify potential risks to the environment, which can feed an organization's incident reporting workflow.

VCF Operations Data and Site Resiliency Monitoring provides centralized visibility into data protection metrics across private clouds, giving organizations awareness of the state of their data protection posture. VCF Automation includes data compliance controls that require explicit consent before exposing user names. This supports organizations that must track access to personally identifiable information.

To satisfy this control fully, organizations must establish incident response procedures that define reporting timelines, designate responsible personnel, identify the authorities and regulators to notify, and specify the content and format of incident reports. VCF provides the detection, correlation, notification delivery, and evidence-gathering layer, but the reporting process itself requires organizational policies, communication channels, and human decision-making.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-1.1-001

When identifying intended purposes, consider factors such as internal vs. external use, narrow vs. broad application scope, fine-tuning, and varieties of data sources (e.g., grounding, retrieval-augmented generation).

SCF Controls: AAT-03

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-1.1-002

Determine and document the expected and acceptable GAI system context of use in collaboration with socio-cultural and other domain experts, by assessing: Assumptions and limitations; Direct value to the organization; Intended operational environment and observed usage patterns; Potential positive and negative impacts to individuals, public safety, groups, communities, organizations, democratic institutions, and the physical environment; Social norms and expectations.

SCF Controls: AAT-03, AAT-07.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-1.1-003

Document risk measurement plans to address identified risks. Plans may include, as applicable: Individual and group cognitive biases (e.g., confirmation bias, funding bias, groupthink) for AI Actors involved in the design, implementation, and use of GAI systems; Known past GAI system incidents and failure modes; In-context use and foreseeable misuse, abuse, and off-label use; Over reliance on quantitative metrics and methodologies without sufficient awareness of their limitations in the context(s) of use; Standard measurement and structured human feedback approaches; Anticipated human-AI configurations.

SCF Controls: AAT-07, AAT-09

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-1.1-004

Identify and document foreseeable illegal uses or applications of the GAI system that surpass organizational risk tolerances.

SCF Controls: AAT-02.1, AAT-09, AAT-27, TDA-01.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-1.2-001

Establish and empower interdisciplinary teams that reflect a wide range of capabilities, competencies, demographic groups, domain expertise, educational backgrounds, lived experiences, professions, and skills across the enterprise to inform and conduct risk measurement and management functions.

SCF Controls: AAT-07

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-1.2-002

Verify that data or benchmarks used in risk measurement, and users, participants, or subjects involved in structured GAI public feedback exercises are representative of diverse in-context user populations.

SCF Controls: AAT-11.1, AAT-11.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-2.1-001

Establish known assumptions and practices for determining data origin and content lineage, for documentation and evaluation purposes.

SCF Controls: AAT-12.1, AAT-12.3, RSK-01.1

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides explicit data source identification and tracking for AI workloads. The Data Indexing and Retrieval service offers a Knowledge Bases interface where users create and manage knowledge bases, and a knowledge base can be linked to up to ten data sources. Each data source is created with a name, description, type, and the required properties for retrieving content from its original location, and a single data source can be linked to multiple knowledge bases. The data source type is fixed at creation. Supported data source types include Microsoft SharePoint sites, Amazon S3 compatible stores, and other enterprise content management systems. Each data source stores the credentials required for retrieval, such as service account key file information when applicable, which creates a documented connection between the AI system and its data origins.

During knowledge base indexing, PAIS tracks document-level detail including crawled documents, new documents found, deleted documents, existing unmodified documents, existing modified documents requiring embedding updates, and skipped documents. It captures size metrics including total megabytes downloaded, new chunks indexed corresponding to inference server invocations for embeddings, and final index size after removal of deleted documents. It records timing metrics for download, embedding generation, and the overall index operation. Document counts returned by retrieval queries are tracked, and counts of documents per media type such as PDF and PPTX are maintained for each knowledge base. Automatic collection and indexing of updates from linked data sources over time keeps these records current.

For runtime visibility, PAIS supports LLM trace collection through an OpenTelemetry Collector by configuring the `spec.observability.llmTraces` section of the `PAISConfiguration` custom resource, with traces transmitted over HTTP/protobuf or gRPC. PAIS metrics, including knowledge base activity, can be visualized in third-party observability platforms such as Grafana. The Data Indexing and Retrieval component also implements a Model Context Protocol (MCP) server that exports a set of search tools for each knowledge base, which gives downstream AI applications a documented programmatic surface tied back to specific data sources. PAIS AI catalog items, including AI Workstation, DSM Database, Triton Inferencing Server, and AI Kubernetes Cluster, are added to the VCF Automation catalog by running the PAIS Quickstart, with required images uploaded in advance to a Harbor registry in disconnected environments.

Organizational decisions about which data sources are approved for AI use, how data lineage is documented outside the platform, and how training data provenance is governed remain the responsibility of the organization. PAIS supplies the technical registration and tracking infrastructure that the organization configures and operates.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-2.1-002

Institute test and evaluation for data and content flows within the GAI system, including but not limited to, original data sources, data transformations, and decision-making criteria.

SCF Controls: AAT-12.1, AAT-12.2

Aggregate Coverage: Contributes

VCF (Contributes)

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Distributed Firewall microsegmentation enforces least-privilege network access at the vNIC level, limiting which workloads and users can reach databases, object stores, and data pipelines used for AI training and operation. Attacks that attempt to contaminate or corrupt AI source data require network access to the stores where training data resides; microsegmentation restricts that access to authorized workloads and identities. Security Intelligence provides visibility into traffic flows between workloads, generates firewall rule recommendations, and validates that security policies are enforced. Security Intelligence administrators can monitor for traffic leakage between environments, establish exceptions for authorized communication paths, and implement lockdown protection rules to control unauthorized inter-environment traffic. Gateway Firewall provides an additional enforcement point at the perimeter of network segments containing sensitive data repositories.

vDefend IDS/IPS inspects traffic to and from data repositories for known exploit patterns, command-and-control traffic, lateral movement, and data exfiltration signatures. The IDS/IPS filestore capability stores matched files to disk when a signature rule fires, supporting forensic analysis and post-incident review of suspected data integrity compromises. IDS/IPS signature-based detection also uses file magic inspection, identifying file types by their binary structure via libmagic, to detect misrepresented or malformed files in transit to data repositories.

Malware Prevention intercepts and analyzes files transferred over the network to data repositories. Profiles can be configured to send files to the cloud for detailed analysis or to process them locally without cloud connectivity. Verdict decisions classify files as Benign, Trusted, Highly Trusted, or Malicious. Security Services Platform Malware Prevention collects file analysis data including file hash submissions, assigns an Analyst UUID to each cloud submission for tracking, and records in-memory script buffer inspection details with associated analysis reports. The Security Services Platform also provides backup and restore mechanisms covering hardware malfunctions, software malfunctions, data corruption, and malware incidents, providing a recovery path if SSP itself is compromised.

vDefend contributes to this control because its microsegmentation, threat detection, and file inspection capabilities reduce the attack surface and detection gap for data poisoning attacks targeting AI workloads. Protecting data integrity at the application layer, including content validation, provenance tracking, and checksums within data stores, is outside vDefend's domain and requires organizational processes and additional tooling.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-2.2-001

Identify and document how the system relies on upstream data sources, including for content provenance, and if it serves as an upstream dependency for other systems.

SCF Controls: AAT-12.1

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides explicit data source identification and tracking for AI workloads. The Data Indexing and Retrieval service offers a Knowledge Bases interface where users create and manage knowledge bases, and a knowledge base can be linked to up to ten data sources. Each data source is created with a name, description, type, and the required properties for retrieving content from its original location, and a single data source can be linked to multiple knowledge bases. The data source type is fixed at creation. Supported data source types include Microsoft SharePoint sites, Amazon S3 compatible stores, and other enterprise content management systems. Each data source stores the credentials required for retrieval, such as service account key file information when applicable, which creates a documented connection between the AI system and its data origins.

During knowledge base indexing, PAIS tracks document-level detail including crawled documents, new documents found, deleted documents, existing unmodified documents, existing modified documents requiring embedding updates, and skipped documents. It captures size metrics including total megabytes downloaded, new chunks indexed corresponding to inference server invocations for embeddings, and final index size after removal of deleted documents. It records timing metrics for download, embedding generation, and the overall index operation. Document counts returned by retrieval queries are tracked, and counts of documents per media type such as PDF and PPTX are maintained for each knowledge base. Automatic collection and indexing of updates from linked data sources over time keeps these records current.

For runtime visibility, PAIS supports LLM trace collection through an OpenTelemetry Collector by configuring the `spec.observability.llmTraces` section of the `PAISConfiguration` custom resource, with traces transmitted over HTTP/protobuf or gRPC. PAIS metrics, including knowledge base activity, can be visualized in third-party observability platforms such as Grafana. The Data Indexing and Retrieval component also implements a Model Context Protocol (MCP) server that exports a set of search tools for each knowledge base, which gives downstream AI applications a documented programmatic surface tied back to specific data sources. PAIS AI catalog items, including AI Workstation, DSM Database, Triton Inferencing Server, and AI Kubernetes Cluster, are added to the VCF Automation catalog by running the PAIS Quickstart, with required images uploaded in advance to a Harbor registry in disconnected environments.

Organizational decisions about which data sources are approved for AI use, how data lineage is documented outside the platform, and how training data provenance is governed remain the responsibility of the organization. PAIS supplies the technical registration and tracking infrastructure that the organization configures and operates.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-2.2-002

Observe and analyze how the GAI system interacts with external networks, and identify any potential for negative externalities, particularly where content provenance might be compromised.

SCF Controls: AAT-20.1, NET-05.1

Aggregate Coverage: Contributes

VCF (Contributes)

VCF provides several boundary protection mechanisms that help organizations enforce policies prohibiting direct connections between sensitive systems and external networks.

NSX, the networking component of VCF, manages external connectivity through Transit Gateways and Edge nodes. NSX Edge nodes provide north-south traffic connectivity between the physical data center networks and the NSX software-defined networks, serving as the boundary protection devices through which all external traffic must pass. The External Connection configuration defines how a Transit Gateway connects to networks outside NSX and which services are available on that connection. This architecture means that workloads on internal segments do not have direct paths to external networks; all external traffic passes through a defined gateway boundary. Administrators configure external connections through the NSX Manager Networking interface, where distributed and centralized connection types control how traffic reaches outside environments. A gateway CIDR IPv4 address must be explicitly configured to define the route through which traffic is forwarded to any external network, providing a controlled and auditable point of egress. External connections are defined at the infrastructure level and require dedicated Edge node deployment, giving administrators a centralized point of control for outbound connectivity.

For workloads that require outside reachability, VCF uses External IPs with 1:1 NAT to grant access to specific virtual machines on private networks from the external environment. Rather than exposing internal addresses directly, this IP translation mechanism acts as a boundary intermediary. A NAT firewall in this path discards unsolicited requests and data packets, blocking communication attempts that were not initiated from the internal side.

VCF provides network isolation capabilities at multiple layers. Network profiles support isolation using subnets, security groups, and VLAN transport zones. Private network profiles using VLAN transport zones restrict workloads to internal-only connectivity by design, with no external path unless one is explicitly provisioned. In VCF Automation, security groups define isolation settings for on-demand private and outbound networks. Network profiles can be configured with isolation policies so that provisioned workloads on private or outbound networks have controlled, one-way access to upstream networks rather than open bidirectional connectivity. The Tier-1 uplink port IP is added to an isolation security group to enable the data path while maintaining segmentation boundaries. VCF Automation also supports a denylist of internal hosts that restricts connection-testing API access, protecting internal infrastructure from probing through external-facing interfaces. At the vSphere Distributed Switch layer, traffic filtering and marking policies can be applied on distributed ports and uplink ports to deny, allow, or tag specific traffic flows, providing a boundary enforcement capability at the virtual networking infrastructure level independent of NSX policy.

For environments with stringent separation requirements, VCF supports the Workload Separation Distributed Switch Model, which provides physical air-gapped separation of network traffic. This approach uses separate distributed virtual switches so that sensitive workloads have no shared network path with less-trusted traffic, enforcing separation at the virtual infrastructure layer.

VCF Operations includes firewall hardening that restricts network access to its internal services, limiting external exposure of the management platform itself. Service connections are classified as Incoming or Outgoing based on whether a remote process is connected to a listening port, providing visibility into the direction and nature of external communications. VCF Operations for Networks supports FIPS validated cryptographic modules for external connections, and administrators can enable FIPS mode to restrict cryptographic modules used on external connections to FIPS validated implementations only. The vSphere security guidance recommends securing management network access through a bastion host that isolates client

traffic, CLI and API traffic, and third-party software traffic from other network traffic, so that the management plane itself is not directly reachable from external networks.

The organizational policy decision of which systems are sensitive and which boundary protection devices are required is outside VCF's scope. VCF provides the technical mechanisms for boundary enforcement, including gateway-based egress control, NAT-based address translation, network isolation profiles, air-gapped switch models, and management plane hardening, but the prohibition policy itself must be defined and enforced through organizational processes.

VMware vDefend (Contributes)

VMware vDefend (VMware vDefend) contributes to prohibiting direct connections from sensitive systems to external networks by providing boundary protection mechanisms that operate at both the gateway and workload levels.

The vDefend gateway firewall operates on NSX Edge nodes and enforces stateful firewall rules on all north-south traffic traversing the boundary between VCF-managed workloads and external networks. Because all external connectivity must pass through Edge nodes, the Gateway Firewall acts as an organization-defined boundary protection device where administrators define allow, drop, or reject actions for traffic crossing the perimeter. Stateful rule processing permits only traffic matching approved flows to pass outbound or inbound. In multi-project deployments, Enterprise Administrators can enforce isolation between projects using centralized transit Gateway Firewall rules that deny all inter-project communication except for specified infrastructure protocols such as DHCP, DNS, NTP, and ICMP.

The vDefend distributed firewall (DFW) extends boundary enforcement to the workload level. DFW rules are enforced at the vNIC of each virtual machine, so even if a workload has a network path to an external segment, DFW policy can block that traffic before it leaves the hypervisor. This provides defense in depth: a sensitive workload can be microsegmented so that its only permitted communication is with internal resources, regardless of the routing topology. The Application Connectivity Strategy can be configured to DROP all unsolicited inbound traffic by default, blocking external ingress to an application group. The "Allow Intra, Deny Other Traffic" connectivity strategy generates a specific intra-group ALLOW rule combined with a default ANY/ANY/DROP rule, isolating the group from all external traffic. DFW also supports applying scoped deny rules targeted to specific infrastructure groups via the "Applied-To" field, enabling phased Zero Trust adoption that restricts external access to critical servers. Security Services Platform segmentation monitoring can configure DFW protection rules with Drop or Reject actions to block unknown flows between environment pairs, such as development and production, that should never communicate.

The Security Services Platform's Secure External Communications project identifies workloads with inbound or outbound connections to external IP addresses, providing visibility into boundary protection gaps that administrators can then remediate through DFW policy.

For environments requiring strict isolation, vDefend Advanced Threat Prevention features including IDS, Network Traffic Analysis, and Network Detection and Response can operate in air-gapped environments where internet connectivity is not permitted. Network Detection and Response supports standalone operation without external network access, executing all detection, correlation, and response activities entirely within the closed network.

The organizational determination of which systems are sensitive and the policy decisions about permitted external connections must be established through organizational processes outside vDefend's scope. vDefend provides the enforcement mechanisms at multiple layers, but the prohibition policy itself must be defined by the organization and then implemented through vDefend's rule sets and segmentation capabilities.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer (Avi) functions as an application-layer boundary protection device for workloads published through it. Virtual Service Network Security Policies evaluate incoming client traffic based on network-level parameters and determine whether a connection should be accepted, discarded, or rate-limited. These policies can block traffic from unauthorized IP addresses by dropping TCP SYN packets before connections are established. When a Network Security Policy rule discards a connection, subsequent HTTP Security, Request, or Response policy evaluation does not occur for that

traffic, stopping further processing at the boundary. Administrators configure Network Security Policies through the Avi Controller UI under the Policies tab of each virtual service or via the CLI `configure networksecuritypolicy` command.

Network Security and HTTP Security policies support IP-based connection control through a Blocklist IP group mechanism. A policy rule that references an IP group named Blocklist discards any connection originating from an address in that group before the request reaches backend pool members. The Avi Controller also exposes a `networksecuritypolicydos` API endpoint with a block action that allows programmatic addition of client IPs to a virtual service's blocked list, supporting automated threat response workflows.

In Kubernetes environments, the Avi Kubernetes Operator (AKO) extends boundary protection to ingress traffic. The `networkSecurityPolicy` setting in an AKO `HostRule` custom resource applies a Network Security Policy to parent virtual services and dedicated virtual services, enforcing connection control at the Kubernetes ingress layer. AKO also supports setting `insecureEdgeTerminationPolicy` to `None` for Secure Routes, which disables insecure edge termination and prevents unprotected connections on routes configured for HTTPS.

Routing all application traffic through the load balancer so that backend servers are not directly reachable from external networks requires network topology and segmentation decisions at the VCF platform level. Avi's documentation explicitly states that Service Engine data plane traffic egressing the Data Plane vNIC is subjected to routing isolation and ACL and firewalling controls present in the external network. VCF network segmentation and vDefend microsegmentation rules are the mechanisms that restrict traffic from bypassing Avi and reaching backend servers directly. Avi's own policy controls operate on traffic already directed to a virtual service listener and do not by themselves control routing paths within the broader network.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-2.3-001

Assess the accuracy, quality, reliability, and authenticity of GAI output by comparing it to a set of known ground truth data and by using a variety of evaluation methods (e.g., human oversight and automated evaluation, proven cryptographic techniques, review of content inputs).

SCF Controls: AAT-16.5

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) supports domain expert validation through a structured role-based workflow built around a validated-model gallery, container-registry model storage, and approval gates for agent tooling. The platform separates responsibilities so that domain-appropriate experts review AI systems at each stage, with MLOps engineers acting as the designated domain expert for model quality and safety within the PAIS workflow.

MLOps engineers validate ML models onboarded to PAIS against the security, privacy, and technical requirements of the organization before the models are made available for downstream use. Validated models are stored in a Harbor registry that acts as the model storage backend, with authentication and certificate-based trust for retrieval; OCI-compliant registries from other vendors are also supported, including in disconnected environments. A central model gallery is the recommended store for ML models, providing centralized management and consistency across deployment targets, which gives reviewers a single inventory to assess and approve before promotion.

For agent-based AI applications, PAIS requires examination and approval of Model Context Protocol (MCP) server tools and capabilities before they are used in agents, and the Agent Builder maintains a list of approved tools for use when connecting to MCP servers. Deep learning VMs give domain experts hands-on environments to prototype, fine-tune, and validate model behavior before promotion. When deploying completion or embedding model endpoints, A/B testing is recommended so that reviewers can compare candidate models in real-world settings. Catalog items for AI applications provisioned through VCF Automation, together with PAIS observability surfaced in platforms such as Grafana, give reviewers the operational signals needed to validate that AI systems perform consistently after release.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-2.3-002

Review and document accuracy, representativeness, relevance, suitability of data used at different stages of AI life cycle.

SCF Controls: AAT-10.13

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-2.3-003

Deploy and document fact-checking techniques to verify the accuracy and veracity of information generated by GAI systems, especially when the information comes from multiple (or unknown) sources.

SCF Controls: AAT-26

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides several infrastructure capabilities that support the development and implementation of fact-checking techniques for AI-generated content. The most direct mechanism is the Retrieval-Augmented Generation (RAG) architecture available through PAIS. RAG workloads combine a large language model with an external knowledge base, grounding model responses in organizational data rather than relying solely on the model's training data. Data scientists can build knowledge bases that link to data sources hosted in external content management systems, including Microsoft SharePoint sites and Amazon S3 compatible stores, with a maximum of 10 data sources per knowledge base and each data source bound to a fixed type at creation. PAIS Data Indexing and Retrieval enables automatic collection and indexing of data updates from linked sources over time, which helps keep the grounding data current. This architecture provides a structural approach to improving the accuracy of AI-generated information by anchoring responses to authoritative source material rather than unconstrained generation.

The validate-then-publish workflow for ML models supports pre-deployment testing that can include accuracy verification. MLOps engineers validate ML models against the security, privacy, and technical requirements of their organization before storing them in PAIS. PAIS uses a Harbor registry service in the Supervisor to store validated ML models, with authentication and certificate-based trust required for model retrieval, and OCI-compliant registries from other vendors are also supported. Storing models in a controlled registry provides traceability for which model version was validated and deployed.

Agent Builder in PAIS integrates with Data Indexing and Retrieval to add business data from knowledge bases to agents, allowing teams to build generative AI applications where agent output is informed by curated, access-controlled knowledge bases. Agent Builder also maintains a list of approved tools for use in AI agents when connecting to MCP servers, and available tools are displayed in the tool gallery for namespace-level examination and approval before an agent can call them. MLOps engineers and application developers can monitor the health, quality, and behavior of models used in agents through observability platforms such as Grafana, and PAIS observability can be activated by updating the PAISConfiguration custom resource with metrics and OpenTelemetry trace parameters. The combination of RAG grounding, model validation, approved-tool gating, and observability provides the technical mechanisms organizations need to implement fact-checking techniques, while the specific fact-checking methodologies and acceptance criteria remain the responsibility of the deploying organization.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-2.3-004

Develop and implement testing techniques to identify GAI produced content (e.g., synthetic media) that might be indistinguishable from human-generated content.

SCF Controls: AAT-26.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-2.3-005

Implement plans for GAI systems to undergo regular adversarial testing to identify vulnerabilities and potential manipulation or misuse.

SCF Controls: AAT-10

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure that supports AI Test, Evaluation, Validation and Verification (TEVV) activities throughout the model lifecycle. Deep learning VMs provisioned from PAIS images provide dedicated testing environments for AI prototyping, fine-tuning, validation, and inference workloads. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so TEVV activities occur in conditions comparable to the deployment environment.

The validate-then-publish workflow creates a formal gate for AI deployment. MLOps engineers validate models for inference functionality using the Triton Inference Server and evaluate performance and safety against business use cases, including examination of inference requests for malicious behavior and functional testing. Models must pass this validation before being uploaded to the Model Gallery for production use. Model integrity is verified through hash code validation, and each revision is assigned a unique digest stored as an OCI artifact with an immutable manifest, giving cryptographic verification that the tested model is the same artifact that gets deployed.

The platform also supports TEVV through its monitoring stack. VCF Operations monitors GPU consumption at cluster, host system, and host properties levels. Each DL VM includes a pre-installed DCGM Exporter with Prometheus collection at 5-second scrape intervals and Grafana visualization, supplying performance telemetry during testing and evaluation. The software stack for running AI applications on top of NVIDIA GPUs is validated in advance, and NVIDIA NGC containers are pre-tested and ready to run, reducing variables during TEVV by providing a known-good baseline.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-3.4-001

Evaluate whether GAI operators and end-users can accurately understand content lineage and origin.

SCF Controls: AAT-12.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-3.4-002

Adapt existing training programs to include modules on digital content transparency.

SCF Controls: SAT-01.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-3.4-003

Develop certification programs that test proficiency in managing GAI risks and interpreting content provenance, relevant to specific industry and context.

SCF Controls: AAT-01, CPL-01.4, TDA-01.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-3.4-004

| Delineate human proficiency tests from tests of GAI capabilities.

SCF Controls: AAT-26.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-3.4-005

Implement systems to continually monitor and track the outcomes of human-GAI configurations for future refinement and improvements.

SCF Controls: AAT-08

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-3.4-006

Involve the end-users, practitioners, and operators in GAI system in prototyping and testing activities. Make sure these tests cover various scenarios, such as crisis situations or ethically sensitive contexts.

SCF Controls: AAT-26.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-4.1-001

Conduct periodic monitoring of AI-generated content for privacy risks; address any possible instances of PII or sensitive data exposure.

SCF Controls: AAT-10.13, PRI-01.6

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-4.1-002

Implement processes for responding to potential intellectual property infringement claims or other rights.

SCF Controls: AAT-12

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-4.1-003

Connect new GAI policies, procedures, and processes to existing model, data, software development, and IT governance and to legal, compliance, and risk management activities.

SCF Controls: AAT-01, TDA-22

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-4.1-004

Document training data curation policies, to the extent possible and according to applicable laws and policies.

SCF Controls: AAT-16.7

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-4.1-005

Establish policies for collection, retention, and minimum quality of data, in consideration of the following risks: Disclosure of inappropriate CBRN information; Use of Illegal or dangerous content; Offensive cyber capabilities; Training data imbalances that could give rise to harmful biases; Leak of personally identifiable information, including facial likenesses of individuals.

SCF Controls: AAT-01, AAT-10.8

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-4.1-006

Implement policies and practices defining how third-party intellectual property and training data will be used, stored, and protected.

SCF Controls: AAT-12, AAT-12.2

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to protecting the integrity of data sources used by AI workloads through storage-level encryption and access controls. vSAN encryption at rest and VM encryption protect AI training data stored on VCF infrastructure from unauthorized access at the storage layer. VMware vCenter RBAC restricts who can access or modify VMs containing AI data. Storage policies control datastore placement and encryption properties for AI workload VMs.

At the Consumption layer, VMware Kubernetes Service (VKS) on vSphere Supervisor extends these protections to containerized AI workloads. Harbor, available both as a Supervisor Service and as a standard package on VKS clusters, is the private container image registry, with images organized by vSphere Namespace. Harbor scans images for known vulnerabilities, enforces content trust so that only signed and verified artifacts are trusted, applies role-based access control to restrict which principals can push or overwrite model artifacts and pipeline images, and uses policy-based replication to control how artifacts move between registries. After Harbor is integrated with the Supervisor, all VKS clusters on the same Supervisor automatically trust it as a source of verified images.

Kubernetes admission controllers, including ValidatingAdmissionPolicy and validating webhook controllers, can be configured on VKS clusters to enforce requirements before workloads are admitted. Relevant policies include restricting image pulls to the trusted Harbor registry, requiring that images carry a valid signature, and blocking workloads that reference unscanned or unapproved artifacts. These controls reduce the risk that a workload carrying corrupted or tampered data processing logic reaches AI training data.

GitOps workflows, supported through the Argo CD Supervisor Service, store all workload and infrastructure configuration declaratively in a Git repository as the single source of truth. Every change to the deployed state flows through Git, creating a durable record of every authorized modification. Immutable infrastructure patterns complement this approach by making unauthorized changes identifiable through drift detection between actual cluster state and the declared configuration.

Kubernetes RBAC restricts which users and service accounts can read from or write to data volumes, ConfigMaps, and Secrets within VKS clusters. Supervisor namespace isolation with resource quotas contains workloads within defined boundaries, reducing the risk that one tenant can access or alter another tenant's data. Antrea-based network policies define namespace-level ingress and egress rules, scoping the network access surface for storage components that hold AI training data or inference inputs.

CSI Volume Snapshots for VKS clusters support crash-consistent backups with deduplication, compression, and encryption. The ability to restore from a known-good snapshot provides a recovery path when data corruption is detected. The VCF CLI binary, used to interact with VCF Services consumption, publishes a sha256sum.txt checksum file alongside each download so that administrators can verify the binary's integrity before use.

Validating the provenance and accuracy of AI training data at the application level is outside VCF's scope. Data poisoning detection, training data lineage, and AI-specific data provenance tracking require application-layer tooling or AI platform components built on top of the VCF platform.

VMware Private AI Services (PAIS) provides integrity controls for both model artifacts and data sources used by AI workloads. Model revisions in the Model Gallery use Harbor as a storage backend, and each push of the same model receives a unique digest so that successive revisions are content-addressed and distinguishable. Before upload, model files should be validated for integrity by checking hash codes, scanned for malware and serialization attacks, and tested for correct inference behavior using Triton Inference Server as part of the sandbox validation process in a Deep Learning VM. This pre-upload validation

provides cryptographic verification and behavioral testing that helps prevent contaminated or tampered model data from entering production.

The Model Gallery uses Harbor project access capabilities to restrict access to sensitive data used to train and tune models. Harbor integration requires authentication and certificate-based trust for model retrieval, and PAIS requires CA trust bundles to be imported for the Private Harbor registry, content management systems used as data sources in Data Indexing and Retrieval, and MCP servers providing tools for agents. These access controls and verified chains of trust restrict unauthorized modification of model and data artifacts. The documentation recommends using the Model Gallery to reduce redundant copies of models and maintain model integrity in a controlled environment.

For knowledge base data sources used by retrieval-augmented generation, PAIS Data Indexing and Retrieval enables automatic collection and indexing of data updates from linked data sources over time, supporting external content management systems such as Microsoft SharePoint and Amazon S3 compatible stores. Each knowledge base data source is created with a specific type that cannot be changed after creation, which helps prevent silent reconfiguration of the underlying source provider. The content management system used as a data source must be supported, and valid credentials are required for access to the content location. SharePoint data sources require user name and password authentication, and service account key file data sources require the information from the service account key file during data source creation. PAIS uses a CA trust bundle for secure communication with external services and databases, establishing verified transport between PAIS and its data sources.

VMware vDefend (Contributes)

VMware vDefend contributes to protecting the integrity of source data used by AI workloads on VCF through network-level security controls that restrict access to data stores and pipelines and detect malicious activity targeting data repositories.

Distributed Firewall microsegmentation enforces least-privilege network access at the vNIC level, limiting which workloads and users can reach databases, object stores, and data pipelines used for AI training and operation. Attacks that attempt to contaminate or corrupt AI source data require network access to the stores where training data resides; microsegmentation restricts that access to authorized workloads and identities. Security Intelligence provides visibility into traffic flows between workloads, generates firewall rule recommendations, and validates that security policies are enforced. Security Intelligence administrators can monitor for traffic leakage between environments, establish exceptions for authorized communication paths, and implement lockdown protection rules to control unauthorized inter-environment traffic. Gateway Firewall provides an additional enforcement point at the perimeter of network segments containing sensitive data repositories.

vDefend IDS/IPS inspects traffic to and from data repositories for known exploit patterns, command-and-control traffic, lateral movement, and data exfiltration signatures. The IDS/IPS filestore capability stores matched files to disk when a signature rule fires, supporting forensic analysis and post-incident review of suspected data integrity compromises. IDS/IPS signature-based detection also uses file magic inspection, identifying file types by their binary structure via libmagic, to detect misrepresented or malformed files in transit to data repositories.

Malware Prevention intercepts and analyzes files transferred over the network to data repositories. Profiles can be configured to send files to the cloud for detailed analysis or to process them locally without cloud connectivity. Verdict decisions classify files as Benign, Trusted, Highly Trusted, or Malicious. Security Services Platform Malware Prevention collects file analysis data including file hash submissions, assigns an Analyst UUID to each cloud submission for tracking, and records in-memory script buffer inspection details with associated analysis reports. The Security Services Platform also provides backup and restore mechanisms covering hardware malfunctions, software malfunctions, data corruption, and malware incidents, providing a recovery path if SSP itself is compromised.

vDefend contributes to this control because its microsegmentation, threat detection, and file inspection capabilities reduce the attack surface and detection gap for data poisoning attacks targeting AI workloads. Protecting data integrity at the application layer, including content validation, provenance tracking, and checksums within data stores, is outside vDefend's domain and requires organizational processes and additional tooling.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-4.1-007

Re-evaluate models that were fine-tuned or enhanced on top of third-party models.

SCF Controls: AAT-10

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure that supports AI Test, Evaluation, Validation and Verification (TEVV) activities throughout the model lifecycle. Deep learning VMs provisioned from PAIS images provide dedicated testing environments for AI prototyping, fine-tuning, validation, and inference workloads. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so TEVV activities occur in conditions comparable to the deployment environment.

The validate-then-publish workflow creates a formal gate for AI deployment. MLOps engineers validate models for inference functionality using the Triton Inference Server and evaluate performance and safety against business use cases, including examination of inference requests for malicious behavior and functional testing. Models must pass this validation before being uploaded to the Model Gallery for production use. Model integrity is verified through hash code validation, and each revision is assigned a unique digest stored as an OCI artifact with an immutable manifest, giving cryptographic verification that the tested model is the same artifact that gets deployed.

The platform also supports TEVV through its monitoring stack. VCF Operations monitors GPU consumption at cluster, host system, and host properties levels. Each DL VM includes a pre-installed DCGM Exporter with Prometheus collection at 5-second scrape intervals and Grafana visualization, supplying performance telemetry during testing and evaluation. The software stack for running AI applications on top of NVIDIA GPUs is validated in advance, and NVIDIA NGC containers are pre-tested and ready to run, reducing variables during TEVV by providing a known-good baseline.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-4.1-008

Re-evaluate risks when adapting GAI models to new domains. Additionally, establish warning systems to determine if a GAI system is being used in a new domain where previous assumptions (relating to context of use or mapped risks such as security, and safety) may no longer hold.

SCF Controls: AAT-07.2, AAT-10

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure that supports AI Test, Evaluation, Validation and Verification (TEVV) activities throughout the model lifecycle. Deep learning VMs provisioned from PAIS images provide dedicated testing environments for AI prototyping, fine-tuning, validation, and inference workloads. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so TEVV activities occur in conditions comparable to the deployment environment.

The validate-then-publish workflow creates a formal gate for AI deployment. MLOps engineers validate models for inference functionality using the Triton Inference Server and evaluate performance and safety against business use cases, including examination of inference requests for malicious behavior and functional testing. Models must pass this validation before being uploaded to the Model Gallery for production use. Model integrity is verified through hash code validation, and each revision is assigned a unique digest stored as an OCI artifact with an immutable manifest, giving cryptographic verification that the tested model is the same artifact that gets deployed.

The platform also supports TEVV through its monitoring stack. VCF Operations monitors GPU consumption at cluster, host system, and host properties levels. Each DL VM includes a pre-installed DCGM Exporter with Prometheus collection at 5-second scrape intervals and Grafana visualization, supplying performance telemetry during testing and evaluation. The software stack for running AI applications on top of NVIDIA GPUs is validated in advance, and NVIDIA NGC containers are pre-tested and ready to run, reducing variables during TEVV by providing a known-good baseline.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-4.1-009

Leverage approaches to detect the presence of PII or sensitive data in generated output text, image, video, or audio.

SCF Controls: AAT-23, AAT-27, SEA-09

Aggregate Coverage: Contributes

VCF (Contributes)

VCF provides several mechanisms across its management, automation, and lifecycle components that contribute to information output filtering and validation, though the control's full scope requires organizational processes beyond what the platform delivers on its own.

VCF Operations includes output filtering capabilities in its dashboard and widget framework. The Output Filter section available across widget types, including the Alert List Widget, Environment Status Widget, Health Widget, Object Relationship Widget, Geo Widget, Scoreboard Widget, Rolling View Chart Widget, Weather Map Widget, and Property List Widget, restricts the data displayed based on selected filter criteria. Advanced output filter options further refine widget data by object type, allowing administrators to control which information is presented and confirm that displayed data aligns with expected content. Tag-based output filtering provides an additional dimension for refining widget data, where multiple values for the same tag include objects with any of those tags applied, and multiple values for different tags include only objects with all tags applied. Views in VCF Operations support multiple criteria sets that enable complex filtering logic, giving operators granular control over how operational data is surfaced.

VCF Operations orchestrator provides a workflow validation tool that identifies errors in workflows and verifies that data flows correctly from one element to the next. Validation elements within orchestrator workflows can be configured to execute external validation actions, returning an error string when validation fails or allowing the workflow to continue when output is consistent with expectations. The input form designer can validate workflow inputs using external actions that execute action scripts and return error messages when input parameter values are invalid. This programmatic validation capability allows organizations to build automated output checks into their operational workflows.

VCF Automation extends output validation to catalog request forms and custom forms. The custom form designer supports data validation by adding constraints to form fields, restricting what values can be submitted. External validations can be attached to catalog request forms, running VCF Operations orchestrator action scripts when a user submits a request from either the UI or the API. These validations check that form data meets defined criteria before the request proceeds. A content validation interface in VCF Automation identifies custom forms containing actions that cannot be validated, helping administrators identify gaps in their validation coverage. Approval policies can be configured so that project administrators review all changes to deployed catalog instances before they take effect, providing a human review layer for output that affects shared resources.

Configuration validation within vCenter examines configuration files for errors and conflicts, returning output that helps administrators avoid loading incorrect or conflicting configurations. The VCF Automation API supports the `$filter` query option to filter API results by predicate expressions using operators including equality (eq), inequality (ne), and logical connectives (and, or), providing programmatic output filtering at the API layer. The vSphere TaskInfo object enables client applications to monitor task completion status and handle errors when tasks do not complete successfully, offering a feedback mechanism for verifying that operations produce expected results. ESX supports configurable log output filtering through the `'logRegexp'` parameter, a Python regular expression that identifies log messages to suppress, alongside `'numLogs'`, which controls how many matching messages are permitted before filtering begins.

vSphere Lifecycle Manager validates software specifications before they are applied. The Image Manager component runs a validation check before a draft software specification is saved to the database, and administrators can validate the compliance of hosts in a cluster or standalone hosts against the desired state before applying changes. The Compliance interface compares the desired state against the current state of each host and calculates compliance, helping ensure that the software configuration output matches what was intended. vCenter also performs an automatic checksum check on all software

manually imported into the vSphere Lifecycle Manager depot and verifies the SHA-256 checksum of downloaded software before it is applied.

These platform-level validation and filtering mechanisms support the control objective. Organizational processes are needed to define validation criteria and review procedures specific to the applications running on VCF, establish acceptance thresholds for output data, and maintain the approval workflows that govern changes to shared resources.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer operates as the application proxy layer for workloads running on VCF and provides application-layer mechanisms for inspecting and validating HTTP response content as traffic passes through the Service Engine.

The DataScript scripting engine is the primary Avi-native mechanism for output inspection and conditional validation. DataScript runs at the proxy boundary and provides an external request validation API that returns a response object containing the HTTP response status code, headers table, and HTTP response body (p.3658). Operators can write DataScript policies that inspect this response content against expected patterns and take conditional action on responses that deviate from expectations, including modifying headers, rewriting content, or terminating connections. DataScript also provides certificate-level validation functions: `avi.ssl.check_client_cert_validity()` enables enforcement of certificate presentation and validity checks in virtual service policies (p.3600), and `avi.ssl.clear_client_cert_verify_error` supports conditional handling of verification errors based on error code evaluation, such as clearing errors when a CRL has expired (p.3605). These functions support inspection and conditional response handling beyond HTTP-layer content.

The Avi WAF Positive Security Model, also called Application Specific Policy, describes expected application behavior and validates inputs by setting an accepted range and length of characters, reporting a policy violation when input falls outside those bounds (p.3535). This positive security approach constrains what output an application can produce by defining what the application is permitted to accept and process, helping identify deviations from expected content patterns.

Avi generates per-virtual-service Application Security Reports in PDF format, documenting application security findings and pre-check validation results (p.3481). Reports are visible on the Reports page of the Controller UI and can be downloaded for external review. This provides operators with a structured artifact for assessing whether application output behavior is consistent with security expectations.

The Avi Kubernetes Operator (AKO) performs syntactical validations at the time of Custom Resource Definition (CRD) object creation to verify configuration correctness and provides effective status messages (p.83, [avi-kubernetes-operator-2-1.pdf](#)). This supports output consistency validation for Kubernetes-native Avi configurations.

The Controller API supports response field filtering through the `fields query` parameter, which restricts API responses to specified fields and reduces information returned to only what is relevant (p.4296). Query parameter-based filtering is available for targeted operations on specific resource subsets (p.4295). Application logs on each Service Engine support filtering using free-form text, keywords, or formal search syntax with field-based filtering using equality operators (p.4109, p.4121), enabling focused review of application output behavior.

Coverage is Contributes because, while Avi provides the proxy-level inspection, WAF validation, and reporting infrastructure, organizations must define what constitutes expected output for each protected application, author the DataScript policies and WAF rules that encode those expectations, and establish a review schedule for Application Security Reports.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-4.1-010

Conduct appropriate diligence on training data use to assess intellectual property, and privacy, risks, including to examine whether use of proprietary or sensitive training data is consistent with applicable laws.

SCF Controls: AAT-12, TDA-22

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-5.1-001

Apply TEVV practices for content provenance (e.g., probing a system's synthetic data generation capabilities for potential misuse or vulnerabilities).

SCF Controls: AAT-10

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure that supports AI Test, Evaluation, Validation and Verification (TEVV) activities throughout the model lifecycle. Deep learning VMs provisioned from PAIS images provide dedicated testing environments for AI prototyping, fine-tuning, validation, and inference workloads. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so TEVV activities occur in conditions comparable to the deployment environment.

The validate-then-publish workflow creates a formal gate for AI deployment. MLOps engineers validate models for inference functionality using the Triton Inference Server and evaluate performance and safety against business use cases, including examination of inference requests for malicious behavior and functional testing. Models must pass this validation before being uploaded to the Model Gallery for production use. Model integrity is verified through hash code validation, and each revision is assigned a unique digest stored as an OCI artifact with an immutable manifest, giving cryptographic verification that the tested model is the same artifact that gets deployed.

The platform also supports TEVV through its monitoring stack. VCF Operations monitors GPU consumption at cluster, host system, and host properties levels. Each DL VM includes a pre-installed DCGM Exporter with Prometheus collection at 5-second scrape intervals and Grafana visualization, supplying performance telemetry during testing and evaluation. The software stack for running AI applications on top of NVIDIA GPUs is validated in advance, and NVIDIA NGC containers are pre-tested and ready to run, reducing variables during TEVV by providing a known-good baseline.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-5.1-002

Identify potential content provenance harms of GAI, such as misinformation or disinformation, deepfakes, including NCII, or tampered content. Enumerate and rank risks based on their likelihood and potential impact, and determine how well provenance solutions address specific risks and/or harms.

SCF Controls: AAT-07, AAT-07.2, AAT-10.17, AAT-12.1, AAT-12.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-5.1-003

Consider disclosing use of GAI to end users in relevant contexts, while considering the objective of disclosure, the context of use, the likelihood and magnitude of the risk posed, the audience of the disclosure, as well as the frequency of the disclosures.

SCF Controls: AAT-22.7, AAT-22.8

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-5.1-004

Prioritize GAI structured public feedback processes based on risk assessment estimates.

SCF Controls: AAT-11, AAT-11.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-5.1-005

Conduct adversarial role-playing exercises, GAI red-teaming, or chaos testing to identify anomalous or unforeseen failure modes.

SCF Controls: VPM-10

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-5.1-006

Profile threats and negative impacts arising from GAI systems interacting with, manipulating, or generating content, and outlining known and potential vulnerabilities and the likelihood of their occurrence.

SCF Controls: AAT-07.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-5.2-001

Determine context-based measures to identify if new impacts are present due to the GAI system, including regular engagements with downstream AI Actors to identify and quantify new contexts of unanticipated impacts of GAI systems.

SCF Controls: AAT-07.1, AAT-07.2, AAT-18

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MP-5.2-002

Plan regular engagements with AI Actors responsible for inputs to GAI systems, including third-party data and algorithms, to review and evaluate unanticipated impacts.

SCF Controls: AAT-07.1, AAT-11

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-1.1-001

Employ methods to trace the origin and modifications of digital content.

SCF Controls: AAT-12.4

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to logging modifications to digital content through vCenter event logging that records VM-level operations (creation, modification, deletion, snapshot, migration) and VCF Log Management that aggregates these events for analysis. Changes to VMs hosting AI workloads or their associated storage are captured in vCenter's event stream with timestamps and user attribution through VCF SSO.

For AI and autonomous technology workloads deployed on VMware Kubernetes Service (VKS) within the VCF Consumption layer, the Kubernetes API server includes a configurable audit subsystem that records all API requests representing content modifications, including create, update, patch, and delete operations. Audit events follow the structure defined by the audit.k8s.io API group, capturing the audit level, a unique audit ID, requesting user, timestamp, and affected resource. Administrators configure audit behavior through an audit policy that determines what events are recorded and at what level of detail. Four audit levels are available: None (matching events are not logged), Metadata (logs request metadata including requesting user, timestamp, and resource), Request (logs metadata and request body), and RequestResponse (logs metadata, request body, and response body). When the `--audit-log-path` flag is omitted from the kube-apiserver configuration, no events are logged; the flag must be set explicitly to activate audit recording and direct logs to the appropriate file path. Audit event truncation is disabled by default and requires explicit enablement through separate API server flags. The Supervisor can also be configured to preserve the source IP addresses of authentication and Kubernetes API server requests, making those addresses available for attribution in audit logs. This configurability allows organizations to capture the full content of modifications to Kubernetes resources used by AI workloads, including ConfigMaps, Secrets, Deployments, and custom resources.

VKS clusters on Supervisor include Fluent Bit logging agents deployed in the `vmware-system-logging` namespace. Fluent Bit processes and tags Kubernetes API server audit logs separately from container logs and Linux daemon logs, enabling targeted forwarding of audit streams to external log management systems. Supported destinations include Syslog, HTTP, Elasticsearch, Kafka, and Splunk logging servers, as well as VCF Log Management using a configurable values file. On the Supervisor control plane, Fluent Bit collects kube-apiserver audit logs from `/var/log/vmware/audit/kube-apiserver.log` using the tag `audit.apiserver.*`, with records buffered locally in `/var/log/vmware/fluentbit/flb_audit_apiserver` before forwarding. The kube-apiserver exposes the `apiserver_audit_event_total` Prometheus metric to allow continuous monitoring of the total audit events exported. A second metric, `apiserver_audit_level_total`, tracks audit events by policy level with one counter increment per request.

For workloads running as vSphere Pods in vSphere Namespaces, the LogAgentConfiguration Custom Resource configures log streaming to a remote Syslog server. Syslog messages from vSphere Pods include metadata fields covering namespace name, pod UID, pod name, container name, and ESX host name, enabling attribution of events to specific workloads and infrastructure components.

For AI workloads where syscall-level modification auditing is needed, Kubernetes seccomp audit profiles can be configured in a pod's security context. The `audit.json` seccomp profile logs all system calls of a process without restricting any calls, directing records to syslog or auditd via `SCMP_ACT_LOG`. VKS clusters also generate component logs from CAPI, CAPV, VM Operator, and VKS Controller Manager, with VM Operator specifically recording reconciliation events for `VirtualMachineService` resources that reflect changes to workload infrastructure.

Logging modifications to AI model content or training data at the application level requires AI-specific tooling beyond what the VCF platform provides.

VMware Private AI Services (PAIS) provides content modification logging for AI knowledge base operations and for model artifacts. During knowledge base indexing, PAIS knowledge base operations track document-level changes including crawled

documents, new documents found, deleted documents, existing unmodified documents, existing modified documents requiring embedding updates, and skipped documents. This document-level change record produces an audit trail of what content was added, modified, or removed from the knowledge base that feeds retrieval-augmented generation agents. Knowledge base operations also capture size metrics such as total megabytes downloaded, number of new chunks indexed corresponding to inference server invocations for embeddings, and final index size after removal of deleted documents, along with timing metrics for file download, embedding generation, and overall index operation duration.

For model artifacts, each push to the Model Gallery creates a new model revision stored as an immutable OCI artifact identified by content digest, with deduplication so that identical model data pushed multiple times results in only one stored revision. The Model Gallery assigns a unique digest to each revision when the same model is pushed multiple times, supporting integrity verification of model content over time. Revisions can be tagged, supporting human-readable identification of model changes alongside the digest.

PAIS observability extends content-modification visibility through OpenTelemetry. The PAISConfiguration custom resource exposes a `spec.observability.llmTraces` section and an optional Prometheus metrics section that can be activated to send LLM traces and metrics for agent execution flow and knowledge base indexing to an OpenTelemetry Collector with configurable storage class and metrics retention period. PAIS Controller metrics are exposed at a Prometheus endpoint on the `pais-controller-manager` service for collection by monitoring systems, and Deep Learning VMs expose vGPU performance metrics through Prometheus. The vLLM inference server logging level is controlled by the `VLLM_LOGGING_LEVEL` environment variable, and per-pod logs for model endpoints can be retrieved with `kubectl` in the `pais` namespace. Deep Learning VMs record cloud-init script logs in `/var/log/dl.log` and GPU driver installation logs in `/var/log/gpu-install.log`.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-1.1-002

Integrate tools designed to analyze content provenance and detect data anomalies, verify the authenticity of digital signatures, and identify patterns associated with misinformation or manipulation.

SCF Controls: AAT-10.17, AAT-26, MON-16

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-1.1-003

Disaggregate evaluation metrics by demographic factors to identify any discrepancies in how content provenance mechanisms work across diverse populations.

SCF Controls: AAT-10.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-1.1-004

Develop a suite of metrics to evaluate structured public feedback exercises informed by representative AI Actors.

SCF Controls: AAT-11.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-1.1-005

Evaluate novel methods and technologies for the measurement of GAI-related risks including in content provenance, offensive cyber, and CBRN, while maintaining the models' ability to produce valid, reliable, and factually accurate outputs.

SCF Controls: AAT-17.4

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-1.1-006

Implement continuous monitoring of GAI system impacts to identify whether GAI outputs are equitable across various sub-populations. Seek active and direct feedback from affected communities via structured feedback mechanisms or red-teaming to monitor and improve outputs.

SCF Controls: AAT-10.13, AAT-11

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-1.1-007

Evaluate the quality and integrity of data used in training and the provenance of AI-generated content, for example by employing techniques like chaos engineering and seeking stakeholder feedback.

SCF Controls: AAT-11, AAT-12.2

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to protecting the integrity of data sources used by AI workloads through storage-level encryption and access controls. vSAN encryption at rest and VM encryption protect AI training data stored on VCF infrastructure from unauthorized access at the storage layer. VMware vCenter RBAC restricts who can access or modify VMs containing AI data. Storage policies control datastore placement and encryption properties for AI workload VMs.

At the Consumption layer, VMware Kubernetes Service (VKS) on vSphere Supervisor extends these protections to containerized AI workloads. Harbor, available both as a Supervisor Service and as a standard package on VKS clusters, is the private container image registry, with images organized by vSphere Namespace. Harbor scans images for known vulnerabilities, enforces content trust so that only signed and verified artifacts are trusted, applies role-based access control to restrict which principals can push or overwrite model artifacts and pipeline images, and uses policy-based replication to control how artifacts move between registries. After Harbor is integrated with the Supervisor, all VKS clusters on the same Supervisor automatically trust it as a source of verified images.

Kubernetes admission controllers, including ValidatingAdmissionPolicy and validating webhook controllers, can be configured on VKS clusters to enforce requirements before workloads are admitted. Relevant policies include restricting image pulls to the trusted Harbor registry, requiring that images carry a valid signature, and blocking workloads that reference unscanned or unapproved artifacts. These controls reduce the risk that a workload carrying corrupted or tampered data processing logic reaches AI training data.

GitOps workflows, supported through the Argo CD Supervisor Service, store all workload and infrastructure configuration declaratively in a Git repository as the single source of truth. Every change to the deployed state flows through Git, creating a durable record of every authorized modification. Immutable infrastructure patterns complement this approach by making unauthorized changes identifiable through drift detection between actual cluster state and the declared configuration.

Kubernetes RBAC restricts which users and service accounts can read from or write to data volumes, ConfigMaps, and Secrets within VKS clusters. Supervisor namespace isolation with resource quotas contains workloads within defined boundaries, reducing the risk that one tenant can access or alter another tenant's data. Antrea-based network policies define namespace-level ingress and egress rules, scoping the network access surface for storage components that hold AI training data or inference inputs.

CSI Volume Snapshots for VKS clusters support crash-consistent backups with deduplication, compression, and encryption. The ability to restore from a known-good snapshot provides a recovery path when data corruption is detected. The VCF CLI binary, used to interact with VCF Services consumption, publishes a sha256sum.txt checksum file alongside each download so that administrators can verify the binary's integrity before use.

Validating the provenance and accuracy of AI training data at the application level is outside VCF's scope. Data poisoning detection, training data lineage, and AI-specific data provenance tracking require application-layer tooling or AI platform components built on top of the VCF platform.

VMware Private AI Services (PAIS) provides integrity controls for both model artifacts and data sources used by AI workloads. Model revisions in the Model Gallery use Harbor as a storage backend, and each push of the same model receives a unique digest so that successive revisions are content-addressed and distinguishable. Before upload, model files should be validated for integrity by checking hash codes, scanned for malware and serialization attacks, and tested for correct inference behavior using Triton Inference Server as part of the sandbox validation process in a Deep Learning VM. This pre-upload validation

provides cryptographic verification and behavioral testing that helps prevent contaminated or tampered model data from entering production.

The Model Gallery uses Harbor project access capabilities to restrict access to sensitive data used to train and tune models. Harbor integration requires authentication and certificate-based trust for model retrieval, and PAIS requires CA trust bundles to be imported for the Private Harbor registry, content management systems used as data sources in Data Indexing and Retrieval, and MCP servers providing tools for agents. These access controls and verified chains of trust restrict unauthorized modification of model and data artifacts. The documentation recommends using the Model Gallery to reduce redundant copies of models and maintain model integrity in a controlled environment.

For knowledge base data sources used by retrieval-augmented generation, PAIS Data Indexing and Retrieval enables automatic collection and indexing of data updates from linked data sources over time, supporting external content management systems such as Microsoft SharePoint and Amazon S3 compatible stores. Each knowledge base data source is created with a specific type that cannot be changed after creation, which helps prevent silent reconfiguration of the underlying source provider. The content management system used as a data source must be supported, and valid credentials are required for access to the content location. SharePoint data sources require user name and password authentication, and service account key file data sources require the information from the service account key file during data source creation. PAIS uses a CA trust bundle for secure communication with external services and databases, establishing verified transport between PAIS and its data sources.

VMware vDefend (Contributes)

VMware vDefend contributes to protecting the integrity of source data used by AI workloads on VCF through network-level security controls that restrict access to data stores and pipelines and detect malicious activity targeting data repositories.

Distributed Firewall microsegmentation enforces least-privilege network access at the vNIC level, limiting which workloads and users can reach databases, object stores, and data pipelines used for AI training and operation. Attacks that attempt to contaminate or corrupt AI source data require network access to the stores where training data resides; microsegmentation restricts that access to authorized workloads and identities. Security Intelligence provides visibility into traffic flows between workloads, generates firewall rule recommendations, and validates that security policies are enforced. Security Intelligence administrators can monitor for traffic leakage between environments, establish exceptions for authorized communication paths, and implement lockdown protection rules to control unauthorized inter-environment traffic. Gateway Firewall provides an additional enforcement point at the perimeter of network segments containing sensitive data repositories.

vDefend IDS/IPS inspects traffic to and from data repositories for known exploit patterns, command-and-control traffic, lateral movement, and data exfiltration signatures. The IDS/IPS filestore capability stores matched files to disk when a signature rule fires, supporting forensic analysis and post-incident review of suspected data integrity compromises. IDS/IPS signature-based detection also uses file magic inspection, identifying file types by their binary structure via libmagic, to detect misrepresented or malformed files in transit to data repositories.

Malware Prevention intercepts and analyzes files transferred over the network to data repositories. Profiles can be configured to send files to the cloud for detailed analysis or to process them locally without cloud connectivity. Verdict decisions classify files as Benign, Trusted, Highly Trusted, or Malicious. Security Services Platform Malware Prevention collects file analysis data including file hash submissions, assigns an Analyst UUID to each cloud submission for tracking, and records in-memory script buffer inspection details with associated analysis reports. The Security Services Platform also provides backup and restore mechanisms covering hardware malfunctions, software malfunctions, data corruption, and malware incidents, providing a recovery path if SSP itself is compromised.

vDefend contributes to this control because its microsegmentation, threat detection, and file inspection capabilities reduce the attack surface and detection gap for data poisoning attacks targeting AI workloads. Protecting data integrity at the application layer, including content validation, provenance tracking, and checksums within data stores, is outside vDefend's domain and requires organizational processes and additional tooling.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-1.1-008

Define use cases, contexts of use, capabilities, and negative impacts where structured human feedback exercises, e.g., GAI red-teaming, would be most beneficial for GAI risk measurement and management based on the context of use.

SCF Controls: AAT-11, AAT-26.3, TDA-01.1, TDA-21, VPM-10

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-1.1-009

Track and document risks or opportunities related to all GAI risks that cannot be measured quantitatively, including explanations as to why some risks cannot be measured (e.g., due to technological limitations, resource constraints, or trustworthy considerations). Include unmeasured risks in marginal risks.

SCF Controls: AAT-16.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-1.3-001

Define relevant groups of interest (e.g., demographic groups, subject matter experts, experience with GAI technology) within the context of use as part of plans for gathering structured public feedback.

SCF Controls: AAT-11

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-1.3-002

Engage in internal and external evaluations, GAI red-teaming, impact assessments, or other structured human feedback exercises in consultation with representative AI Actors with expertise and familiarity in the context of use, and/or who are representative of the populations associated with the context of use.

SCF Controls: AAT-07.1, AAT-10, AAT-11, VPM-10

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure that supports AI Test, Evaluation, Validation and Verification (TEVV) activities throughout the model lifecycle. Deep learning VMs provisioned from PAIS images provide dedicated testing environments for AI prototyping, fine-tuning, validation, and inference workloads. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so TEVV activities occur in conditions comparable to the deployment environment.

The validate-then-publish workflow creates a formal gate for AI deployment. MLOps engineers validate models for inference functionality using the Triton Inference Server and evaluate performance and safety against business use cases, including examination of inference requests for malicious behavior and functional testing. Models must pass this validation before being uploaded to the Model Gallery for production use. Model integrity is verified through hash code validation, and each revision is assigned a unique digest stored as an OCI artifact with an immutable manifest, giving cryptographic verification that the tested model is the same artifact that gets deployed.

The platform also supports TEVV through its monitoring stack. VCF Operations monitors GPU consumption at cluster, host system, and host properties levels. Each DL VM includes a pre-installed DCGM Exporter with Prometheus collection at 5-second scrape intervals and Grafana visualization, supplying performance telemetry during testing and evaluation. The software stack for running AI applications on top of NVIDIA GPUs is validated in advance, and NVIDIA NGC containers are pre-tested and ready to run, reducing variables during TEVV by providing a known-good baseline.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-1.3-003

Verify those conducting structured human feedback exercises are not directly involved in system development tasks for the same GAI model.

SCF Controls: HRS-11

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.2-001

Assess and manage statistical biases related to GAI content provenance through techniques such as re-sampling, re-weighting, or adversarial training.

SCF Controls: AAT-06, AAT-10.8, AAT-12.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.2-002

Document how content provenance data is tracked and how that data interacts with privacy and security.
Consider: Anonymizing data to protect the privacy of human subjects; Leveraging privacy output filters;
Removing any personally identifiable information (PII) to prevent potential harm or misuse.

SCF Controls: AAT-12.1, AAT-26, DCH-23

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides explicit data source identification and tracking for AI workloads. The Data Indexing and Retrieval service offers a Knowledge Bases interface where users create and manage knowledge bases, and a knowledge base can be linked to up to ten data sources. Each data source is created with a name, description, type, and the required properties for retrieving content from its original location, and a single data source can be linked to multiple knowledge bases. The data source type is fixed at creation. Supported data source types include Microsoft SharePoint sites, Amazon S3 compatible stores, and other enterprise content management systems. Each data source stores the credentials required for retrieval, such as service account key file information when applicable, which creates a documented connection between the AI system and its data origins.

During knowledge base indexing, PAIS tracks document-level detail including crawled documents, new documents found, deleted documents, existing unmodified documents, existing modified documents requiring embedding updates, and skipped documents. It captures size metrics including total megabytes downloaded, new chunks indexed corresponding to inference server invocations for embeddings, and final index size after removal of deleted documents. It records timing metrics for download, embedding generation, and the overall index operation. Document counts returned by retrieval queries are tracked, and counts of documents per media type such as PDF and PPTX are maintained for each knowledge base. Automatic collection and indexing of updates from linked data sources over time keeps these records current.

For runtime visibility, PAIS supports LLM trace collection through an OpenTelemetry Collector by configuring the `spec.observability.llmTraces` section of the `PAISConfiguration` custom resource, with traces transmitted over HTTP/protobuf or gRPC. PAIS metrics, including knowledge base activity, can be visualized in third-party observability platforms such as Grafana. The Data Indexing and Retrieval component also implements a Model Context Protocol (MCP) server that exports a set of search tools for each knowledge base, which gives downstream AI applications a documented programmatic surface tied back to specific data sources. PAIS AI catalog items, including AI Workstation, DSM Database, Triton Inferencing Server, and AI Kubernetes Cluster, are added to the VCF Automation catalog by running the PAIS Quickstart, with required images uploaded in advance to a Harbor registry in disconnected environments.

Organizational decisions about which data sources are approved for AI use, how data lineage is documented outside the platform, and how training data provenance is governed remain the responsibility of the organization. PAIS supplies the technical registration and tracking infrastructure that the organization configures and operates.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.2-003

Provide human subjects with options to withdraw participation or revoke their consent for present or future use of their data in GAI applications.

SCF Controls: PRI-03.4

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.2-004

Use techniques such as anonymization, differential privacy or other privacy-enhancing technologies to minimize the risks associated with linking AI-generated content back to individual human subjects.

SCF Controls: DCH-23

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.3-001

Consider baseline model performance on suites of benchmarks when selecting a model for fine tuning or enhancement with retrieval-augmented generation.

SCF Controls: CFG-02

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.3-002

Evaluate claims of model capabilities using empirically validated methods.

SCF Controls: AAT-10.16

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.3-003

Share results of pre-deployment testing with relevant GAI Actors, such as those with system release approval authority.

SCF Controls: AAT-10.15

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.3-004

Utilize a purpose-built testing environment such as NIST Dioptra to empirically evaluate GAI trustworthy characteristics.

SCF Controls: TDA-07

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.5-001

Avoid extrapolating GAI system performance or capabilities from narrow, non- systematic, and anecdotal assessments.

SCF Controls: AAT-10.16

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.5-002

Document the extent to which human domain knowledge is employed to improve GAI system performance, via, e.g., RLHF, fine-tuning, retrieval-augmented generation, content moderation, business rules.

SCF Controls: AAT-20.3

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure for several of the human domain knowledge mechanisms listed in this control, giving organizations the platform through which domain expertise is incorporated into AI system performance.

For retrieval-augmented generation, PAIS includes the Data Indexing and Retrieval service, which provides a Knowledge Bases interface for creating and managing knowledge bases. A knowledge base is built from text chunks and vector embeddings generated using an embedding model endpoint deployed through PAIS Model Endpoints, and a single knowledge base can be linked to as many as 10 data sources. Supported data sources include Microsoft SharePoint sites, Amazon S3 compatible stores, and other external content management systems, allowing domain experts to curate and update the knowledge that augments model responses without modifying the underlying model. Knowledge base operations record document counts, embedding update activity, total megabytes downloaded, chunk counts, and timing metrics across the download, embedding, and indexing phases, producing technical artifacts that an organization can reference when documenting how RAG augments its AI systems.

For business rules style direction, PAIS deploys agents in Agent Builder that consume completion model endpoints. An agent can be configured with static tool execution to access knowledge bases in Data Indexing and Retrieval over MCP when a tool-calling model is unavailable. Agents are deployed alongside model endpoints, RAG applications, and knowledge bases within an organization namespace once PAIS has been activated by an organization administrator or MLOps engineer.

PAIS is consumed through VCF Automation. AI catalog items are added by running the PAIS Quickstart, allowing users in the organization to provision AI workloads including model endpoints, agents, and RAG applications from a curated catalog. The catalog items give an organization a defined provisioning surface from which AI workloads with documented RAG and agent configurations are deployed.

PAIS does not provide reinforcement learning from human feedback or built-in content moderation features. Organizations requiring those mechanisms would implement them on Deep Learning VMs or through external tooling. The platform's contribution is the infrastructure and workflow through which RAG knowledge curation, agent instructions, and catalog-driven provisioning of AI workloads are applied and tracked.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.5-003

Review and verify sources and citations in GAI system outputs during pre- deployment risk measurement and ongoing monitoring activities.

SCF Controls: AAT-12.1, AAT-26

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides explicit data source identification and tracking for AI workloads. The Data Indexing and Retrieval service offers a Knowledge Bases interface where users create and manage knowledge bases, and a knowledge base can be linked to up to ten data sources. Each data source is created with a name, description, type, and the required properties for retrieving content from its original location, and a single data source can be linked to multiple knowledge bases. The data source type is fixed at creation. Supported data source types include Microsoft SharePoint sites, Amazon S3 compatible stores, and other enterprise content management systems. Each data source stores the credentials required for retrieval, such as service account key file information when applicable, which creates a documented connection between the AI system and its data origins.

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For runtime visibility, PAIS supports LLM trace collection through an OpenTelemetry Collector by configuring the `spec.observability.llmTraces` section of the `PAISConfiguration` custom resource, with traces transmitted over HTTP/protobuf or gRPC. PAIS metrics, including knowledge base activity, can be visualized in third-party observability platforms such as Grafana. The Data Indexing and Retrieval component also implements a Model Context Protocol (MCP) server that exports a set of search tools for each knowledge base, which gives downstream AI applications a documented programmatic surface tied back to specific data sources. PAIS AI catalog items, including AI Workstation, DSM Database, Triton Inferencing Server, and AI Kubernetes Cluster, are added to the VCF Automation catalog by running the PAIS Quickstart, with required images uploaded in advance to a Harbor registry in disconnected environments.

Organizational decisions about which data sources are approved for AI use, how data lineage is documented outside the platform, and how training data provenance is governed remain the responsibility of the organization. PAIS supplies the technical registration and tracking infrastructure that the organization configures and operates.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.5-004

Track and document instances of anthropomorphization (e.g., human images, mentions of human feelings, cyborg imagery or motifs) in GAI system interfaces.

SCF Controls: AAT-26.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.5-005

Verify GAI system training data and TEVV data provenance, and that fine-tuning or retrieval-augmented generation data is grounded.

SCF Controls: AAT-12.1

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides explicit data source identification and tracking for AI workloads. The Data Indexing and Retrieval service offers a Knowledge Bases interface where users create and manage knowledge bases, and a knowledge base can be linked to up to ten data sources. Each data source is created with a name, description, type, and the required properties for retrieving content from its original location, and a single data source can be linked to multiple knowledge bases. The data source type is fixed at creation. Supported data source types include Microsoft SharePoint sites, Amazon S3 compatible stores, and other enterprise content management systems. Each data source stores the credentials required for retrieval, such as service account key file information when applicable, which creates a documented connection between the AI system and its data origins.

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Organizational decisions about which data sources are approved for AI use, how data lineage is documented outside the platform, and how training data provenance is governed remain the responsibility of the organization. PAIS supplies the technical registration and tracking infrastructure that the organization configures and operates.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.5-006

Regularly review security and safety guardrails, especially if the GAI system is being operated in novel circumstances. This includes reviewing reasons why the GAI system was initially assessed as being safe to deploy.

SCF Controls: AAT-01

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.6-001

Assess adverse impacts, including health and wellbeing impacts for value chain or other AI Actors that are exposed to sexually explicit, offensive, or violent information during GAI training and maintenance.

SCF Controls: AAT-17.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.6-002

Assess existence or levels of harmful bias, intellectual property infringement, data privacy violations, obscenity, extremism, violence, or CBRN information in system training data.

SCF Controls: AAT-12, AAT-17.1, AAT-17.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.6-003

Re-evaluate safety features of fine-tuned models when the negative risk exceeds organizational risk tolerance.

SCF Controls: AAT-10, AAT-10.10

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure that supports AI Test, Evaluation, Validation and Verification (TEVV) activities throughout the model lifecycle. Deep learning VMs provisioned from PAIS images provide dedicated testing environments for AI prototyping, fine-tuning, validation, and inference workloads. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so TEVV activities occur in conditions comparable to the deployment environment.

The validate-then-publish workflow creates a formal gate for AI deployment. MLOps engineers validate models for inference functionality using the Triton Inference Server and evaluate performance and safety against business use cases, including examination of inference requests for malicious behavior and functional testing. Models must pass this validation before being uploaded to the Model Gallery for production use. Model integrity is verified through hash code validation, and each revision is assigned a unique digest stored as an OCI artifact with an immutable manifest, giving cryptographic verification that the tested model is the same artifact that gets deployed.

The platform also supports TEVV through its monitoring stack. VCF Operations monitors GPU consumption at cluster, host system, and host properties levels. Each DL VM includes a pre-installed DCGM Exporter with Prometheus collection at 5-second scrape intervals and Grafana visualization, supplying performance telemetry during testing and evaluation. The software stack for running AI applications on top of NVIDIA GPUs is validated in advance, and NVIDIA NGC containers are pre-tested and ready to run, reducing variables during TEVV by providing a known-good baseline.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.6-004

Review GAI system outputs for validity and safety: Review generated code to assess risks that may arise from unreliable downstream decision-making.

SCF Controls: AAT-26

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.6-005

Verify that GAI system architecture can monitor outputs and performance, and handle, recover from, and repair errors when security anomalies, threats and impacts are detected.

SCF Controls: AAT-19

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.6-006

Verify that systems properly handle queries that may give rise to inappropriate, malicious, or illegal usage, including facilitating manipulation, extortion, targeted impersonation, cyber-attacks, and weapons creation.

SCF Controls: TDA-09.4

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.6-007

Regularly evaluate GAI system vulnerabilities to possible circumvention of safety measures.

SCF Controls: VPM-06

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.7-001

Apply established security measures to: Assess likelihood and magnitude of vulnerabilities and threats such as backdoors, compromised dependencies, data breaches, eavesdropping, man-in-the-middle attacks, reverse engineering, autonomous agents, model theft or exposure of model weights, AI inference, bypass, extraction, and other baseline security concerns.

SCF Controls: AAT-10.5

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to evaluating the security and resilience of AI workloads through VCF Operations compliance dashboards that assess infrastructure security posture, configuration drift detection, and health monitoring across the compute, storage, and networking layers that AI workloads depend on. VCF Log Management aggregates event data that supports security assessment. The Security Configuration Guide and DISA STIG provide hardening baselines that can be validated against the infrastructure hosting AI workloads.

The Consumption layer of the VCF platform, delivered through VMware Kubernetes Service (VKS) and vSphere Supervisor, provides pre-deployment evaluation mechanisms for AI and machine learning workloads. DISA STIGs are available for both the Supervisor control plane and VKS cluster VKr images, supplying a structured body of security requirements against which AI workload infrastructure can be evaluated before production deployment. AI and machine learning workloads on Supervisor run on Ubuntu-based Kubernetes release images (VKrs), which are the designated runtime for such workloads in VKS clusters. Cluster readiness assessment is built into the VKS Controller, which reports status conditions covering cluster phase and component health; administrators can query these conditions using kubectl to verify that the infrastructure meets defined readiness criteria before AI workloads are scheduled.

Pod Security Admission is enforced on VKS clusters and provides a pre-execution gate for workload pods. Starting with VKr v1.25, the `kubernetes.security.podSecurityStandard` configuration variable selects Baseline, Restricted, or Privileged profiles to constrain what workload pods can execute. VKS Cluster Management adds a fleet-wide policy governance layer powered by OPA Gatekeeper, with out-of-box security policy templates that mirror industry-standard Pod Security Standards profiles; policies applied at the organization or project level are automatically inherited by all child resources, providing scalable governance across the VKS cluster estate. The VKS Cluster Management Policy Insights capability surfaces health alerts for infrastructure-level failures that would prevent policy enforcement, such as an unhealthy policy operator or a failed Gatekeeper installation, giving administrators visibility into the operational state of the security control.

The Kubernetes security documentation available through Supervisor includes guidance for pre-deployment security evaluation: threat modeling to identify trust boundaries and risks, cloud-native security lifecycle checklists organized around the develop phase, and recommendations for advanced security assessment techniques such as fuzzing and security chaos engineering where appropriate. The Kubernetes Security Response Committee publishes CVE feeds and tracks vulnerabilities in the Kubernetes platform, acknowledging reports within three working days, which gives teams visibility into known security issues in the AI and machine learning runtime infrastructure. For high-trust environments where AI workloads handle sensitive data, Supervisor supports deploying workloads on confidential virtual machines, adding hardware-level memory isolation as an additional element of the security profile that can be evaluated during a pre-deployment review.

The broader AI TEVV program, including evaluation criteria, result interpretation, and model-level testing for safety, fairness, and adversarial robustness, is an organizational responsibility. VCF provides the observable, policy-governed runtime environment that supports those activities but does not perform AI-specific evaluation itself.

VMware Private AI Services (PAIS) supports pre-deployment security and resilience evaluation of AI workloads through PAIS-native mechanisms in the model lifecycle workflow. MLOps engineers validate ML models against the organization's security, privacy, and technical requirements before onboarding them to the PAIS model gallery, with deep learning VMs and the PAIS model runtime available as the validation environment for functional and behavioral testing.

PAIS observability integrates with platforms such as Grafana and VCF Operations so that MLOps engineers, application developers, and VI administrators have ongoing visibility into model and component health, quality, and behavior in deployed AI applications. Metrics collection is activated through the PAISConfiguration custom resource, with a Prometheus runtime, a configurable storage class, a configurable retention period, and optional LLM traces forwarded through the OpenTelemetry (OTel) Collector. This telemetry provides the data inputs for continued security and resilience review of models in production, including tracking of anomalous model behavior over time.

PAIS deployment leases bound how long an AI workload deployment can exist, providing a control for time-limited evaluation environments where candidate models can be exercised before broader rollout. PAIS supports both connected and disconnected (air-gapped) deployment models, allowing organizations to perform security evaluation of AI workloads in isolated environments. A Harbor or other OCI-compliant registry is used as the artifact source for models and container images, giving organizations a controlled location from which validated artifacts are retrieved.

Resilience evaluation for AI workloads depends on the underlying VCF platform, which provides vSphere High Availability for automatic VM restart on host failure, Distributed Resource Scheduler for workload balancing, and vMotion for live migration of GPU-accelerated VMs. These capabilities are VCF platform mechanisms; see VCF Coverage for the underlying resilience infrastructure.

VMware vDefend (Contributes)

VMware vDefend contributes to evaluating the security and resilience of AI and autonomous technology workloads by providing continuous network-level security assessment capabilities that operate during and after deployment.

The vDefend distributed firewall includes a dedicated Security Assessment workflow that generates Security Segmentation Reports and Blast Radius Reports. These reports identify at-risk workloads, communication chains, and traffic flows, providing structured input for evaluating whether network isolation is correctly applied to AI workload segments and identifying gaps in protection coverage.

Security Intelligence offers traffic flow analysis, a quantitative Security Segmentation Score, and environment-level monitoring that can inform resilience assessments. The Security Segmentation Score calculates posture by evaluating Distributed Firewall configuration and traffic rules; score values from 0 to 50 indicate low security, 50 to 80 indicate medium security, and values above 80 indicate a highly secure environment. The score breakdown report details total environments and environment pairs, protection rule actions, and highlights environment pairs lacking inter-environment policies. Security Intelligence generates policy recommendations by analyzing VM traffic flows over specified time periods and boundary definitions, and monitors inter-environment traffic to detect leakage and support lockdown rule implementation. The Security Intelligence dashboard displays environments, environment pairs, and segmentation state to help administrators understand current security posture.

The Security Services Platform collects and surfaces IDS/IPS readiness and performance metrics. Administrators can view intrusion details including severity, CVE identifiers, CVSS scores, and affected VMs through vDefend IDS/IPS to inform mitigation planning. Intelligent Assist assists with IDS policies, rules, and security profiles during remediation operations. The vDefend Threat Intelligence Service Portal supplies application-level threat intelligence that can inform distributed and gateway firewall policies targeting lateral and perimeter network threats.

The broader TEVV program for AI security and resilience, including defining evaluation criteria, establishing test plans, and interpreting results against organizational risk thresholds, is an organizational responsibility outside vDefend's scope.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.7-002

Benchmark GAI system security and resilience related to content provenance against industry standards and best practices. Compare GAI system security features and content provenance methods against industry state-of-the-art.

SCF Controls: AAT-10.17

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.7-003

Conduct user surveys to gather user satisfaction with the AI-generated content and user perceptions of content authenticity. Analyze user feedback to identify concerns and/or current literacy levels related to content provenance and understanding of labels on content.

SCF Controls: AAT-11.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.7-004

Identify metrics that reflect the effectiveness of security measures, such as data provenance, the number of unauthorized access attempts, inference, bypass, extraction, penetrations, or provenance verification.

SCF Controls: GOV-05

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.7-005

Measure reliability of content authentication methods, such as watermarking, cryptographic signatures, digital fingerprints, as well as access controls, conformity assessment, and model integrity verification, which can help support the effective implementation of content provenance techniques. Evaluate the rate of false positives and false negatives in content provenance, as well as true positives and true negatives for verification.

SCF Controls: AAT-10.17, AAT-12.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.7-006

Measure the rate at which recommendations from security checks and incidents are implemented. Assess how quickly the AI system can adapt and improve based on lessons learned from security incidents and feedback.

SCF Controls: VPM-05.3

Aggregate Coverage: Contributes

VCF (Contributes)

VCF provides mechanisms to track the effectiveness of remediation operations through metrics reporting, compliance management, diagnostics, and lifecycle management capabilities.

VCF Operations includes a compliance management function that detects violations, highlights risk areas, and provides actionable remediation recommendations. This gives administrators visibility into the current compliance posture of the environment and surfaces specific items requiring remediation. Each metric panel within VCF Operations includes a Check Recommendations button that provides guidance on remediation steps, linking metrics directly to corrective actions. Alert definitions can be configured with symptom conditions based on metrics or properties, each with associated remediation recommendations. When alerts trigger, VCF Operations maps out-of-the-box workflows to those recommendations and associates them with specific alerts for automated remediation. Actions can be defined to run on target objects and added as recommendations for alert definitions configured across different base objects, tying remediation activities directly to the alerts and metrics that triggered them. The Recommended Actions widget on the VCF Operations Home dashboard displays how many objects are in a critical state and how many require immediate attention, giving administrators a real-time summary of outstanding remediation work.

During incident investigation, administrators can flag individual metrics and mark specific metrics as root cause, creating a traceable link between the detected issue and the remediation action taken. This supports post-remediation analysis by recording which metrics drove the remediation and how they were resolved. The Recent Tasks screen allows administrators to confirm that optimization and remediation actions have completed and to filter actions by their status, identifying actions that succeeded, failed, or reached maximum time limits.

VCF Operations Diagnostics automatically detects known issues and critical vulnerabilities within the VCF software stack, providing detailed descriptions, root cause insights, and actionable remediation steps. This integrated diagnostics capability builds on several legacy tools and operational data to provide issue detection and remediation across the platform. Efficiency alerts identify resources that can be reclaimed, and system audit reports provide visibility into super metrics across the environment. The metric chart widget enables creation of ad-hoc reports when analyzing metrics, supporting custom views that track remediation trends and measure time to resolution.

Workload Optimization generates exact moves planned for compute and storage resources as part of optimization recommendations. The Workload Optimization UI pages enable monitoring of optimizing moves in fully automated systems and can be used to conduct research and run actions directly in systems that are not fully automated, allowing administrators to track whether automated or manual remediation actions achieved their intended results.

For host-level remediation, vSphere Lifecycle Manager (vLCM) tracks the current state of each host against the desired cluster specification. vLCM enables cluster remediation by applying a desired software state on each host whose current state differs from the desired specification. After remediation operations, administrators can check the compliance of a cluster or host against the desired state to determine whether additional remediation steps are required. When remediation fails, vSphere Lifecycle Manager provides information about the specific reasons for failure, supporting root cause analysis and follow-up corrective action. Performance metrics can be configured with redInterval and yellowInterval thresholds that trigger defined actions when sustained beyond specified durations, providing time-based tracking of whether remediation brought metrics back within acceptable ranges. For CPU resources, vSphere documentation identifies that CPU usage above 90 percent combined with a CPU ready value above 20 percent indicates that performance is being impacted and corrective action should be considered, providing a documented threshold that connects observed metrics to remediation decisions.

Custom dashboards with role-based access control allow organizations to build remediation-specific views. Dashboards support differentiated visibility for various user roles and can be assigned to specific user groups. Automated reports provide the ability to track remediation metrics over time, giving visibility into remediation trends and outcomes across the infrastructure.

VMware vDefend (Contributes)

VMware vDefend (vDefend) contributes to tracking the effectiveness of remediation operations through its network security metrics, remediation workflow tracking, and operational dashboards across its distributed firewall, IDS/IPS, malware prevention, and network detection capabilities.

When IDS/IPS detects intrusion events, administrators can initiate remediation using either a Targeted Strategy for surgical, safer-to-deploy policies with potentially lower effectiveness, or a Comprehensive Strategy for broader remediation with potentially lower safety. Intelligent Assist tracks these remediation actions by adding them as rules to the Intelligent Assist Remediation policy using the naming convention prefix 'ai-', providing a structured record of remediation actions taken. Objects generated during remediation that encounter errors require manual cleanup, meaning tracking must account for failed remediation attempts.

The Security Services Platform (SSP) provides a metrics service with Top N analysis, lifetime counters, and historical trend statistics for monitoring firewall operational state. The Metrics feature collects data on entities in NSX and SSP environments and presents it through the Platform dashboard. Metrics APIs support advanced server-side filtering, multi-dimensional aggregation, grouping, and related-metrics queries. Two metric models are available: a Status model that retains and overwrites the last reported value only when it changes, and a Point-in-Time model used in Top-N analysis where the last value is overwritten with new data points and times out after 24 hours. Bare Metal Servers report rule statistics to the Metrics Server every five minutes, extending rule-level visibility to bare metal infrastructure in addition to virtualized workloads. The Flow Retention metric displays the number of days of flow data retained based on ingestion configuration, giving operators visibility into the historical depth of network telemetry available for retrospective analysis.

The SSP also tracks operational data management metrics including records purged from Malware Prevention Service tables, covering INSPECTION_EVENTS, INTERCEPTED_ENTITIES, and MALWARE_CLASS categories. The Process Analysis Report includes links to artifact reports on the VirusTotal portal, supporting cross-reference of malware findings against external threat intelligence.

The Security Segmentation Report provides security operations personnel with an assessment of the organization's segmentation status and a basis for prioritizing remediation efforts. Security Intelligence supports prioritizing critical workloads and throttling the ingestion of lower-priority flows to maintain system stability under load. In the NDR component, campaigns with higher impact scores are identified as higher priority to address and triage, enabling severity-based remediation tracking.

Under memory pressure, vDefend IDS/IPS follows a prioritized approach to event handling, dropping non-critical severity data (Low and Informational) first, followed by Medium, then High and Critical severity events and packet captures only as a last resort. This behavior affects the metrics available for remediation effectiveness analysis under high-load conditions.

These capabilities address the network security dimension of remediation metrics. Tracking of patching, compute, and infrastructure remediation is handled by other VCF components such as VCF Operations and vSphere Lifecycle Manager.

Not applicable for this control: VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.7-007

Perform AI red-teaming to assess resilience against: Abuse to facilitate attacks on other systems (e.g., malicious code generation, enhanced phishing content), GAI attacks (e.g., prompt injection), ML attacks (e.g., adversarial examples/prompts, data poisoning, membership inference, model extraction, sponge examples).

SCF Controls: AAT-12.2, VPM-10

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.7-008

Verify fine-tuning does not compromise safety and security controls.

SCF Controls: AAT-17.5

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to preserving security controls during AI model fine-tuning and updates by providing VM snapshots that capture point-in-time state before changes, resource pools that maintain resource boundaries during updates, and storage policies that persist encryption and placement properties regardless of workload changes. vCenter RBAC controls who can modify AI workload VMs during update operations.

For AI and machine learning workloads deployed through VMware Kubernetes Service (VKS) on the vSphere Supervisor, the Consumption layer adds enforcement mechanisms that operate at the Kubernetes API level. Kubernetes admission controllers, including validating and mutating webhooks, intercept API requests before workloads are scheduled. MutatingAdmissionPolicy resources allow platform teams to define declarative policies that automatically validate or transform AI workload configurations at admission time, with reinvocationPolicy controlling whether mutations apply once or conditionally on subsequent plugin modifications. MutatingAdmissionPolicyBinding objects scope enforcement to specific API kinds and namespaces, so admission controls can be targeted at the namespaces where fine-tuning jobs run. VKS Cluster Management extends this with OPA Gatekeeper-driven security guardrails that restrict pod admission to compliant configurations, and VCF Automation mutation policies with the pod-security template can automatically remediate non-compliant pods before admission rather than merely blocking them.

Pod Security Admission (PSA) enforces restrictions on what containerized workloads can do within a namespace. Administrators can configure the podSecurityStandard.enforce setting to restricted, which limits privilege escalation and restricts Linux capabilities for workloads in a given namespace; audit and warn modes can be applied simultaneously to observe policy deviations without interrupting workloads.

Kubernetes RBAC governs who can create, modify, or delete workload resources, including those used in fine-tuning operations. Roles and RoleBindings can be scoped to the namespaces hosting AI workloads, limiting the blast radius of any misconfiguration or unauthorized action. Platform teams should review default RBAC rights for hardening opportunities.

Kubernetes audit logging records all API requests, including creation and modification of AI workload resources, providing a trail of actions taken during fine-tuning operations. The immutable infrastructure pattern is supported in VKS: unauthorized changes can be overwritten by automated processes or blocked at the admission control layer, reducing the risk that side effects of fine-tuning operations modify cluster configuration in ways that affect other security controls.

VKS also supports MachineHealthCheck, a Cluster API resource that specifies node conditions such as NotReady status, MemoryPressure, DiskPressure, and PIDPressure, along with a timeout duration for each. When a node meets the unhealthy criteria, the controller triggers automatic remediation. For resource-intensive fine-tuning workloads that may degrade node health, MachineHealthCheck helps maintain the resilience of cluster infrastructure by returning unhealthy nodes to a compliant state without requiring manual intervention.

Harbor, included with Supervisor, serves as a private container image registry organized by vSphere namespace. VKS clusters on the same Supervisor automatically trust Harbor and can pull container images from it, supporting controlled supply chain integrity for AI workload images. In air-gapped deployments, Harbor Supervisor Service must be configured as the private registry for VKS images.

Managing AI model-level risks during fine-tuning (output quality, safety properties) is outside VCF's scope.

VMware Private AI Services (PAIS) provides AI-infrastructure controls that help protect existing safety and security properties during model fine-tuning. Deep learning VMs are the designated fine-tuning environment, where data scientists download models from public or organizational registries and validate or fine-tune them before storing the result in the Model Gallery. PAIS guidance is explicit that models distributed within an organization should be those validated on a deep learning VM

rather than models pulled directly from the Internet, since Internet-sourced models can contain malicious code or be tuned for malicious behavior.

The validate-then-publish workflow acts as a safety gate after fine-tuning. MLOps engineers validate ML models against the security, privacy, and technical requirements of the organization before storing them in the Model Gallery. Sandbox validation in deep learning VMs checks model file integrity by verifying hash codes, scans the files for malware, scans them for serialization attacks, and tests inference behavior using the Triton Inference Server. Model performance and safety are evaluated for the intended business use case by examining inference requests for malicious behavior and performing hands-on functional tests. A fine-tuned model passes the same validation gate as any new model before reaching production model endpoints.

The Model Gallery is the controlled repository for validated models, reducing redundant copies and maintaining model integrity across the environment. Hash code validation confirms that the revision deployed to a production endpoint is the one that passed sandbox validation, and the same hash references support a rollback path: if a fine-tuned model exhibits degraded safety or compliance behavior after deployment, operators can revert to a previously validated revision in the gallery. When PAIS is deployed in a disconnected air-gapped environment, the validated Model Gallery and a Harbor or OCI-compliant registry remain the only paths by which models reach production endpoints, which keeps fine-tuning activity inside the controlled boundary.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.7-009

Regularly assess and verify that security measures remain effective and have not been compromised.

SCF Controls: AAT-11.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.8-001

Compile statistics on actual policy violations, take-down requests, and intellectual property infringement for organizational GAI systems: Analyze transparency reports across demographic groups, languages groups.

SCF Controls: AAT-07, AAT-12

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.8-002

Document the instructions given to data annotators or AI red-teamers.

SCF Controls: AAT-26.4

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.8-003

Use digital content transparency solutions to enable the documentation of each instance where content is generated, modified, or shared to provide a tamper-proof history of the content, promote transparency, and enable traceability. Robust version control systems can also be applied to track changes across the AI lifecycle over time.

SCF Controls: AAT-20.1, AAT-26.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.8-004

Verify adequacy of GAI system user instructions through user testing.

SCF Controls: AAT-17.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.9-001

Apply and document ML explanation results such as: Analysis of embeddings, Counterfactual prompts, Gradient-based attributions, Model compression/surrogate models, Occlusion/term reduction.

SCF Controls: AAT-26, TDA-22

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.9-002

Document GAI model details including: Proposed use and organizational value; Assumptions and limitations, Data collection methodologies; Data provenance; Data quality; Model architecture (e.g., convolutional neural network, transformers, etc.); Optimization objectives; Training algorithms; RLHF approaches; Fine-tuning or retrieval-augmented generation approaches; Evaluation data; Ethical considerations; Legal and regulatory requirements.

SCF Controls: AAT-01.1, AAT-04.1, AAT-10, AAT-20.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.10-001

Conduct AI red-teaming to assess issues such as: Outputting of training data samples, and subsequent reverse engineering, model extraction, and membership inference risks; Revealing biometric, confidential, copyrighted, licensed, patented, personal, proprietary, sensitive, or trade-marked information; Tracking or revealing location information of users or members of training datasets.

SCF Controls: AAT-19, AAT-19.8, VPM-10

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.10-002

Engage directly with end-users and other stakeholders to understand their expectations and concerns regarding content provenance. Use this feedback to guide the design of provenance data-tracking techniques.

SCF Controls: AAT-11.1, AAT-11.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.10-003

Verify deduplication of GAI training data samples, particularly regarding synthetic data.

SCF Controls: AAT-10.18, AAT-12.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.11-001

Apply use-case appropriate benchmarks (e.g., Bias Benchmark Questions, Real Hateful or Harmful Prompts, Winogender Schemas¹⁵) to quantify systemic bias, stereotyping, denigration, and hateful content in GAI system outputs; Document assumptions and limitations of benchmarks, including any actual or possible training/test data cross contamination, relative to in-context deployment environment.

SCF Controls: AAT-06, AAT-10.8, AAT-26

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.11-002

Conduct fairness assessments to measure systemic bias. Measure GAI system performance across demographic groups and subgroups, addressing both quality of service and any allocation of services and resources. Quantify harms using: field testing with sub-group populations to determine likelihood of exposure to generated content exhibiting harmful bias, AI red-teaming with counterfactual and low-context (e.g., “leader,” “bad guys”) prompts. For ML pipelines or business processes with categorical or numeric outcomes that rely on GAI, apply general fairness metrics (e.g., demographic parity, equalized odds, equal opportunity, statistical hypothesis tests), to the pipeline or business outcome where appropriate; Custom, context-specific metrics developed in collaboration with domain experts and affected communities; Measurements of the prevalence of denigration in generated content in deployment (e.g., sub- sampling a fraction of traffic and manually annotating denigrating content).

SCF Controls: AAT-10.8

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.11-003

Identify the classes of individuals, groups, or environmental ecosystems which might be impacted by GAI systems through direct engagement with potentially impacted communities.

SCF Controls: AAT-07

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.11-004

Review, document, and measure sources of bias in GAI training and TEVV data: Differences in distributions of outcomes across and within groups, including intersecting groups; Completeness, representativeness, and balance of data sources; demographic group and subgroup coverage in GAI system training data; Forms of latent systemic bias in images, text, audio, embeddings, or other complex or unstructured data; Input data features that may serve as proxies for demographic group membership (i.e., image metadata, language dialect) or otherwise give rise to emergent bias within GAI systems; The extent to which the digital divide may negatively impact representativeness in GAI system training and TEVV data; Filtering of hate speech or content in GAI system training data; Prevalence of GAI-generated data in GAI system training data.

SCF Controls: AAT-10.8

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.11-005

Assess the proportion of synthetic to non-synthetic training data and verify training data is not overly homogenous or GAI-produced to mitigate concerns of model collapse.

SCF Controls: AAT-10.18

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.12-001

Assess safety to physical environments when deploying GAI systems.

SCF Controls: AAT-17, AAT-17.1, EMB-15

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.12-002

Document anticipated environmental impacts of model development, maintenance, and deployment in product design decisions.

SCF Controls: AAT-17.2

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides PAIS-native mechanisms that help organizations assess the environmental footprint of AI workloads. Metrics collection for PAIS is activated by uncommenting the `spec.observability.prometheusRuntime` section of the `PAISConfiguration` custom resource, after which DevOps and MLOps engineers can visualize PAIS health and performance metrics for components running in a VCF Automation namespace through observability platforms such as Grafana. The same observability pipeline lets MLOps engineers and application developers monitor models used in agents, giving operators visibility into the operational characteristics of individual AI workloads across deep learning VMs, VKS clusters, and RAG applications. This telemetry is the raw operational data that organizations use to quantify the energy and resource footprint of AI activity.

PAIS also supports resource efficiency through deployment leases. A lease controls how long a PAIS deployment can exist, so that AI workloads provisioned from VCF Automation catalog items are reclaimed at the end of a defined period rather than continuing to consume GPU and compute resources indefinitely. Disconnected (air-gapped) and connected deployment modes both rely on the same observability and lease controls, so the same sustainability-relevant data and lifecycle controls apply regardless of connectivity model.

Broader sustainability and green-metric monitoring at the cluster and host level is a VCF platform capability and is covered under VCF Coverage. The organizational activity of setting sustainability targets, analyzing collected metrics, and documenting environmental impact is a process obligation outside PAIS itself, which is why the rating is Contributes rather than Meets.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.12-003

Measure or estimate environmental impacts (e.g., energy and water consumption) for training, fine tuning, and deploying models: Verify tradeoffs between resources used at inference time versus additional resources required at training time.

SCF Controls: AAT-17.2

Aggregate Coverage: Contributes

VCF (Contributes)

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Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-2.12-004

Verify effectiveness of carbon capture or offset programs for GAI training and applications, and address green-washing concerns.

SCF Controls: AAT-17.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-3.2-001

Establish processes for identifying emergent GAI system risks including consulting with external AI Actors.

SCF Controls: AAT-07

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-3.3-001

Conduct impact assessments on how AI-generated content might affect different social, economic, and cultural groups.

SCF Controls: AAT-07.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-3.3-003

Evaluate potential biases and stereotypes that could emerge from the AI- generated content using appropriate methodologies including computational testing methods as well as evaluating structured feedback input.

SCF Controls: AAT-06

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-3.3-004

Provide input for training materials about the capabilities and limitations of GAI systems related to digital content transparency for AI Actors, other professionals, and the public about the societal impacts of AI and the role of diverse and inclusive content generation.

SCF Controls: AAT-05

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-3.3-005

Record and integrate structured feedback about content provenance from operators, users, and potentially impacted communities through the use of methods such as user research studies, focus groups, or community forums. Actively seek feedback on generated content quality and potential biases. Assess the general awareness among end users and impacted communities about the availability of these feedback channels.

SCF Controls: AAT-10.8, AAT-11

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-4.2-001

Conduct adversarial testing at a regular cadence to map and measure GAI risks, including tests to address attempts to deceive or manipulate the application of provenance techniques or other misuses. Identify vulnerabilities and understand potential misuse scenarios and unintended outputs.

SCF Controls: AAT-10, AAT-26

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides infrastructure that supports AI Test, Evaluation, Validation and Verification (TEVV) activities throughout the model lifecycle. Deep learning VMs provisioned from PAIS images provide dedicated testing environments for AI prototyping, fine-tuning, validation, and inference workloads. These DL VMs run on the same GPU-accelerated workload domains, vGPU profiles, and VM classes used in production, so TEVV activities occur in conditions comparable to the deployment environment.

The validate-then-publish workflow creates a formal gate for AI deployment. MLOps engineers validate models for inference functionality using the Triton Inference Server and evaluate performance and safety against business use cases, including examination of inference requests for malicious behavior and functional testing. Models must pass this validation before being uploaded to the Model Gallery for production use. Model integrity is verified through hash code validation, and each revision is assigned a unique digest stored as an OCI artifact with an immutable manifest, giving cryptographic verification that the tested model is the same artifact that gets deployed.

The platform also supports TEVV through its monitoring stack. VCF Operations monitors GPU consumption at cluster, host system, and host properties levels. Each DL VM includes a pre-installed DCGM Exporter with Prometheus collection at 5-second scrape intervals and Grafana visualization, supplying performance telemetry during testing and evaluation. The software stack for running AI applications on top of NVIDIA GPUs is validated in advance, and NVIDIA NGC containers are pre-tested and ready to run, reducing variables during TEVV by providing a known-good baseline.

Not applicable for this control: VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-4.2-002

Evaluate GAI system performance in real-world scenarios to observe its behavior in practical environments and reveal issues that might not surface in controlled and optimized testing environments.

SCF Controls: AAT-25

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-4.2-003

Implement interpretability and explainability methods to evaluate GAI system decisions and verify alignment with intended purpose.

SCF Controls: AAT-20.1

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-4.2-004

- Monitor and document instances where human operators or other systems override the GAI's decisions.
- Evaluate these cases to understand if the overrides are linked to issues related to content provenance.

SCF Controls: AAT-15.2

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control MS-4.2-005

Verify and document the incorporation of results of structured public feedback exercises into design, implementation, deployment approval (“go”/“no-go” decisions), monitoring, and decommission decisions.

SCF Controls: AAT-11.1, AAT-11.3

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

